Main Routine

EDC Equipment Driver
Sends Fault, Quality and Trend to the EDC Gateway

Configure Driver in the following Routines:
_090_USER_Config
_092_USER_LogixRemote
_093_USER_LogixFaults
_094_USER_LogixTools
_095_USER_LogixTrends
_096_USER_LogixCycles

Reference the EDC Driver Configuration document for additional details.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 1
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Main Routine

v1.4 - add Program Version and Time Synch
Manually Toggle UpdateVersion to send Version to Gateway
Manually Toggle UpdateTimeSync to Sync Time from Gateway
v1.11 DHF018 - Refresh EQ Driver Version and Controller Hardware/Firmware Version
v1.16 DHF024 Add Quality Targets
v1.17 DHF029 Add Alert Status to PreDefined Fault Events
v1.18 DHF024 Quality Target Update to support Scaled and Unscaled Target Values
v1.19 DHF031 Duplicate Message EN Inhibit on Startup
v2.00 EDC2 New Features
v2.01 EDC2 PreDefined Changes
v2.02 Trend/Cycle Changes
v2.03 FEC_Share and Toyopuc Share Changes
v2.04 Driver Reset cleared FEC DateTime on EQID change
v2.05 Remove Pre.06 from MaintActive; move Toyopuc TimeSync to FEC
v2.06 TpucEnable - use Toyopuc Module Instance to make Program Uninhibit asynchronous with other Programs
v2.08 - expand Tpuc Tools from 120 to 255
v2.09 - prep for K1A deploy
v2.10 - Tool and Trend Judgement 1-5
v3.02 - Ring Buffer Read Message Delay Time Optimization; Data Refresh
v3.50 Bulk Fault Message (10 Records / Message)

2

S:FS

Equipment Data Collection - User Config Toggle ON to Update Program Version and send to Gateway
UserCfg.UpdateVersion

Message Control Send Data Initiate
VersionMsgCtl.SendInit

Program Version

aoiEDC_ProgVersion	
aoiEDC_ProgVersion	aoiProgVersion ...
Major	3
Minor	5
Build	0
Patch	0
Version	Prog.Version

Reset / Configuration / Initialization Routines

3

Configuration and Initialization Routine

JSR	
Routine Name	_009_ConfigInit

Main Routine

v2.01 Move Version and TImeSync below ConfigInit
ConfigInit was clearing Version and TimeSync CTL words on initial EQID setup

4

S:FS

Message Control Send Data
VersionMsgCtl.Send
L

Equipment Data Collection - User Config Toggle ON to Update Program Version and send to Gateway
UserCfg.UpdateVersion
U

Equipment Data Collection - User Config Toggle ON to Update Program Version and send to Gateway
UserCfg.UpdateVersion

S:FS

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get
L

Equipment Data Collection - User Config Toggle ON to Update Time Sync from Gateway
UserCfg.UpdateTimeSync
U

Equipment Data Collection - User Config Toggle ON to Update Time Sync from Gateway
UserCfg.UpdateTimeSync

CLx Controller Functions

5

Monitors CLx Status
JSR
Routine Name _030_CLxStatus

Constants
(ASCII Table Initialization, Strings, etc.)

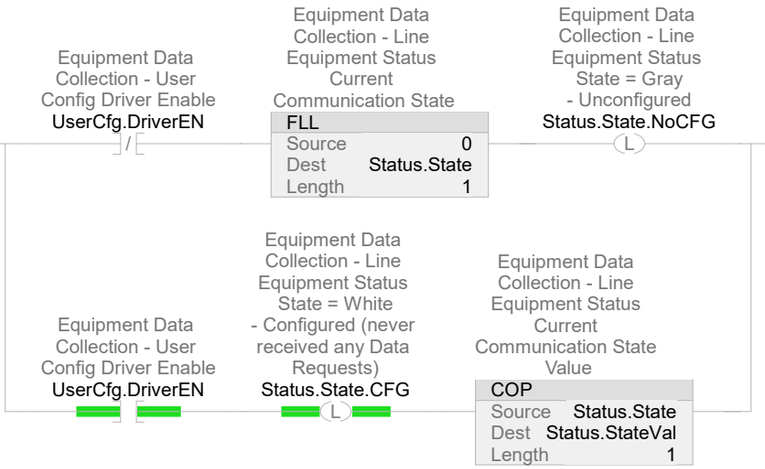
6

Define ASCII Table
JSR
Routine Name _031_AsciiTable

Main Routine

Initialize Status to NoCFG or CFG

7



Main Routine

v1.8 Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)

After power-up, block Driver Messages and monitor the Message control bits for activity
Keep Alive Time (5 minutes) + 30 seconds provides enough time to see activity during non-production windows

8

S:FS

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR
RES

Duplicate Message Detection Control Duplicate Message Delay Timer

ADD	
Source A	CFG.LiveMsgResendTime 31000
Source B	30000
Dest	DuplicateMsg.DelayTMR.PRE 61000

Duplicate Message Detection Control Duplicate Message Delay Timer

TON		
Timer	DuplicateMsg.DelayTMR	EN
Preset	61000	DN
Accum	61000	

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN
/

Listen for Duplicate Message Tag activity before enabling Driver Messages

JSR	
Routine Name	_050_DuplicateMsgDetect

v2.09 Data Refresh signal (daily, pre-shift)
Send all configured Fault Packets to refresh any bits stuck on or off after communication outages, buffer overruns, etc.
(this has mostly been an issue after shutdowns)

EqDriver uses its EQID and the Scan Counter to randomize the sending of the refresh data.
10,000 Scans take 3-4 minutes at ~20msec scan

9

Scheduled Time for Data Refresh 1-shot
DataRefreshTime

Data Refresh Setup (anti-repeat)
DataRefreshSetup
L

Data Refresh Setup (anti-repeat)
DataRefreshSetup

Scan Count 0-9999

MOD	
Source A	CLX.ScanCount 1106690484
Source B	10000
Dest	DataRefreshScanCount 6

Scan Count 0-9999

EQ	
Source A	DataRefreshScanCount 6
Source B	UserCfg.EQID 6

Diagnostic manual reset

ADD	
Source A	DataRefreshCount 97
Source B	1
Dest	DataRefreshCount 97

Data Refresh 1-shot
DataRefresh

Data Refresh Setup (anti-repeat)
DataRefreshSetup
U

Main Routine

User Configuration

10

Equipment Driver
User Configuration

JSR

Routine Name _090_USER_Config

Main Routine

USER Mapping of Data from Driver Source
Toyopuc I/O Module, CIP Remote Messages (i.e. CompactLogix, PLC-5) or embedded driver in Logix

11

Equipment Data Collection - User Config Toyopuc Driver Enable
UserCfg.TpucDriverEN

NOP

Equipment Data Collection - User Config Logix Driver Enable
UserCfg.LogixDriverEN

JSR
Routine Name _092_USER_LogixRemote

Equipment Data Collection - User Config Remote Driver Enable
UserCfg.RemDriverEN

Equipment Data Collection - User Config Logix Driver Enable
UserCfg.LogixDriverEN

Simultation for Fault Data Enable Simulation
SimFault.EN

EqDriver embedded in Logix Controller
JSR
Routine Name _093_USER_LogixFaults

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN

Equipment Quality/Health Data Mapping
JSR
Routine Name _094_USER_LogixTools

Equipment Data Collection - User Config Trend Message Enable
UserCfg.TrendMsgEN

Equipment Trend Data Mapping
JSR
Routine Name _095_USER_LogixTrends

Equipment Data Collection - User Config Cycle Message Enable
UserCfg.CycleMsgEN

Equipment Cycle Data Mapping
JSR
Routine Name _096_USER_LogixCycles

Equipment Data Collection - User Config Remote Driver Enable
UserCfg.RemDriverEN

NOP

Main Routine

Send Live Callout Data to EDC Gateway
v1.5 Separate EqFault and PreFault routines - scan EqFault first to update before evaluating for Fault/Alarm status
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

Collect Data Packets in reverse order so Packet 0 w/ PreDefined is updated and loaded last to RingBuffer -- after all other packets have been processed
(this should make Fault, Alarm, Alert events Start/Stop times align with the underlying events)

12

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Collect Fault/Status Data
FOR
Routine Name _121_FaultDataCollect
Index PKTi
-1
Initial Value CFG.FaultEq.ei
Terminal Value 0
Step Size -1

Message Control Send Data
CalloutMsgCtl.Send

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Data Change
NE
Source A FaultMsgCtl.Change
0
Source B 0

Callout Data Load for Message
FOR
Routine Name _112_CalloutDataLoad
Index PKTi
-1
Initial Value 0
Terminal Value CFG.FaultEq.ei
Step Size 1

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Callout Data Message
JSR
Routine Name _114_CalloutDataMsg

Main Routine

Send Buffered Fault/Status Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

13

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN

Load Read Records from RingBuffer to Gateway Message
JSR
Routine Name _123_FaultRingBuffer

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Fault Data Ring Buffer and Message
JSR
Routine Name _124_FaultDataMsg

Send Buffered Quality/Health Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete
v1.16 DHF024 Add Quality Targets
v2.00 Remove Quality Targets - configured on server

14

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN

Load Read Records from RingBuffer to Gateway Message
JSR
Routine Name _133_QualityRingBuffer

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Quality Data Ring Buffer and Message
JSR
Routine Name _134_QualityDataMsg

Main Routine

Send Buffered Trend Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

15

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Trend Message Enable
UserCfg.TrendMsgEN

Load Read Records from RingBuffer to Gateway Message
JSR
Routine Name _143_TrendRingBuffer

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Trend Data Ring Buffer and Message
JSR
Routine Name _144_TrendDataMsg

Send Version and Sync Time every morning before 1st Shfit
use EQID as a seconds-delay so all Equpment don't sync at same time
v1.10 - Moved Time Sync from 5am to 6am to avoid changing during 2nd Shift
v1.10 Changed EQID multiplier from 1000 to 100 (allow 10 EQ/sec instead of 1 EQ/sec - finishes updates in 3 minutes instead of 30)
v1.11 DHF018 - Refresh EQ Driver Version and Controller Hardware/Firmware Version

16

CLx Module Status CLx System Time (from WALLCLOCKTIME)
EQ
Source A CLX.SysTime[3] 15
Source B 6

CLx Module Status CLx System Time (from WALLCLOCKTIME)
EQ
Source A CLX.SysTime[4] 24
Source B 0

CLx Module Status CLx System Time (from WALLCLOCKTIME)
EQ
Source A CLX.SysTime[5] 52
Source B 0

Time Sync Message Control Get Data Initiate
TimeSyncMsgCtl.GetInit
ONS

Message Control Send Data Initiate
VersionMsgCtl.SendInit

Time Sync Message Control Message Delay Timer
MUL
Source A UserCfg.EQID 6
Source B 100
Dest TimeSyncMsgCtl.DelayTMR.PRE 600

Time Sync Message Control Message Delay Timer
TimeSyncMsgCtl.DelayTMR.EN

Time Sync Message Control Message Delay Timer
TON
Timer TimeSyncMsgCtl.DelayTMR
Preset 600
Accum 0
EN
DN

Main Routine

Send Version and Sync Time every morning before 1st Shfit
use EQID as a seconds-delay so all Equpment don't sync at same time
v1.7 - Time Sync if invalid TransTime to avoid bad Fault Event time stamps
v1.16 DHF024 Add Quality Target Periodic Update
v2.00 Remove Quality Target

17

Time Sync Message Control Message Delay Timer
TimeSyncMsgCtl.DelayTMR.DN

Time Sync Message Control Message Delay Timer
TimeSyncMsgCtl.DelayTMR
RES

Equipment Data Collection - User Config Time Synchronization Message Enable
UserCfg.TimeSyncEN

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get
L

Equipment Data Collection - User Config Version Message Enable
UserCfg.VersionEN

Message Control Send Data
VersionMsgCtl.Send
L

CLx Module Status CLx Transaction Time Invalid
CLX.TransTimeInvalid

Equipment Data Collection - User Config Time Synchronization Message Enable
UserCfg.TimeSyncEN

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get
L

Send Buffered Version Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

18

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Version Message Enable
UserCfg.VersionEN

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Message Control Send Data
VersionMsgCtl.Send

Message Control Write Data
VersionMsgCtl.Write
/

Version Data Load for Message
JSR
Routine Name _152_VersionDataLoad

Logix and Code Version Data Message
JSR
Routine Name _154_VersionDataMsg

Main Routine

Synchronizae Time from EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

19

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Time Synchronization Message Enable
UserCfg.TimeSyncEN

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Time Synchronization Data Load for Message
JSR
Routine Name _162_TimeSyncDataLoad

Time Synchronization Data Message
JSR
Routine Name _164_TimeSyncMsg

Send Buffered Cycle Data to EDC Gateway
v1.8 Check for Duplicate Message -- block Message until Message Delay is complete

20

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Cycle Message Enable
UserCfg.CycleMsgEN

Load Read Records from RingBuffer to Gateway Message
JSR
Routine Name _173_CycleRingBuffer

Duplicate Message Detection Control Duplicate Message Delay Timer
DuplicateMsg.DelayTMR.DN

Trend Data Ring Buffer and Message
JSR
Routine Name _174_CycleDataMsg

Fault Event Simulation Mode
blocks EqDriver mapped inputs and generates a bit pattern for PreDefined and EqDefined Faults

21

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Simultation for Fault Data Enable Simulation
SimFault.EN

Simulation of Fault Data
JSR
Routine Name _800_SimFault

Main Routine

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

22

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault

Transaction Log Control SBR External Load -- Optional Parameters

MOVE		
Source	Prog.Log[Prog.LOGi].Routine	0
Dest	Trans.Ctl.Opt[1]	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE		
Source	Prog.Log[Prog.LOGi].Fault.Type	0
Dest	Trans.Ctl.Opt[2]	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE		
Source	Prog.Log[Prog.LOGi].Fault.Code	0
Dest	Trans.Ctl.Opt[3]	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE		
Source	Prog.FaultCount	0
Dest	Trans.Ctl.Opt[4]	-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE		
Source	Prog.LOGi	0
Dest	Trans.Ctl.Opt[5]	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	903
Input Par	0
Input Par	0

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
(U)

Capture Last Scan Time for this Progam
(Common Routine in all Progams)

(this routine must be after Controller Status for Program Scan total to be correct)

23

CLx Module Status Clear Scan History
CLX.ProgScan.Clear

Local Program Scan Time & Fault Routine Clear Scan History
Prog.Scan.Clear
(L)

Program Scan

aoiEDC_ProgScan	
aoiEDC_ProgScan	aoiProgScan ...
Scan	Prog.Scan
ScanTemp	CLX.ProgScanTemp

Main Routine

Program Scan Exceeds High Limit (configured for 100 msec)

24

Local Program Scan Time & Fault Routine Last Scan

GT
Source A: Prog.Scan.Last 144
Source B: CLxScanHighLimit 100000

Program High Scan Count

ADD
Source A: ProgScanHighCount 0
Source B: 1
Dest: ProgScanHighCount 0

Program High Scan Events

CLxProgScanHigh.31 (L)

CLx Module Status Clear Scan History

CLX.ProgScan.Clear

Program High Scan Count

CLR
Dest: ProgScanHighCount 0

(End)

Fault Routine

Program Fault Routine
(Common Routine in all Progams)

This routine is invoked by CLx Major Faults. It logs the Fault, clears the Fault and attempts to resume program scan.

0

NOP

Get Fault Record and Increment Fault Count

1

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

GSV

Class Name	Program
Instance Name	THIS
Attribute Name	MAJORFAULTRECORD
Dest	Prog.Fault.TimeLow
	0

Local Program Scan Time & Fault Routine Program Fault Occurrences

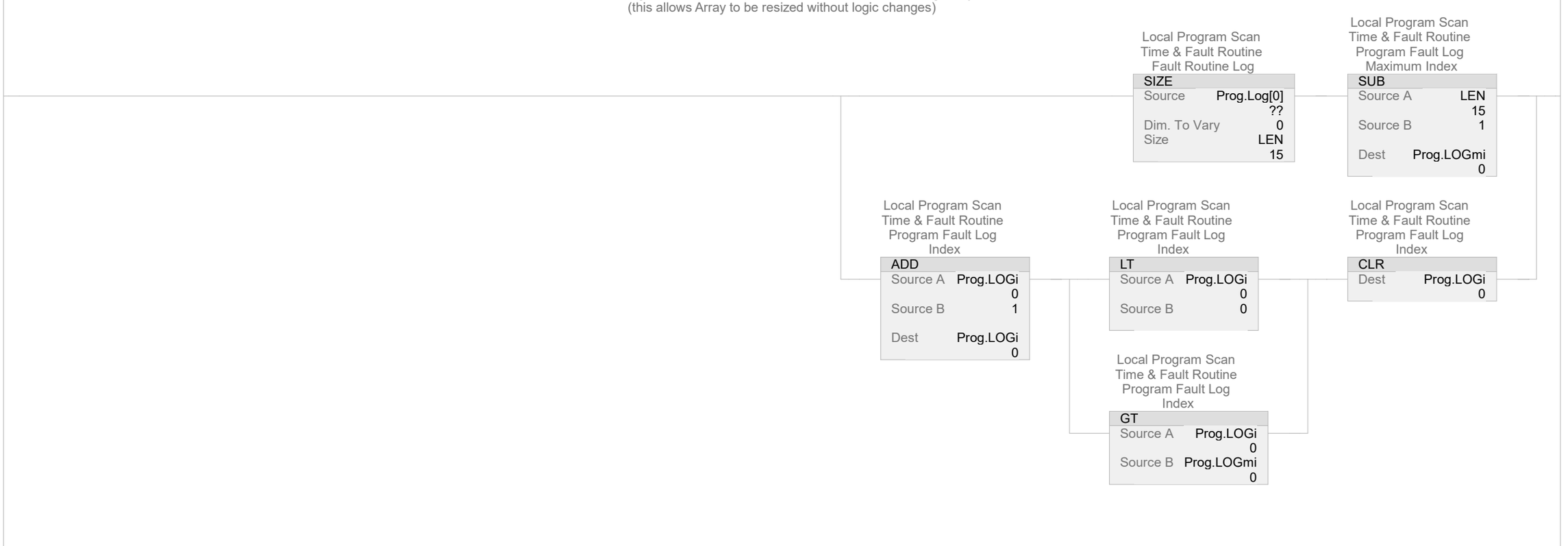
ADD

Source A	Prog.FaultCount
	0
Source B	1
Dest	Prog.FaultCount
	0

Fault Routine

Increment Log Fault Index -- Determine Index limits based on Log Array Size
(this allows Array to be resized without logic changes)

2



Fault Routine

Log Fault in Local Program Log
Optional Parameters are for the Indexes used in this Program -- the likely cause of the Major Faults.

3

Local Program Scan Time & Fault Routine Fault Record
COP
Source Prog.Fault
Dest Prog.Log[Prog.LOGi].Fault
Length 1

Local Program Scan Time & Fault Routine Fault Counter
MOVE
Source Prog.FaultCount
0
Dest Prog.Log[Prog.LOGi].Count
0

Local Program Scan Time & Fault Routine Faulted Routine Number
MOVE
Source Prog.Routine
154
Dest Prog.Log[Prog.LOGi].Routine
0

Local Program Scan Time & Fault Routine Transaction Scan Couner
MOVE
Source CLX.ScanCount
1106690484
Dest Prog.Log[Prog.LOGi].ScanCount
0

Local Program Scan Time & Fault Routine Transaction Date (YYMMDD)
MOVE
Source CLX.SysTime[8]
20261030
Dest Prog.Log[Prog.LOGi].Date
0

Local Program Scan Time & Fault Routine Transaction Time (HHMMSS)
MOVE
Source CLX.SysTime[9]
152452
Dest Prog.Log[Prog.LOGi].Time
0

Local Program Scan Time & Fault Routine Transaction Time (microseconds)
MOVE
Source CLX.SysTime[11]
273
Dest Prog.Log[Prog.LOGi].TimeuSec
0

Local Program Scan Time & Fault Routine Optional Parameter 1 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt1
0

Local Program Scan Time & Fault Routine Optional Parameter 2 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt2
0

Local Program Scan Time & Fault Routine Optional Parameter 3 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt3
0

Local Program Scan Time & Fault Routine Optional Parameter 4 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt4
0

Local Program Scan Time & Fault Routine Optional Parameter 5 (defined by Transaction)
MOVE
Source 0
Dest Prog.Log[Prog.LOGi].Opt5
0

Log Fault Routine Event to Transaction Log
Unable to call TransactionLogSBR from Fault Routine. Main Routine logs the Fault after Program Execution has resumed.
(Only one Fault Routine event can be logged per scan -- but LOGi and Local Fault Log will indicate multiple Events.)

4

Local Program Scan Time & Fault Routine Fault Routine Executed -- Log in Transaction Log
Prog.LogFault
L

Fault Routine

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

5

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 2
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Clear Fault to Resume Program Scan
(Disable Fault Handler to allow Major Controller Faults to occur for debug.)

6

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler

Local Program Scan Time & Fault Routine Program Clear Fault Record

COP
Source Prog.Fault
Dest Prog.ClearFault
Length 1

Local Program Scan Time & Fault Routine Fault Type (Program, I/O, etc.)

MOVE
Source 0
Dest Prog.ClearFault.Type 0

Local Program Scan Time & Fault Routine Unique (by Type) Code for Fault.

MOVE
Source 0
Dest Prog.ClearFault.Code 0

Local Program Scan Time & Fault Routine Low 32 bits of fault timestamp value.

SSV
Class Name Program
Instance Name THIS
Attribute Name MAJORFAULTRECORD
Source Prog.ClearFault.TimeLow 0

(End)

Configuration and Initialization Routine

EDC Driver Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 9
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Reset Program Scan Clear after all Programs have been reset

2

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
U

Reset ALL Structures Prior to Configuration & Initialization
v1.4 Move UserCfg.UpdateEQID from Config to Reset to force all Messages to abort and reload updated configuration

3

EDC Config Manual Reset -- Clears all Data Structures (use this to create Clean file before configuration)
CFG.ResetManual

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID

Reset all Structures (manually invoked to create clean Master for distribution)

JSR
Routine Name _010_Reset

EDC Config Manual Reset -- Clears all Data Structures (use this to create Clean file before configuration)
CFG.ResetManual
U

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID
U

EDC Config Reset ONS (Log Transaction)
CFG.ResetONS
( )

Configuration and Initialization Routine

Configure EDC Parameters
ConfigAuto - set by Download (also performs ConfigInit automatically)
ConfigManual - set manually to force Configuration
v1.4 Move UserCfg.UpdateEQID from Config to Reset to force all Messages to abort and reload updated configuration

4

EDC Config Auto Configuration -- Loads Driver Configuration after Download CFG.ConfigAuto

EDC Config Auto Initialization -- Initializes Driver after Download CFG.InitAuto (L)

EDC Config Manual Configuration -- Loads Driver Configuration CFG.ConfigManual

EDC Config Manual Initialization -- Initializes Driver CFG.InitManual (L)

EDC Config Reset ONS (Log Transaction) CFG.ResetONS

S:FS

EDC Config Auto Configuration -- Loads Driver Configuration after Download CFG.ConfigAuto (U)

EDC Config Manual Configuration -- Loads Driver Configuration CFG.ConfigManual (U)

EDC Config Configuration ONS (Log Transaction) CFG.ConfigONS

Configuration
JSR
Routine Name _011_Config

Initiaize EDC Parameters

5

EDC Config Auto Initialization -- Initializes Driver after Download CFG.InitAuto

EDC Config Manual Initialization -- Initializes Driver CFG.InitManual

EDC Config Reset ONS (Log Transaction) CFG.ResetONS

S:FS

EDC Config Auto Initialization -- Initializes Driver after Download CFG.InitAuto (U)

EDC Config Manual Initialization -- Initializes Driver CFG.InitManual (U)

EDC Config Initialization ONS (Log Transaction) CFG.InitONS

Initialize CLx Module
JSR
Routine Name _012_Initialize

(End)

Reset all Structures
(manually invoked to create clean Master for distribution)

Reset ALL Structures Prior to Configuration & Initialization

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 10
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Reset all Structures
(manually invoked to create clean Master for distribution)

Clear Message Data
v1.3 Replaced CFG value with LEN-1 to set Fault, Quality and Trend ReadPtr - can't rely on CFG values if never set

2

EDC Message Tx Data
FLL
Source 0
Dest EqTxData
<EQ0006_TxData>
Length 1

EDC Message Rx Data
FLL
Source 0
Dest EqRxData
<EQ0006_RxData>
Length 1

Equipment Data Callout/Status Message
FLL
Source 0
Dest CalloutMsg
Length 1

Message Control
FLL
Source 0
Dest CalloutMsgCtl
Length 1

Equipment Data Fault/Status Message
FLL
Source 0
Dest FaultMsg
Length 1

Fault Ring Buffer Message Control for multiplexed packets
FLL
Source 0
Dest FaultMsgCtl
Length 1

EDC Ring Buffer Control
FLL
Source 0
Dest FaultRBCtl
Length 1

Equipment Data Fault/Status Message
SIZE
Source FaultRB[0]
??
Dim. To Vary 0
Size LEN
15

Equipment Data Fault/Status Message
FLL
Source 0
Dest FaultRB[0]
Length LEN

EDC Ring Buffer Control Next Read Record
SUB
Source A LEN
15
Source B 1
Dest FaultRBCtl.ReadPtr
140

Equipment Data Quality/Health Message
FLL
Source 0
Dest QualityMsg
Length 1

Ring Buffer Message Control
FLL
Source 0
Dest QualityMsgCtl
Length 1

EDC Ring Buffer Control
FLL
Source 0
Dest QualityRBCtl
Length 1

Equipment Data Quality/Health Message
SIZE
Source QualityRB[0]
??
Dim. To Vary 0
Size LEN
15

Equipment Data Quality/Health Message
FLL
Source 0
Dest QualityRB[0]
Length LEN

EDC Ring Buffer Control Next Read Record
SUB
Source A LEN
15
Source B 1
Dest QualityRBCtl.ReadPtr
42

Reset all Structures
(manually invoked to create clean Master for distribution)

Equipment Data Trend Message

FLL	
Source	0
Dest	TrendMsg
Length	1

EDC Gateway Trend Message Control

FLL	
Source	0
Dest	TrendMsgCtl
Length	1

EDC Ring Buffer Control

FLL	
Source	0
Dest	TrendRBCtl
Length	1

Equipment Data Trend Message

SIZE	
Source	TrendRB[0] ??
Dim. To Vary	0
Size	LEN 15

Equipment Data Trend Message

FLL	
Source	0
Dest	TrendRB[0]
Length	LEN

EDC Ring Buffer Control Next Read Record

SUB	
Source A	LEN 15
Source B	1
Dest	TrendRBCtl.ReadPtr 49

Equipment Data Cycle Message

FLL	
Source	0
Dest	CycleMsg
Length	1

Ring Buffer Message Control

FLL	
Source	0
Dest	CycleMsgCtl
Length	1

EDC Ring Buffer Control

FLL	
Source	0
Dest	CycleRBCtl
Length	1

Equipment Data Cycle Message

SIZE	
Source	CycleRB[0] ??
Dim. To Vary	0
Size	LEN 15

Equipment Data Cycle Message

FLL	
Source	0
Dest	CycleRB[0]
Length	LEN

EDC Ring Buffer Control Next Read Record

SUB	
Source A	LEN 15
Source B	1
Dest	CycleRBCtl.ReadPtr 19

Equipment Data Version Message Format

FLL	
Source	0
Dest	VersionMsg
Length	1

Message Control

FLL	
Source	0
Dest	VersionMsgCtl
Length	1

Time Sync Message Control

FLL	
Source	0
Dest	TimeSyncMsgCtl
Length	1

Reset all Structures
(manually invoked to create clean Master for distribution)

v1.5 Clear Message Errors

Reset of MsgCtl clears any existing Retry - also need to clear MsgER or Message won't refire.

3

EDC Gateway Callout Write Message CalloutWriteMsg.EN <EQ0006_CalloutWriteMsg.EN>	EDC Gateway Callout Write Message CalloutWriteMsg.ER <EQ0006_CalloutWriteMsg.ER>	EDC Gateway Callout Write Message CalloutWriteMsg.ER <EQ0006_CalloutWriteMsg.ER>
EDC Gateway Fault Read Message FaultReadMsg.EN <EQ0006_FaultReadMsg.EN>	EDC Gateway Fault Read Message FaultReadMsg.ER <EQ0006_FaultReadMsg.ER>	EDC Gateway Fault Read Message FaultReadMsg.ER <EQ0006_FaultReadMsg.ER>
EDC Gateway Fault Write Message FaultWriteMsg.EN <EQ0006_FaultWriteMsg.EN>	EDC Gateway Fault Write Message FaultWriteMsg.ER <EQ0006_FaultWriteMsg.ER>	EDC Gateway Fault Write Message FaultWriteMsg.ER <EQ0006_FaultWriteMsg.ER>
EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>	EDC Gateway Quality Read Message QualityReadMsg.ER <EQ0006_QualityReadMsg.ER>	EDC Gateway Quality Read Message QualityReadMsg.ER <EQ0006_QualityReadMsg.ER>
EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>	EDC Gateway Quality Write Message QualityWriteMsg.ER <EQ0006_QualityWriteMsg.ER>	EDC Gateway Quality Write Message QualityWriteMsg.ER <EQ0006_QualityWriteMsg.ER>
EDC Gateway Trend Read Message TrendReadMsg.EN <EQ0006_TrendReadMsg.EN>	EDC Gateway Trend Read Message TrendReadMsg.ER <EQ0006_TrendReadMsg.ER>	EDC Gateway Trend Read Message TrendReadMsg.ER <EQ0006_TrendReadMsg.ER>
EDC Gateway Trend Write Message TrendWriteMsg.EN <EQ0006_TrendWriteMsg.EN>	EDC Gateway Trend Write Message TrendWriteMsg.ER <EQ0006_TrendWriteMsg.ER>	EDC Gateway Trend Write Message TrendWriteMsg.ER <EQ0006_TrendWriteMsg.ER>
EDC Gateway Cycle Read Message CycleReadMsg.EN <EQ0006_CycleReadMsg.EN>	EDC Gateway Cycle Read Message CycleReadMsg.ER <EQ0006_CycleReadMsg.ER>	EDC Gateway Cycle Read Message CycleReadMsg.ER <EQ0006_CycleReadMsg.ER>
EDC Gateway Cycle Write Message CycleWriteMsg.EN <EQ0006_CycleWriteMsg.EN>	EDC Gateway Cycle Write Message CycleWriteMsg.ER <EQ0006_CycleWriteMsg.ER>	EDC Gateway Cycle Write Message CycleWriteMsg.ER <EQ0006_CycleWriteMsg.ER>
EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN <EQ0006_TimeSyncReadMsg.EN>	EDC Gateway Time Sync Read Message TimeSyncReadMsg.ER <EQ0006_TimeSyncReadMsg.ER>	EDC Gateway Time Sync Read Message TimeSyncReadMsg.ER <EQ0006_TimeSyncReadMsg.ER>
EDC Gateway Version Write Message	EDC Gateway Version Write Message	EDC Gateway Version Write Message

Reset all Structures
(manually invoked to create clean Master for distribution)

Write Message
VersionWriteMsg.EN
<EQ0006_VersionWriteMsg.EN>

Write Message
VersionWriteMsg.ER
<EQ0006_VersionWriteMsg.ER>

Write Message
VersionWriteMsg.ER
<EQ0006_VersionWriteMsg.ER>
(U)

Clear EDC Control Structures
block reset of UserCfg when reset by UpdateEQID -- caused EQID 0000 during Config/Init

4

EDC Config
FLL
Source 0
Dest CFG
Length 1

CLx Module Status
FLL
Source 0
Dest CLX
Length 1

Local Program Scan Time & Fault Routine
FLL
Source 0
Dest Prog
Length 1

Equipment Data Collection - Line Equipment Status
FLL
Source 0
Dest Status
Length 1

Transaction Log Control
FLL
Source 0
Dest Trans
Length 1

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID
/

Equipment Data Collection - User Config
FLL
Source 0
Dest UserCfg
Length 1

Clear Message Data Miscellaneous Structures

5

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults Last
FLL
Source 0
Dest FaultMsgCtl.PreDefLast
Length 1

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined
SIZE
Source FaultMsgCtl.EqDefLast[0]
??
Dim. To Vary 0
Size LEN
15

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined
FLL
Source 0
Dest FaultMsgCtl.EqDef[0]
Length LEN

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined
FLL
Source 0
Dest FaultMsgCtl.EqDefLast[0]
Length LEN

Equipment Data Trend Message Data
FLL
Source 0
Dest TrendCollect
Length 1

Trend Collect Control
FLL
Source 0
Dest TrendCollectCtl
Length 1

Trend Data Collection
SIZE
Source TrendCfg[0]
??
Dim. To Vary 0
Size LEN
15

Trend Data Collection
FLL
Source 0
Dest TrendCfg[0]
Length LEN

Reset all Structures
(manually invoked to create clean Master for distribution)

Clear Transaction Logs

6

Transaction Log Control Transaction Log

SIZE	
Source	Trans.Err[0] ??
Dim. To Vary	0
Size	LEN 15

Transaction Log Control Transaction Log

FLL	
Source	0
Dest	Trans.Err[0]
Length	LEN

Transaction Log Control Transaction Log

FLL	
Source	0
Dest	Trans.ErrSnap[0]
Length	LEN

Transaction Log Control Transaction Log

SIZE	
Source	Trans.Log[0] ??
Dim. To Vary	0
Size	LEN 15

Transaction Log Control Transaction Log

FLL	
Source	0
Dest	Trans.Log[0]
Length	LEN

Transaction Log Control Transaction Log

FLL	
Source	0
Dest	Trans.LogSnap[0]
Length	LEN

v1.8 Duplicate Message Check
v1.10 -- DHF011 UserCfg.UpdateEQID was clearing the DuplicateMsg DnONS and ErONS bits
This caused a nuisance Duplicate Detection if any Msg.DN or Msg.ER bits were on.

7

AFI

Duplicate Message Detection Control

FLL	
Source	0
Dest	DuplicateMsg
Length	1

Reset all Structures
(manually invoked to create clean Master for distribution)

8

Initialize Event Masks

EqDefined Ignore Mask from Server Equipment Data Fault/Status Equipment Defined

SIZE	
Source	EqDefIgnoreMask[0] ??
Dim. To Vary	0
Size	LEN 15

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

SIZE	
Source	EqDefIgnoreMask[0].Word[0] 16#ffff_ffff
Dim. To Vary	0
Size	LEN2 8

Temp Result for Array Size

MUL	
Source A	LEN 15
Source B	LEN2 8
Dest	LEN 15

EqDefined Ignore Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	16#ffff_ffff
Dest	EqDefIgnoreMask[0]
Length	LEN

EqDefined Fault Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	16#ffff_ffff
Dest	EqDefFaultMask[0]
Length	LEN

EqDefined Maint Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	16#ffff_ffff
Dest	EqDefMaintMask[0]
Length	LEN

EqDefined Alarm Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	0
Dest	EqDefAlarmMask[0]
Length	LEN

EqDefined Alert Mask from Server Equipment Data Fault/Status Equipment Defined

FLL	
Source	0
Dest	EqDefAlertMask[0]
Length	LEN

(End)

Configuration

EDC Driver Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 11
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Re-enable Fault Handlers

2

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
U

Message Transaction Number Rollover

3

EDC Config Message Transaction Number Rollover

MOVE
Source 9999
Dest CFG.MsgTransNumRollover 9999

Configuration

Message Delay Timer Preset
Message ER Retry Timer Preset

4

EDC Config Message Delay Timer Preset

MOVE	
Source	250
Dest	CFG.MsgDelayTime
	250

EDC Config Message Error Retry Timer Preset

MOVE	
Source	5000
Dest	CFG.MsgRetryTime
	5000

Ring Buffer Read Message Delay Time Optimization presets

The Limits and Time increase/decrease values should be skewed to quickly increase the delay when Gateway Read notOKs and slowly decrease the delay when Gateway Read OK
The intent is to find the preset that minimizes wasted traffic while maximizing the throughput without too-frequent changes.

5

EDC Config RingBuffer Read OK Count for reducing Delay Time

MOVE	
Source	5
Dest	CfgRB.ReadOkLimit
	5

EDC Config RingBuffer Read Not OK Count for increasing Delay Time

MOVE	
Source	2
Dest	CfgRB.ReadNotOkLimit
	2

EDC Config RingBuffer Read OK Delay Subtract (msec)

MOVE	
Source	10
Dest	CfgRB.ReadOkSub
	10

EDC Config RingBuffer Read Not OK Delay Add (msec)

MOVE	
Source	20
Dest	CfgRB.ReadNotOkAdd
	20

EDC Config RingBuffer Read Delay Time Minimum (msec)

MOVE	
Source	100
Dest	CfgRB.ReadDelayMin
	100

EDC Config RingBuffer Read Delay Time Maximum (msec)

MOVE	
Source	500
Dest	CfgRB.ReadDelayMax
	500

Message Resend Timer Preset
Keep Alive
different times for Live Callout Message and Buffered Fault, Quality, Trend Messages
Callout/Fault are naturally synched by Fault/Status Event changes -- use odd presets to force Keep Alive out of synch
(i.e. 31 seconds and 301 seconds -- not 30 and 300)
v3.02 Reduce Callout Resend from 121 seconds to 31 seconds for faster Mask Download and Data Refresh

6

EDC Config Live Message Keep Alive Resend Timer Preset

MOVE	
Source	31000
Dest	CFG.LiveMsgResendTime
	31000

EDC Config Buffer Message Keep Alive Resend Timer Preset

MOVE	
Source	301000
Dest	CFG.BuffMsgResendTime
	301000

Configuration

Configure Message Timeout
This forces a Timeout for Unconnected or Connected Timeout failure.
(On occasion, have seen Messages hang indefinitely.)

This should be set higher than Unconnected and Connected Timeouts above.

7

EDC Config CLx Message Timeout (msec)

MOVE
Source 5000
Dest CFG.CLx.MsgTimeout 5000

Configure Unconnected & Connected Timeout for ENET Socket Messages
Unconnected: 4,000,000 = 4.0 seconds (CLx default = 30000000 = 30 seconds)
Connection Rate: 1,000,000 = 4.0 seconds (CLx default = 7500000 = 7.5 seconds)
Connection Rate Timeout Multiplier: 0 = *4 (CLx default = 0 = *4 multiplier)

8

EDC Config CLx Unconnected Message Timeout

MOVE
Source 4000000
Dest CFG.CLx.UC_Timeout 4000000

EDC Config CLx Connected Message Timeout Rate

DIV
Source A CFG.CLx.UC_Timeout 4000000
Source B 4
Dest CFG.CLx.ConnectionRate 1000000

EDC Config CLx Connected Message Timeout Multiplier

MOVE
Source 0
Dest CFG.CLx.ConnectionMult 0

Transaction Log Filters
-- Filters are OFF by default -- manually toggle EN and set filter values for Code, LNi and EQi

9

Transaction Log Control Enable Code Filters
Trans.Ctl.Filter.CodeEN
U

Transaction Log Control Code Filter Low Value

MOVE
Source 0
Dest Trans.Ctl.Filter.CodeLow 0

Transaction Log Control Code Filter High Value

MOVE
Source 0
Dest Trans.Ctl.Filter.CodeHigh 0

Configuration

Fault/Status Equipment Defined Array Indexes
Array for multiplexed data Packets

10

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined

SIZE
Source FaultMsgCtl.EqDef[0] ??
Dim. To Vary 0
Size CFG.FaultEq.n8

EDC Config Array Max Index

SUB
Source A CFG.FaultEq.n8
Source B 1
Dest CFG.FaultEq.mi7

EDC Config Array Start Index

MOVE
Source 0
Dest CFG.FaultEq.si0

EDC Config Array End Index

MOVE
Source 0
Dest CFG.FaultEq.ei2

Fault EqDefined Packet Index

MOVE
Source 0
Dest PKTi-1

Configuration

Fault/Status Equipment Defined Array Indexes
Array for multiplexed data Packets
Use Packet Config BIts to determine End Index

11

lbl_FaultEqLoop
LBL

Equipment Data Collection - User Config Equipment Defined Packet EN bit array
UserCfg.EqDefPacketEN.[PKTi]

EDC Config Array End Index
MOVE
Source PKTi -1
Dest CFG.FaultEq.ei 2

Fault EqDefined Packet Index
ADD
Source A PKTi -1
Source B 1
Dest PKTi -1

Fault EqDefined Packet Index
LE
Source A PKTi -1
Source B CFG.FaultEq.mi 7

lbl_FaultEqLoop
JMP

Fault/Status RB Array Indexes

12

Equipment Data Fault/Status Message
SIZE
Source FaultRB[0] ??
Dim. To Vary 0
Size CFG.FaultRB.n 250

EDC Config Array Max Index
SUB
Source A CFG.FaultRB.n 250
Source B 1
Dest CFG.FaultRB.mi 249

EDC Config Array Start Index
MOVE
Source 0
Dest CFG.FaultRB.si 0

EDC Config Array End Index
MOVE
Source CFG.FaultRB.mi 249
Dest CFG.FaultRB.ei 249

Configuration

Quality/Health RB Array Indexes

13

Equipment Data Quality/Health Message

SIZE
Source QualityRB[0] ??
Dim. To Vary 0
Size CFG.QualityRB.n 100

EDC Config Array Max Index

SUB
Source A CFG.QualityRB.n 100
Source B 1
Dest CFG.QualityRB.mi 99

EDC Config Array Start Index

MOVE
Source 0
Dest CFG.QualityRB.si 0

EDC Config Array End Index

MOVE
Source CFG.QualityRB.mi 99
Dest CFG.QualityRB.ei 99

Trend RB Array Indexes

14

Equipment Data Trend Message

SIZE
Source TrendRB[0] ??
Dim. To Vary 0
Size CFG.TrendRB.n 50

EDC Config Array Max Index

SUB
Source A CFG.TrendRB.n 50
Source B 1
Dest CFG.TrendRB.mi 49

EDC Config Array Start Index

MOVE
Source 0
Dest CFG.TrendRB.si 0

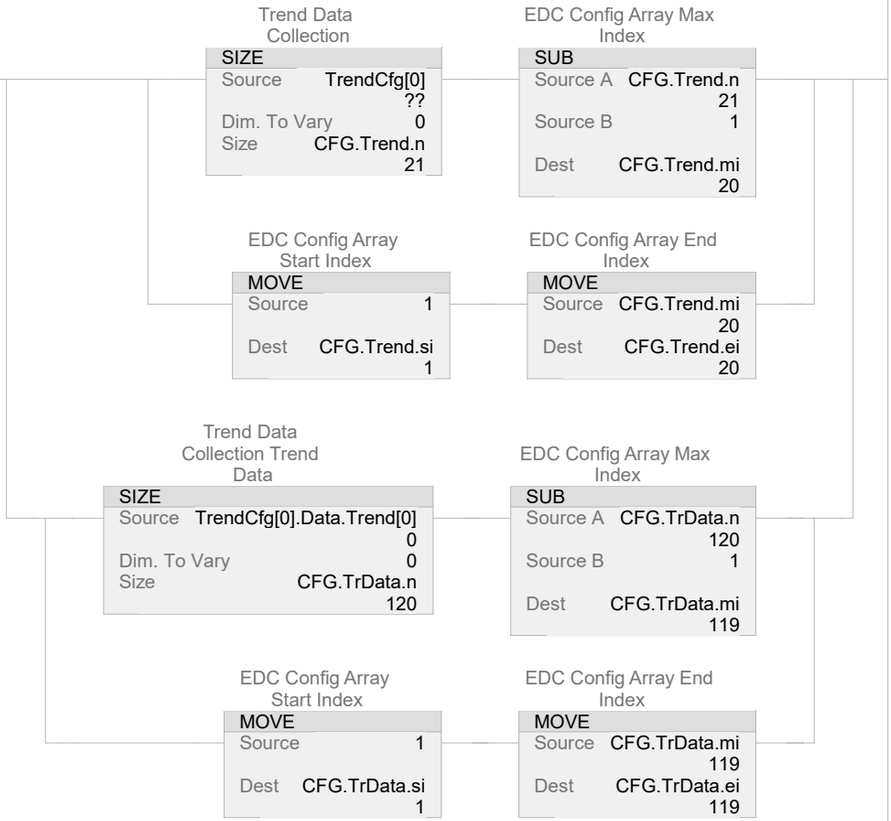
EDC Config Array End Index

MOVE
Source CFG.TrendRB.mi 49
Dest CFG.TrendRB.ei 49

Configuration

Trend Array Indexes

15



Configuration

Cycle RB Array Indexes

16

Equipment Data Cycle Message

SIZE	
Source	CycleRB[0] ??
Dim. To Vary	0
Size	CFG.CycleRB.n 20

EDC Config Array Max Index

SUB	
Source A	CFG.CycleRB.n 20
Source B	1
Dest	CFG.CycleRB.mi 19

EDC Config Array Start Index

MOVE	
Source	0
Dest	CFG.CycleRB.si 0

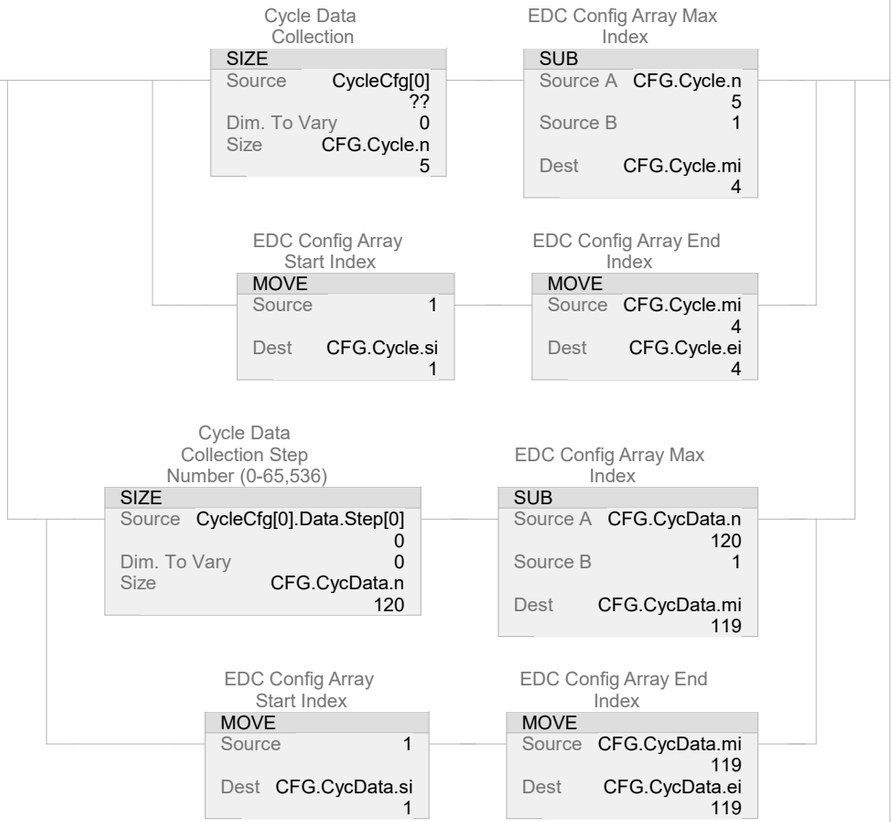
EDC Config Array End Index

MOVE	
Source	CFG.CycleRB.mi 19
Dest	CFG.CycleRB.ei 19

Configuration

Cycle Array Indexes

17



Configuration

18

Callout Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

EDC Message Tx Data Callout/Status Receive Message

SIZE
Source EqTxData.Callout[0] <EQ0006_TxData.Callout[0]> 9993
Dim. To Vary 0
Size CFG.CalloutMSG.n 14

EDC Config Array Max Index

SUB
Source A CFG.CalloutMSG.n 14
Source B 1
Dest CFG.CalloutMSG.mi 13

EDC Config Array End Count (udt size)

MOVE
Source 14
Dest CFG.CalloutMSG.e 14

EDC Config Array End Count (udt size)

GT
Source A CFG.CalloutMSG.e 14
Source B CFG.CalloutMSG.n 14

EDC Config Array End Count (udt size)

MOVE
Source CFG.CalloutMSG.n 14
Dest CFG.CalloutMSG.e 14

EDC Config Array End Index (used for copies)

SUB
Source A CFG.CalloutMSG.e 14
Source B 1
Dest CFG.CalloutMSG.ei 13

Configuration

19

Bulk Fault Record & Message Arrays
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

EDC Message Tx Data Bulk Fault/Status Receive Message

SIZE
Source EqTxData.Fault[0] <EQ0006_TxData.Fault[0]> 3758
Dim. To Vary 0
Size CFG.FaultMSG.n 143

EDC Config Array Max Index

SUB
Source A CFG.FaultMSG.n 143
Source B 1
Dest CFG.FaultMSG.mi 142

EDC Config Array End Count (udt size)

MOVE
Source 143
Dest CFG.FaultMSG.e 143

EDC Config Array End Count (udt size)

GT
Source A CFG.FaultMSG.e 143
Source B CFG.FaultMSG.n 143

EDC Config Array End Count (udt size)

MOVE
Source CFG.FaultMSG.n 143
Dest CFG.FaultMSG.e 143

EDC Config Array End Index (used for copies)

SUB
Source A CFG.FaultMSG.e 143
Source B 1
Dest CFG.FaultMSG.ei 142

EDC Config Array Size

MOVE
Source 14
Dest CFG.FaultRec.n 14

EDC Config Array Max Index

SUB
Source A CFG.FaultRec.n 14
Source B 1
Dest CFG.FaultRec.mi 13

EDC Config Array End Count (udt size)

MOVE
Source 14
Dest CFG.FaultRec.e 14

EDC Config Array End Count (udt size)

GT
Source A CFG.FaultRec.e 14
Source B CFG.FaultRec.n 14

EDC Config Array End Count (udt size)

MOVE
Source CFG.FaultRec.n 14
Dest CFG.FaultRec.e 14

EDC Config Array End Index (used for copies)

SUB
Source A CFG.FaultRec.e 14

Configuration

Source B	1
Dest	CFG.FaultRec.ei 13

Quality Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array
v1.6 Increased array from 29 to 34 DINTs for 10 Quality Data

20

EDC Message Tx Data Quality/Health Receive Message

SIZE

Source	EqTxData.Quality[0] <EQ0006_TxData.Quality[0]> 4157
Dim. To Vary	0
Size	CFG.QualityMSG.n 39

EDC Config Array Max Index

SUB

Source A	CFG.QualityMSG.n 39
Source B	1
Dest	CFG.QualityMSG.mi 38

EDC Config Array End Count (udt size)

MOVE

Source	39
Dest	CFG.QualityMSG.e 39

EDC Config Array End Count (udt size)

GT

Source A	CFG.QualityMSG.e 39
Source B	CFG.QualityMSG.n 39

EDC Config Array End Count (udt size)

MOVE

Source	CFG.QualityMSG.n 39
Dest	CFG.QualityMSG.e 39

EDC Config Array End Index (used for copies)

SUB

Source A	CFG.QualityMSG.e 39
Source B	1
Dest	CFG.QualityMSG.ei 38

Configuration

Trend Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

21

EDC Message Tx Data Trend Receive Message

SIZE
Source EqTxData.Trend[0] <EQ0006_TxData.Trend[0]>
0
Dim. To Vary 0
Size CFG.TrendMSG.n 70

EDC Config Array Max Index

SUB
Source A CFG.TrendMSG.n 70
Source B 1
Dest CFG.TrendMSG.mi 69

EDC Config Array End Count (udt size)

MOVE
Source 70
Dest CFG.TrendMSG.e 70

EDC Config Array End Count (udt size)

GT
Source A CFG.TrendMSG.e 70
Source B CFG.TrendMSG.n 70

EDC Config Array End Count (udt size)

MOVE
Source CFG.TrendMSG.n 70
Dest CFG.TrendMSG.e 70

EDC Config Array End Index (used for copies)

SUB
Source A CFG.TrendMSG.e 70
Source B 1
Dest CFG.TrendMSG.ei 69

Configuration

22

Version Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

EDC Message Tx Data Version Receive Message

SIZE
Source EqTxData.Version[0] <EQ0006_TxData.Version[0]> 677
Dim. To Vary 0
Size CFG.VersionMSG.n 30

EDC Config Array Max Index

SUB
Source A CFG.VersionMSG.n 30
Source B 1
Dest CFG.VersionMSG.mi 29

EDC Config Array End Count (udt size)

MOVE
Source 30
Dest CFG.VersionMSG.e 30

EDC Config Array End Count (udt size)

GT
Source A CFG.VersionMSG.e 30
Source B CFG.VersionMSG.n 30

EDC Config Array End Count (udt size)

MOVE
Source CFG.VersionMSG.n 30
Dest CFG.VersionMSG.e 30

EDC Config Array End Index (used for copies)

SUB
Source A CFG.VersionMSG.e 30
Source B 1
Dest CFG.VersionMSG.ei 29

Configuration

Cycle Message Array
Configure Array Indexes
e used for UDT-DINT copies -- must match DINT count of UDT
ei used to locate Footer Transaction Number in DINT array

23

EDC Message Tx Data Cycle Receive Message

SIZE
Source: EqTxData.Cycle[0] <EQ0006_TxData.Cycle[0]>
0
Dim. To Vary: 0
Size: CFG.CycleMSG.n 134

EDC Config Array Max Index

SUB
Source A: CFG.CycleMSG.n 134
Source B: 1
Dest: CFG.CycleMSG.mi 133

EDC Config Array End Count (udt size)

MOVE
Source: 134
Dest: CFG.CycleMSG.e 134

EDC Config Array End Count (udt size)

GT
Source A: CFG.CycleMSG.e 134
Source B: CFG.CycleMSG.n 134

EDC Config Array End Count (udt size)

MOVE
Source: CFG.CycleMSG.n 134
Dest: CFG.CycleMSG.e 134

EDC Config Array End Index (used for copies)

SUB
Source A: CFG.CycleMSG.e 134
Source B: 1
Dest: CFG.CycleMSG.ei 133

Configuration

Transaction Log
-- TransactionLogEN is OFF by default -- manually toggle ON to log Transactions
-- TransLogAll is OFF by default -- manually toggle ON to log Keep Alive Messages during debug/deployment.
-- Initialize Size
Initialize Transaction Error Log Size

24

EDC Config Enable Transaction Logging
CFG.TransLogEN
U

EDC Config Log all Transactions (includes all Keep Alives)
CFG.TransLogAll
U

Transaction Log Control Transaction Log
SIZE
Source Trans.Log[0] ??
Dim. To Vary 0
Size CFG.TransLOG.n 150

EDC Config Array Max Index
SUB
Source A CFG.TransLOG.n 150
Source B 1
Dest CFG.TransLOG.mi 149

EDC Config Array Start Index
MOVE
Source 0
Dest CFG.TransLOG.si 0

EDC Config Array End Index
MOVE
Source CFG.TransLOG.mi 149
Dest CFG.TransLOG.ei 149

Transaction Log Control Transaction Log
SIZE
Source Trans.Err[0] ??
Dim. To Vary 0
Size CFG.TransERR.n 50

EDC Config Array Max Index
SUB
Source A CFG.TransERR.n 50
Source B 1
Dest CFG.TransERR.mi 49

EDC Config Array Start Index
MOVE
Source 0
Dest CFG.TransERR.si 0

EDC Config Array End Index
MOVE
Source CFG.TransERR.mi 49
Dest CFG.TransERR.ei 49

Transaction Log Control SBR External Load -- Optional Parameters
SIZE
Source Trans.Ctl.Opt[0] -1
Dim. To Vary 0
Size Trans.Ctl.OptN

Configuration

Size	Trans.Ctl.OptN 6

Calculate Line and Equipment Indexes from EQID

25

EDC Config Line Index from EQID

CPT

Dest	CFG.MsgDest.LNi 0
Expression	TRUNC(UserCfg.EQID/100)

EDC Config Equipment Index from EQID

MOD

Source A	UserCfg.EQID 6
Source B	100
Dest	CFG.MsgDest.EQi 6

Configuration

Gateway Message Destination Tag EDC_EqRxBuffer.

26

EDC Config Message Remote Destination Tag
MOVE
Source 15
Dest CFG.MsgDest.Tag.LEN 15

EDC Config Message Remote Destination Tag
MOVE
Source ascii.E 'E'
Dest CFG.MsgDest.Tag.DATA[0] 'E'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.D 'D'
Dest CFG.MsgDest.Tag.DATA[1] 'D'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.C 'C'
Dest CFG.MsgDest.Tag.DATA[2] 'C'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.Uscore ' _'
Dest CFG.MsgDest.Tag.DATA[3] ' _'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.E 'E'
Dest CFG.MsgDest.Tag.DATA[4] 'E'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._q 'q'
Dest CFG.MsgDest.Tag.DATA[5] 'q'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.R 'R'
Dest CFG.MsgDest.Tag.DATA[6] 'R'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._x 'x'
Dest CFG.MsgDest.Tag.DATA[7] 'x'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.B 'B'
Dest CFG.MsgDest.Tag.DATA[8] 'B'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._u 'u'
Dest CFG.MsgDest.Tag.DATA[9] 'u'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._f 'f'
Dest CFG.MsgDest.Tag.DATA[10] 'f'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._f 'f'
Dest CFG.MsgDest.Tag.DATA[11] 'f'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._e 'e'
Dest CFG.MsgDest.Tag.DATA[12] 'e'

EDC Config Message Remote Destination Tag
MOVE
Source ascii._r 'r'
Dest CFG.MsgDest.Tag.DATA[13] 'r'

EDC Config Message Remote Destination Tag
MOVE
Source ascii.Period '.'
Dest CFG.MsgDest.Tag.DATA[14] '.'

Configuration

27

Gateway Message Destination Tag
EDC_EqRxBufferAux.

EDC Config Message Remote Destination Tag

COP
Source CFG.MsgDest.Tag
Dest CFG.MsgDest.TagAux
Length 1

EDC Config Message Remote Destination Tag

MOVE
Source 18
Dest CFG.MsgDest.TagAux.LEN
18

EDC Config Message Remote Destination Tag

MOVE
Source ascii.A
'A'
Dest CFG.MsgDest.TagAux.DATA[14]
'A'

EDC Config Message Remote Destination Tag

MOVE
Source ascii._u
'u'
Dest CFG.MsgDest.TagAux.DATA[15]
'u'

EDC Config Message Remote Destination Tag

MOVE
Source ascii._x
'x'
Dest CFG.MsgDest.TagAux.DATA[16]
'x'

EDC Config Message Remote Destination Tag

MOVE
Source ascii.Period
'.'
Dest CFG.MsgDest.TagAux.DATA[17]
'.'

28

Gateway Message Destination Tag
EDC_EqRxBufferFault.

EDC Config Message Remote Destination Tag

COP
Source CFG.MsgDest.Tag
Dest CFG.MsgDest.TagFault
Length 1

EDC Config Message Remote Destination Tag

MOVE
Source 20
Dest CFG.MsgDest.TagFault.LEN
20

EDC Config Message Remote Destination Tag

MOVE
Source ascii.F
'F'
Dest CFG.MsgDest.TagFault.DATA[14]
'F'

EDC Config Message Remote Destination Tag

MOVE
Source ascii._a
'a'
Dest CFG.MsgDest.TagFault.DATA[15]
'a'

EDC Config Message Remote Destination Tag

MOVE
Source ascii._u
'u'
Dest CFG.MsgDest.TagFault.DATA[16]
'u'

EDC Config Message Remote Destination Tag

MOVE
Source ascii._l
'l'
Dest CFG.MsgDest.TagFault.DATA[17]
'l'

EDC Config Message Remote Destination Tag

MOVE
Source ascii._t
't'
Dest CFG.MsgDest.TagFault.DATA[18]
't'

EDC Config Message Remote Destination Tag

MOVE
Source ascii.Period
'.'
Dest CFG.MsgDest.TagFault.DATA[19]
'.'

Configuration

29

Gateway Message Destination Tag Line[nn].

EDC Config Message Remote Destination Line

MOVE
Source 7
Dest CFG.MsgDest.Line.LEN 8

EDC Config Message Remote Destination Line

MOVE
Source ascii.L 'L'
Dest CFG.MsgDest.Line.DATA[0] 'L'

EDC Config Message Remote Destination Line

MOVE
Source ascii._i 'i'
Dest CFG.MsgDest.Line.DATA[1] 'i'

EDC Config Message Remote Destination Line

MOVE
Source ascii._n 'n'
Dest CFG.MsgDest.Line.DATA[2] 'n'

EDC Config Message Remote Destination Line

MOVE
Source ascii._e 'e'
Dest CFG.MsgDest.Line.DATA[3] 'e'

EDC Config Message Remote Destination Line

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Line.DATA[4] '['

EDC Config Message Remote Destination Line

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Line.DATA[5] '0'

EDC Config Message Remote Destination Line

MOVE
Source ascii.Period '.'
Dest CFG.MsgDest.Line.DATA[6] ']'

EDC Config Line Index from EQID

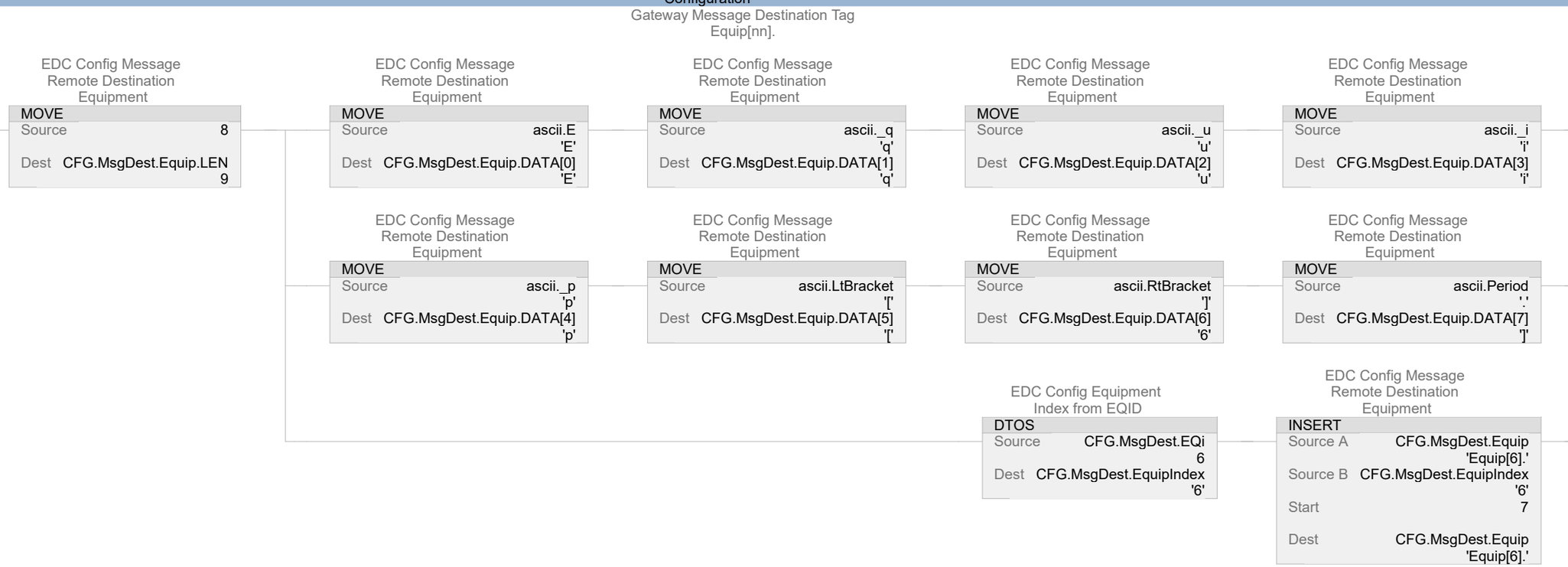
DTOS
Source CFG.MsgDest.LNi 0
Dest CFG.MsgDest.LineIndex '0'

EDC Config Message Remote Destination Line

INSERT
Source A CFG.MsgDest.Line 'Line[0].'
Source B CFG.MsgDest.LineIndex '0'
Start 6
Dest CFG.MsgDest.Line 'Line[0].'

Configuration

30



Configuration

Gateway Message Destination Tag Callout[0]

31

EDC Config Message Remote Destination Element

MOVE
Source 10
Dest CFG.MsgDest.Callout.LEN 10

EDC Config Message Remote Destination Element

MOVE
Source ascii.C 'C'
Dest CFG.MsgDest.Callout.DATA[0] 'C'

EDC Config Message Remote Destination Element

MOVE
Source ascii._a 'a'
Dest CFG.MsgDest.Callout.DATA[1] 'a'

EDC Config Message Remote Destination Element

MOVE
Source ascii._l 'l'
Dest CFG.MsgDest.Callout.DATA[2] 'l'

EDC Config Message Remote Destination Element

MOVE
Source ascii._l 'l'
Dest CFG.MsgDest.Callout.DATA[3] 'l'

EDC Config Message Remote Destination Element

MOVE
Source ascii._o 'o'
Dest CFG.MsgDest.Callout.DATA[4] 'o'

EDC Config Message Remote Destination Element

MOVE
Source ascii._u 'u'
Dest CFG.MsgDest.Callout.DATA[5] 'u'

EDC Config Message Remote Destination Element

MOVE
Source ascii._t 't'
Dest CFG.MsgDest.Callout.DATA[6] 't'

EDC Config Message Remote Destination Element

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Callout.DATA[7] '['

EDC Config Message Remote Destination Element

MOVE
Source ascii._0 '0'
Dest CFG.MsgDest.Callout.DATA[8] '0'

EDC Config Message Remote Destination Element

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Callout.DATA[9] ']'

Configuration

32

Gateway Message Destination Tag
Fault[0]

EDC Config Message
Remote Destination
Element

MOVE
Source 8
Dest CFG.MsgDest.Fault.LEN
8

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.F
'F'
Dest CFG.MsgDest.Fault.DATA[0]
'F'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._a
'a'
Dest CFG.MsgDest.Fault.DATA[1]
'a'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._u
'u'
Dest CFG.MsgDest.Fault.DATA[2]
'u'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._l
'l'
Dest CFG.MsgDest.Fault.DATA[3]
'l'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._t
't'
Dest CFG.MsgDest.Fault.DATA[4]
't'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.LtBracket
'['
Dest CFG.MsgDest.Fault.DATA[5]
'['

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._0
'0'
Dest CFG.MsgDest.Fault.DATA[6]
'0'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.RtBracket
']'
Dest CFG.MsgDest.Fault.DATA[7]
']'

Configuration

Gateway Message Destination Tag Quality[0]

33

EDC Config Message Remote Destination Element

MOVE
Source 10
Dest CFG.MsgDest.Quality.LEN 10

EDC Config Message Remote Destination Element

MOVE
Source ascii.Q 'Q'
Dest CFG.MsgDest.Quality.DATA[0] 'Q'

EDC Config Message Remote Destination Element

MOVE
Source ascii._u 'u'
Dest CFG.MsgDest.Quality.DATA[1] 'u'

EDC Config Message Remote Destination Element

MOVE
Source ascii._a 'a'
Dest CFG.MsgDest.Quality.DATA[2] 'a'

EDC Config Message Remote Destination Element

MOVE
Source ascii._l 'l'
Dest CFG.MsgDest.Quality.DATA[3] 'l'

EDC Config Message Remote Destination Element

MOVE
Source ascii._i 'i'
Dest CFG.MsgDest.Quality.DATA[4] 'i'

EDC Config Message Remote Destination Element

MOVE
Source ascii._t 't'
Dest CFG.MsgDest.Quality.DATA[5] 't'

EDC Config Message Remote Destination Element

MOVE
Source ascii._y 'y'
Dest CFG.MsgDest.Quality.DATA[6] 'y'

EDC Config Message Remote Destination Element

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Quality.DATA[7] '['

EDC Config Message Remote Destination Element

MOVE
Source ascii._0 '0'
Dest CFG.MsgDest.Quality.DATA[8] '0'

EDC Config Message Remote Destination Element

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Quality.DATA[9] ']'

Configuration

Gateway Message Destination Tag
Trend[0]

34

EDC Config Message
Remote Destination
Element

MOVE
Source 8
Dest CFG.MsgDest.Trend.LEN
8

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.T
'T'
Dest CFG.MsgDest.Trend.DATA[0]
'T'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._r
'r'
Dest CFG.MsgDest.Trend.DATA[1]
'r'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._e
'e'
Dest CFG.MsgDest.Trend.DATA[2]
'e'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._n
'n'
Dest CFG.MsgDest.Trend.DATA[3]
'n'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._d
'd'
Dest CFG.MsgDest.Trend.DATA[4]
'd'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.LtBracket
'['
Dest CFG.MsgDest.Trend.DATA[5]
'['

EDC Config Message
Remote Destination
Element

MOVE
Source ascii._0
'0'
Dest CFG.MsgDest.Trend.DATA[6]
'0'

EDC Config Message
Remote Destination
Element

MOVE
Source ascii.RtBracket
']'
Dest CFG.MsgDest.Trend.DATA[7]
']'

Configuration

35

Gateway Message Destination Tag Version[0]

EDC Config Message Remote Destination Element

MOVE
Source 10
Dest CFG.MsgDest.Version.LEN 10

EDC Config Message Remote Destination Element

MOVE
Source ascii.V 'V'
Dest CFG.MsgDest.Version.DATA[0] 'V'

EDC Config Message Remote Destination Element

MOVE
Source ascii._e 'e'
Dest CFG.MsgDest.Version.DATA[1] 'e'

EDC Config Message Remote Destination Element

MOVE
Source ascii._r 'r'
Dest CFG.MsgDest.Version.DATA[2] 'r'

EDC Config Message Remote Destination Element

MOVE
Source ascii._s 's'
Dest CFG.MsgDest.Version.DATA[3] 's'

EDC Config Message Remote Destination Element

MOVE
Source ascii._i 'i'
Dest CFG.MsgDest.Version.DATA[4] 'i'

EDC Config Message Remote Destination Element

MOVE
Source ascii._o 'o'
Dest CFG.MsgDest.Version.DATA[5] 'o'

EDC Config Message Remote Destination Element

MOVE
Source ascii._n 'n'
Dest CFG.MsgDest.Version.DATA[6] 'n'

EDC Config Message Remote Destination Element

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Version.DATA[7] '['

EDC Config Message Remote Destination Element

MOVE
Source ascii._0 '0'
Dest CFG.MsgDest.Version.DATA[8] '0'

EDC Config Message Remote Destination Element

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Version.DATA[9] ']'

Configuration

Gateway Message Destination Tag Cycle[0]

36

EDC Config Message Remote Destination Element

MOVE
Source 8
Dest CFG.MsgDest.Cycle.LEN 8

EDC Config Message Remote Destination Element

MOVE
Source ascii.C 'C'
Dest CFG.MsgDest.Cycle.DATA[0] 'C'

EDC Config Message Remote Destination Element

MOVE
Source ascii._y 'y'
Dest CFG.MsgDest.Cycle.DATA[1] 'y'

EDC Config Message Remote Destination Element

MOVE
Source ascii._c 'c'
Dest CFG.MsgDest.Cycle.DATA[2] 'c'

EDC Config Message Remote Destination Element

MOVE
Source ascii._l 'l'
Dest CFG.MsgDest.Cycle.DATA[3] 'l'

EDC Config Message Remote Destination Element

MOVE
Source ascii._e 'e'
Dest CFG.MsgDest.Cycle.DATA[4] 'e'

EDC Config Message Remote Destination Element

MOVE
Source ascii.LtBracket '['
Dest CFG.MsgDest.Cycle.DATA[5] '['

EDC Config Message Remote Destination Element

MOVE
Source ascii._0 '0'
Dest CFG.MsgDest.Cycle.DATA[6] '0'

EDC Config Message Remote Destination Element

MOVE
Source ascii.RtBracket ']'
Dest CFG.MsgDest.Cycle.DATA[7] ']'

Gateway Message Destination Tag Callout

37

EDC Config Message Remote Destination Tag

CONCAT
Source A CFG.MsgDest.Tag 'EDC_EqRxBuffer.'
Source B CFG.MsgDest.Line 'Line[0].'
Dest CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT
Source A CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['
Source B CFG.MsgDest.Equip 'Equip[6].'
Dest CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT
Source A CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['
Source B CFG.MsgDest.Callout 'Callout[0]'
Dest CFG.MsgDest.CalloutDest 'EDC_EqRxBuffer.Line['

Configuration

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Gateway Message Destination Tag Bulk Fault

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.TagFault 'EDC_EqRxBufferFault.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'
Source B	CFG.MsgDest.Fault 'Fault[0]'
Dest	CFG.MsgDest.FaultDest 'EDC_EqRxBufferFault.'

39

Gateway Message Destination Tag Quality

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.Tag 'EDC_EqRxBuffer.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Quality 'Quality[0]'
Dest	CFG.MsgDest.QualityDest 'EDC_EqRxBuffer.Line['

40

Gateway Message Destination Tag Trend

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.Tag 'EDC_EqRxBuffer.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Trend 'Trend[0]'
Dest	CFG.MsgDest.TrendDest 'EDC_EqRxBuffer.Line['

Configuration

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Gateway Message Destination Tag Version

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.Tag 'EDC_EqRxBuffer.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['
Source B	CFG.MsgDest.Version 'Version[0]'
Dest	CFG.MsgDest.VersionDest 'EDC_EqRxBuffer.Line['

42

Gateway Message Destination Tag Cycle
** Cycle uses the EDC_EqRxBufferAux tag **

EDC Config Message Remote Destination Tag

CONCAT	
Source A	CFG.MsgDest.TagAux 'EDC_EqRxBufferAux.'
Source B	CFG.MsgDest.Line 'Line[0].'
Dest	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'
Source B	CFG.MsgDest.Equip 'Equip[6].'
Dest	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'

EDC Config Message Remote Destination Element

CONCAT	
Source A	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'
Source B	CFG.MsgDest.Cycle 'Cycle[0]'
Dest	CFG.MsgDest.CycleDest 'EDC_EqRxBufferAux.Li'

(End)

Initialize CLx Module

Initialize CLx Module

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 12
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Re-enable any Disabled features

2

EDC Config Disable CLx Processor Fault Handler (for Testing Only)
CFG.DisableFaultHandler
U

Initialize and Start Module & Scan Diagnostics

3

CLx Module Status Clear Scan History
CLX.TaskScan.Clear
L

CLx Module Status Clear Scan History
CLX.ProgScan.Clear
L

Clear Performance Data

4

Equipment Data Collection - Line Equipment Status

FLL
Source 0
Dest Status
Length 1

Initialize CLx Module

v1.3 Fault Data Change
Initialize Callout/Fault Message Data Packet 0
(ensures good data for Keep Alive when no Faults occur)
v3.02 Init Data Refresh Packets 0-7 (block on first scan when all EqDrv use time-randomized refresh of all packets)

5

S:FS

Data Refresh By Packet	Data Refresh By Packet	Data Refresh By Packet	Data Refresh By Packet	Data Refresh By Packet	Data Refresh By Packet	Data Refresh By Packet	Data Refresh By Packet
DataRefreshPacket.0	DataRefreshPacket.1	DataRefreshPacket.2	DataRefreshPacket.3	DataRefreshPacket.4	DataRefreshPacket.5	DataRefreshPacket.6	DataRefreshPacket.7
L	L	L	L	L	L	L	L

v1.8 Duplicate Message Check
(only initialize on First Scan -- no need to repeat check for Configuration changes)

v1.10 -- DHF011 Clear entire DuplicateMsg structure to initialize Diagnostic Counts, ONS, etc.
(moved from Reset routine to avoid nuisance DuplicateMsg detects during UserCfg.UpdateEQID)

6

S:FS

Duplicate Message Detection Control

FLL	
Source	0
Dest	DuplicateMsg
Length	1

(End)

Monitors CLx Status

Service CLx Controller Functions

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 30
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Controller Product, Revision, Serial Number, Name
(Note: Revision Major/Minor of 20/11 = v20.011)

2

S:FS

CLx Module Status Controller Type (56=L63; see GSV Help for Lookup Table)

GSV
Class Name ControllerDevice
Instance Name
Attribute Name ProductCode
Dest CLX.Version.ProductCode 309

High Byte = Minor
Low Byte = Major

GSV
Class Name ControllerDevice
Instance Name
Attribute Name ProductRev
Dest ProductRev 16#0b26

CLx Module Status Controller Serial Number (Hex)

GSV
Class Name ControllerDevice
Instance Name
Attribute Name SerialNumber
Dest CLX.Version.SerialNumber 16#016b_5c03

CLx Module Status Controller Name

GSV
Class Name Controller
Instance Name
Attribute Name Name
Dest CLX.Version.Name 'eAssy_EDC_Gway'

CLx Module Status Controller Major Product Revision

CLR
Dest CLX.Version.ProductRevMajor 38

CLx Module Status Controller Minor Product Revision

CLR
Dest CLX.Version.ProductRevMinor 11

CLx Module Status Controller Major Product Revision

BTD
Source ProductRev 16#0b26
Source Bit 0
Dest CLX.Version.ProductRevMajor 38
Dest Bit 0
Length 8

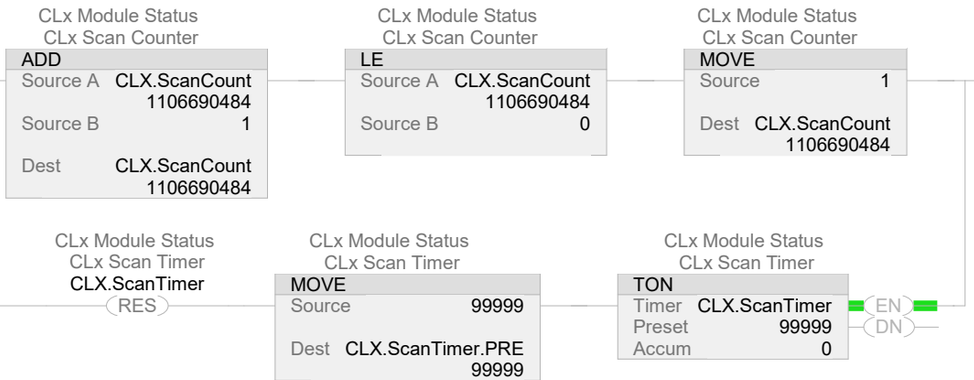
CLx Module Status Controller Minor Product Revision

BTD
Source ProductRev 16#0b26
Source Bit 8
Dest CLX.Version.ProductRevMinor 11
Dest Bit 0
Length 8

Monitors CLx Status

Increment Scan Counter and Initialize Scan Timer
v3.02 Data Refresh only works with positive scan counts

3



Monitors CLx Status

Create DateStamp & TimeStamp DINTs for Transaction Logs
Note: All events on same scan will have same Date/Time Stamp

[0]-[2] = Year YYYY, Month MM, Day DD
[3]-[6] = Hour HH, Minute MM, Second SS, microsec UUUUUU
[7] = YY, [8] = YYYYMMDD, [9] = HHMMSS
[10] = last SS, [11] = msec, [12] = fractional sec .ss

4

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

GSV
Class Name WallClockTime
Instance Name
Attribute Name LocalDateTime
Dest CLX.SysTime[0]
2026

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DIV
Source A CLX.SysTime[6]
273675
Source B 1000
Dest CLX.SysTime[11]
273

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

DIV
Source A CLX.SysTime[11]
273
Source B 10
Dest CLX.SysTime[12]
27

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

NE
Source A CLX.SysTime[5]
52
Source B CLX.SysTime[10]
52

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

MOVE
Source CLX.SysTime[5]
52
Dest CLX.SysTime[10]
52

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

MOD
Source A CLX.SysTime[0]
2026
Source B 100
Dest CLX.SysTime[7]
26

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

CPT
Dest CLX.SysTime[8]
20261030
Expression CLX.SysTime[0]*10000+(CLX.SysTime[1]*100)+CLX.SysTime[2]

CLx Module Status
CLx System Time
(from WALLCLOCKTIME)

CPT
Dest CLX.SysTime[9]
152452
Expression CLX.SysTime[3]*10000+(CLX.SysTime[4]*100)+CLX.SysTime[5]

Monitors CLx Status

Create DateTime Stamp STRING from SysTime arrays
-- Clear & Initialize to 0's for auto-padding of single-digit Month, Day, etc.
-- Insert each Date/Time field based on string length to use 0-pad

- Date (yyyymmdd)
- Time (hhmmss)
- millisec (sss)

5

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

FLL
Source 0
Dest CLX.DateTime
Length 1

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

MOVE
Source 17
Dest CLX.DateTime.LEN
17

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

FLL
Source ascii._0
Dest CLX.DateTime.DATA[0]
Length CLX.DateTime.LEN

CLx Module Status CLx System Time (from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[8]
20261030
Dest TempString
'390'

Temp Result for Array Size

SUB
Source A 9
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
'20261030152452273'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
'20261030152452273'
Source B TempString
'390'
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status CLx System Time (from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[9]
152452
Dest TempString
'390'

Temp Result for Array Size

SUB
Source A 15
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
'20261030152452273'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
'20261030152452273'
Source B TempString
'390'
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status CLx System Time (from WALLCLOCKTIME)

DTOS
Source CLX.SysTime[11]
273
Dest TempString
'390'

Temp Result for Array Size

SUB
Source A 18
Source B TempString.LEN
3
Dest LEN
15

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

DELETE
Source CLX.DateTime
'20261030152452273'
Qty TempString.LEN
3
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

CLx Module Status CLx System DateTime (YYYYMMDDHHMMSSsss)

INSERT
Source A CLX.DateTime
'20261030152452273'
Source B TempString
'390'
Start LEN
15
Dest CLX.DateTime
'20261030152452273'

Monitors CLx Status

DHF014 -- External Point Status Monitor

Calcluate seconds in hour for use as Heartbeat (0-3600)

6

CLx Module Status
CLx System Time
Seconds in Current
Hour (mm*60)+ss

CPT
Dest CLX.SecondsInHour 1492
Expression CLX.SysTime[4]*60+CLX.SysTime[5]

CLx Transaction Timestamp for Messages
v1.7 - Invalid Transaction Time if Year<2017

7

CLx Module Status
Year (YYYY)

MOVE
Source CLX.SysTime[0] 2026
Dest CLX.TransTime.Year 2026

CLx Module Status
Month (MM)

MOVE
Source CLX.SysTime[1] 10
Dest CLX.TransTime.Month 10

CLx Module Status
Day (DD)

MOVE
Source CLX.SysTime[2] 30
Dest CLX.TransTime.Day 30

CLx Module Status
Hour (HH)

MOVE
Source CLX.SysTime[3] 15
Dest CLX.TransTime.Hour 15

CLx Module Status
Minute (MM)

MOVE
Source CLX.SysTime[4] 24
Dest CLX.TransTime.Min 24

CLx Module Status
Second (SS)

MOVE
Source CLX.SysTime[5] 52
Dest CLX.TransTime.Sec 52

CLx Module Status
Fractional Seconds
(.ss)

MOVE
Source CLX.SysTime[12] 27
Dest CLX.TransTime.FracSec 27

CLx Module Status
Year (YYYY)

LT
Source A CLX.TransTime.Year 2026
Source B 2017

CLx Module Status
CLx Transaction Time
Invalid
CLX.TransTimeInvalid

Monitors CLx Status

CLx Task Scan Time
-- Clear Values
-- Initialize Array Length

8

CLx Module Status
Clear Scan History
CLX.TaskScan.Clear

S:FS

CLx Module Status
CLx Task Scan Time

FLL	
Source	0
Dest	CLX.TaskScan
Length	1

CLx Module Status
Enable Scan Monitor
CLX.TaskScan.EN

L

CLx Module Status
History Array

SIZE	
Source	CLX.TaskScan.History[0]
	15370
Dim. To Vary	0
Size	LEN
	15

CLx Module Status
Avg Scan CONTROL

MOVE	
Source	LEN
	15
Dest	CLX.TaskScan.AvgCTL.LEN
	100

v2.09 Data Refresh signal (daily, pre-shift)
Send all configured Fault Packets to refresh any bits stuck on or off after communication outages, buffer overruns, etc.
(this has mostly been an issue after shutdowns)

DataRefreshTime is located in CLxStatus because it's related to CLxTime and is a common routine for the FEC and Logix EqDriver. (avoid duplicate code in every FEC EqDriver)
each Refresh reloads the RefreshTime configuration

9

CLx Module Status
Hour (HH)

EQ	
Source A	CLX.TransTime.Hour
	15
Source B	DataRefreshHour
	5

CLx Module Status
Minute (MM)

EQ	
Source A	CLX.TransTime.Min
	24
Source B	DataRefreshMin
	30

CLx Module Status
Second (SS)

EQ	
Source A	CLX.TransTime.Sec
	52
Source B	DataRefreshSec
	0

DataRefreshTimeONS

ONS

s:fs

Scheduled Time
for Data Refresh
1-shot
DataRefreshTime

MOVE	
Source	5
Dest	DataRefreshHour
	5

MOVE	
Source	30
Dest	DataRefreshMin
	30

MOVE	
Source	0
Dest	DataRefreshSec
	0

Monitors CLx Status

CLx Task Scan Time
-- Store New Value in History Array
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value

10

CLx Module Status
Enable Scan Monitor
CLX.TaskScan.EN

CLx Module Status
Last Scan

GSV	
Class Name	Task
Instance Name	THIS
Attribute Name	LastScanTime
Dest	CLX.TaskScan.Last 13553

CLx Module Status
History Array

MOVE	
Source	CLX.TaskScan.Last 13553
Dest	CLX.TaskScan.History[CLX.TaskScan.i] 15370

CLx Module Status
Last Scan

GT	
Source A	CLX.TaskScan.Last 13553
Source B	CLX.TaskScan.Max 1883372

CLx Module Status
Max Scan

MOVE	
Source	CLX.TaskScan.Last 13553
Dest	CLX.TaskScan.Max 1883372

CLx Module Status
Last Scan

LT	
Source A	CLX.TaskScan.Last 13553
Source B	CLX.TaskScan.Min 598

CLx Module Status
Last Scan

GT	
Source A	CLX.TaskScan.Last 13553
Source B	0

CLx Module Status
Min Scan

MOVE	
Source	CLX.TaskScan.Last 13553
Dest	CLX.TaskScan.Min 598

CLx Module Status
Min Scan

EQ	
Source A	CLX.TaskScan.Min 598
Source B	0

CLx Module Status
History Array Index

ADD	
Source A	CLX.TaskScan.i 0
Source B	1
Dest	CLX.TaskScan.i 0

CLx Module Status
History Array Index

GE	
Source A	CLX.TaskScan.i 0
Source B	CLX.TaskScan.AvgCTL.LEN 100

CLx Module Status
History Array Index

CLR	
Dest	CLX.TaskScan.i 0

CLx Module Status
History Array Index

LT	
Source A	CLX.TaskScan.i 0
Source B	0

CLx Module Status
Avg Scan CONTROL
CLX.TaskScan.AvgCTL
RES

CLx Module Status
Avg Scan CONTROL

AVE		
Array	CLX.TaskScan.History[0]	EN
Dim. To Vary	0	
Dest	CLX.TaskScan.Avg 14242	DN
Control	CLX.TaskScan.AvgCTL	ER
Length	100	
Position	99	

Monitors CLx Status

11

CLx Configuration & Initialization Program Scan Time
-- Clear Values
-- Initialize Array Length

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear

S:FS

CLx Module Status
CLx Total Program
Scan Time (All
Programs)

FLL
Source 0
Dest CLX.ProgScan
Length 1

CLx Module Status
Clear Scan History
CLX.ProgScan.Clear

L

CLx Module Status
Enable Scan Monitor
CLX.ProgScan.EN

L

CLx Module Status
History Array

SIZE
Source CLX.ProgScan.History[0]
148
Dim. To Vary 0
Size LEN
15

CLx Module Status
Avg Scan CONTROL

MOVE
Source LEN
15
Dest CLX.ProgScan.AvgCTL.LEN
100

Monitors CLx Status

CLx Configuration & Initialization Program Scan Time
-- Last Value & History Array are summed by logic in each Program
-- Calculate Running Average
-- Increment History Array Index
-- Store Max Value
-- Store Min Value

12

S:FS /

CLx Module Status Enable Scan Monitor
CLX.ProgScan.EN

CLx Module Status Last Scan
MOVE
Source CLX.ProgScanTemp 140
Dest CLX.ProgScan.Last 141

CLx Module Status History Array
MOVE
Source CLX.ProgScan.Last 141
Dest CLX.ProgScan.History[Prog.Scan.i] 144

CLx Module Status Last Scan
GT
Source A CLX.ProgScan.Last 141
Source B CLX.ProgScan.Max 305

CLx Module Status Max Scan
MOVE
Source CLX.ProgScan.Last 141
Dest CLX.ProgScan.Max 305

CLx Module Status Last Scan
LT
Source A CLX.ProgScan.Last 141
Source B CLX.ProgScan.Min 47

CLx Module Status Last Scan
GT
Source A CLX.ProgScan.Last 141
Source B 0

CLx Module Status Min Scan
MOVE
Source CLX.ProgScan.Last 141
Dest CLX.ProgScan.Min 47

CLx Module Status Min Scan
EQ
Source A CLX.ProgScan.Min 47
Source B 0

CLx Module Status History Array Index
ADD
Source A CLX.ProgScan.i 99
Source B 1
Dest CLX.ProgScan.i 99

CLx Module Status History Array Index
GE
Source A CLX.ProgScan.i 99
Source B CLX.ProgScan.AvgCTL.LEN 100

CLx Module Status History Array Index
CLR
Dest CLX.ProgScan.i 99

CLx Module Status History Array Index
LT
Source A CLX.ProgScan.i 99
Source B 0

CLx Module Status Avg Scan CONTROL
CLX.ProgScan.AvgCTL
RES

CLx Module Status Avg Scan CONTROL
AVE
Array CLX.ProgScan.History[0]
Dim. To Vary 0
Dest CLX.ProgScan.Avg 143
Control CLX.ProgScan.AvgCTL
Length 100
Position 0
EN
DN
ER

Monitors CLx Status

13

Log Transactions from Configuration Program

(Clear Trans.LOGi when TransLog is initialized on Reset)

EDC Config Reset ONS (Log Transaction)
CFG.ResetONS

Transaction Log Control -- Transaction Log Index
CLR
Dest Trans.Ctl.LOGi
0

Transaction Log Control -- Transaction Error Log Index
CLR
Dest Trans.Ctl.ERRi
0

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 9000
Input Par 0
Input Par -1

EDC Config Configuration ONS (Log Transaction)
CFG.ConfigONS

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 9001
Input Par 0
Input Par -1

EDC Config Initialization ONS (Log Transaction)
CFG.InitONS

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 9002
Input Par 0
Input Par -1

14

DHF010 -- Log Transaction for Task Scan Exceeds Threshold (hard-coded to 100 msec)

CLx Module Status Last Scan
GT
Source A CLX.TaskScan.Last
13553
Source B 100000

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source CLX.TaskScan.Last
13553
Dest Trans.Ctl.Opt[1]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 9004
Input Par 0
Input Par -1

Monitors CLx Status

Clear Total Program Scan & History Array for next Scan values

CLx Module Status
CLx Total Program
Scan Time - Temp
Summation

CLR
Dest CLX.ProgScanTemp
140

15

(End)

Define ASCII Table

ASCII Table Enumeration
2008 Aug 22 -- Initial Release by Dan Guendelsberger (Rockwell Automation)
2009 Jun 11 -- Updated by Dan Guendelsberger
2010 Apr 08 -- Converted to Ladder Logic by Dan Guendelsberger

Creates udt-based ASCII Table for use in enumerations to avoid ASCII table lookups (and mistakes).
ascii._1 = 48
ascii._2 = 49
etc.
ascii.A = 65
ascii.B = 66
etc.
ascii._a = 97
ascii._b = 98
etc.

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 31
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Skip Value Initialize Loop if already initialized
2009Jun11 -- execute loop once to save scan time -- about 1 msec

2

ASCII Table Enumeration Control Structure SINT Array of Values 0-127 for loading ASCII udt

EQ
Source A asciiTable.Values[127] 127
Source B 127

lbl_InitDone
JMP

Define ASCII Table

Reset Index for Initialization Loop

3

ASCII Table Enumeration Control Structure Table Loop Index

CLR
Dest asciiTable.LOOPi 128

Initialize value array (0-127)

4

lbl_Init
LBL

ASCII Table Enumeration Control Structure SINT Array of Values 0-127 for loading ASCII udt

MOVE
Source asciiTable.LOOPi 128
Dest asciiTable.Values[asciiTable.LOOPi] ??

ASCII Table Enumeration Control Structure Table Loop Index

ADD
Source A asciiTable.LOOPi 128
Source B 1
Dest asciiTable.LOOPi 128

ASCII Table Enumeration Control Structure Table Loop Index

LE
Source A asciiTable.LOOPi 128
Source B 127

lbl_Init
JMP

copy value array (0-127) to the ASCII Table udt
2009Jun11 -- continuously execute to ensure ASCII values aren't mistakenly overwritten by users

5

lbl_InitDone
LBL

ASCII Table Enumeration

COP
Source asciiTable.Values[0]
Dest ascii
Length 1

(End)

Listen for Duplicate Message Tag activity before enabling Driver Messages

v1.8 Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 50
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

2

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

Duplicate Message Detection Control Duplicate Callout Read Enable 1-Shot
DuplicateMsg.CalloutReadEnONS
ONS

EDC Gateway Callout Read Message
CalloutReadMsg.DN
<EQ0006_CalloutReadMsg.DN>

Duplicate Message Detection Control Duplicate Callout Read Done 1-Shot
DuplicateMsg.CalloutReadDnONS
ONS

EDC Gateway Callout Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

Duplicate Message Detection Control Duplicate Callout Read Done 1-Shot
DuplicateMsg.CalloutReadErONS
ONS

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

Duplicate Message Detection Control Duplicate Callout Write Enable 1-Shot
DuplicateMsg.CalloutWriteEnONS
ONS

EDC Gateway Callout Write Message
CalloutWriteMsg.DN
<EQ0006_CalloutWriteMsg.DN>

Duplicate Message Detection Control Duplicate Callout Write Done 1-Shot
DuplicateMsg.CalloutWriteDnONS
ONS

EDC Gateway Callout Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Duplicate Message Detection Control Duplicate Callout Write Done 1-Shot
DuplicateMsg.CalloutWriteErONS
ONS

S:FS
/

S:FS
/

Duplicate Message Detection Control Duplicate Callout Read Count

ADD
Source A DuplicateMsg.CalloutReadCount
0
Source B 1
Dest DuplicateMsg.CalloutReadCount
0

Duplicate Message Detection Control Duplicate Callout Write Count

ADD
Source A DuplicateMsg.CalloutWriteCount
0
Source B 1
Dest DuplicateMsg.CalloutWriteCount
0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 135
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 145
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found
DuplicateMsg.FD
L

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

3

EDC Gateway Fault Read Message
FaultReadMsg.EN
<EQ0006_FaultReadMsg.EN>

Duplicate Message Detection Control Duplicate Fault Read Enable 1-Shot
DuplicateMsg.FaultReadEnONS
ONS

EDC Gateway Fault Read Message
FaultReadMsg.DN
<EQ0006_FaultReadMsg.DN>

Duplicate Message Detection Control Duplicate Fault Read Done 1-Shot
DuplicateMsg.FaultReadDnONS
ONS

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

Duplicate Message Detection Control Duplicate Fault Read Done 1-Shot
DuplicateMsg.FaultReadErONS
ONS

EDC Gateway Fault Write Message
FaultWriteMsg.EN
<EQ0006_FaultWriteMsg.EN>

Duplicate Message Detection Control Duplicate Fault Write Enable 1-Shot
DuplicateMsg.FaultWriteEnONS
ONS

EDC Gateway Fault Write Message
FaultWriteMsg.DN
<EQ0006_FaultWriteMsg.DN>

Duplicate Message Detection Control Duplicate Fault Write Done 1-Shot
DuplicateMsg.FaultWriteDnONS
ONS

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Duplicate Message Detection Control Duplicate Fault Write Done 1-Shot
DuplicateMsg.FaultWriteErONS
ONS

S:FS
/

Duplicate Message Detection Control Duplicate Fault Read Count

ADD
Source A DuplicateMsg.FaultReadCount
0
Source B 1
Dest DuplicateMsg.FaultReadCount
0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 235
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

S:FS
/

Duplicate Message Detection Control Duplicate Fault Write Count

ADD
Source A DuplicateMsg.FaultWriteCount
0
Source B 1
Dest DuplicateMsg.FaultWriteCount
0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 245
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found
DuplicateMsg.FD
L

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

4

EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>

Duplicate Message Detection Control Duplicate Quality Read Enable 1-Shot DuplicateMsg.QualityReadEnONS ONS

EDC Gateway Quality Read Message QualityReadMsg.DN <EQ0006_QualityReadMsg.DN>

Duplicate Message Detection Control Duplicate Quality Read Done 1-Shot DuplicateMsg.QualityReadDnONS ONS

EDC Gateway Quality Read Message QualityReadMsg.ER <EQ0006_QualityReadMsg.ER>

Duplicate Message Detection Control Duplicate Quality Read Done 1-Shot DuplicateMsg.QualityReadErONS ONS

S:FS /

Duplicate Message Detection Control Duplicate Quality Read Count

ADD
Source A DuplicateMsg.QualityReadCount 0
Source B 1
Dest DuplicateMsg.QualityReadCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 335
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD (L)

EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>

Duplicate Message Detection Control Duplicate Quality Write Enable 1-Shot DuplicateMsg.QualityWriteEnONS ONS

EDC Gateway Quality Write Message QualityWriteMsg.DN <EQ0006_QualityWriteMsg.DN>

Duplicate Message Detection Control Duplicate Quality Write Done 1-Shot DuplicateMsg.QualityWriteDnONS ONS

EDC Gateway Quality Write Message QualityWriteMsg.ER <EQ0006_QualityWriteMsg.ER>

Duplicate Message Detection Control Duplicate Quality Write Done 1-Shot DuplicateMsg.QualityWriteErONS ONS

S:FS /

Duplicate Message Detection Control Duplicate Quality Write Count

ADD
Source A DuplicateMsg.QualityWriteCount 0
Source B 1
Dest DuplicateMsg.QualityWriteCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 345
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

5

EDC Gateway Trend Read Message TrendReadMsg.EN <EQ0006_TrendReadMsg.EN>
Duplicate Message Detection Control Duplicate Trend Read Enable 1-Shot DuplicateMsg.TrendReadEnONS
ONS

EDC Gateway Trend Read Message TrendReadMsg.DN <EQ0006_TrendReadMsg.DN>
Duplicate Message Detection Control Duplicate Trend Read Done 1-Shot DuplicateMsg.TrendReadDnONS
ONS

EDC Gateway Trend Read Message TrendReadMsg.ER <EQ0006_TrendReadMsg.ER>
Duplicate Message Detection Control Duplicate Trend Read Done 1-Shot DuplicateMsg.TrendReadErONS
ONS

S:FS
/

Duplicate Message Detection Control Duplicate Trend Read Count

ADD
Source A DuplicateMsg.TrendReadCount 0
Source B 1
Dest DuplicateMsg.TrendReadCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 435
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD
L

EDC Gateway Trend Write Message TrendWriteMsg.EN <EQ0006_TrendWriteMsg.EN>
Duplicate Message Detection Control Duplicate Trend Write Enable 1-Shot DuplicateMsg.TrendWriteEnONS
ONS

EDC Gateway Trend Write Message TrendWriteMsg.DN <EQ0006_TrendWriteMsg.DN>
Duplicate Message Detection Control Duplicate Trend Write Done 1-Shot DuplicateMsg.TrendWriteDnONS
ONS

EDC Gateway Trend Write Message TrendWriteMsg.ER <EQ0006_TrendWriteMsg.ER>
Duplicate Message Detection Control Duplicate Trend Write Done 1-Shot DuplicateMsg.TrendWriteErONS
ONS

S:FS
/

Duplicate Message Detection Control Duplicate Trend Write Count

ADD
Source A DuplicateMsg.TrendWriteCount 0
Source B 1
Dest DuplicateMsg.TrendWriteCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 445
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

6

EDC Gateway Version Write Message VersionWriteMsg.EN <EQ0006_VersionWriteMsg.EN>
Duplicate Message Detection Control Duplicate Version Write Enable 1-Shot DuplicateMsg.VersionWriteEnONS
ONS

EDC Gateway Version Write Message VersionWriteMsg.DN <EQ0006_VersionWriteMsg.DN>
Duplicate Message Detection Control Duplicate Version Write Done 1-Shot DuplicateMsg.VersionWriteDnONS
ONS

EDC Gateway Version Write Message VersionWriteMsg.ER <EQ0006_VersionWriteMsg.ER>
Duplicate Message Detection Control Duplicate Version Write Done 1-Shot DuplicateMsg.VersionWriteErONS
ONS

S:FS /

Duplicate Message Detection Control Duplicate Version Write Count

ADD
Source A DuplicateMsg.VersionWriteCount 0
Source B 1
Dest DuplicateMsg.VersionWriteCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 545
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD
L

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

7

EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN <EQ0006_TimeSyncReadMsg.EN>
Duplicate Message Detection Control Duplicate TimeSync Read Enable 1-Shot DuplicateMsg.TimeSyncReadEnONS
ONS

EDC Gateway Time Sync Read Message TimeSyncReadMsg.DN <EQ0006_TimeSyncReadMsg.DN>
Duplicate Message Detection Control Duplicate TimeSync Read Done 1-Shot DuplicateMsg.TimeSyncReadDnONS
ONS

EDC Gateway Time Sync Read Message TimeSyncReadMsg.ER <EQ0006_TimeSyncReadMsg.ER>
Duplicate Message Detection Control Duplicate TimeSync Read Done 1-Shot DuplicateMsg.TimeSyncReadErONS
ONS

S:FS /

Duplicate Message Detection Control Duplicate TimeSync Read Count

ADD
Source A DuplicateMsg.TimeSyncReadCount 0
Source B 1
Dest DuplicateMsg.TimeSyncReadCount 0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 635
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD
L

Listen for Duplicate Message Tag activity before enabling Driver Messages

Check for Duplicate Message Tag Alias in another EQ Driver
(Duplicate Message Tags will lock up the controller -- block all Messages until Message Delay is complete)
v1.19 DHF031 Duplicate Message EN Inhibit - add ONS

8

EDC Gateway Cycle Read Message CycleReadMsg.EN <EQ0006_CycleReadMsg.EN>

Duplicate Message Detection Control Duplicate Cycle Read Enable 1-Shot DuplicateMsg.CycleReadEnONS

ONS

EDC Gateway Cycle Read Message CycleReadMsg.DN <EQ0006_CycleReadMsg.DN>

Duplicate Message Detection Control Duplicate Cycle Read Done 1-Shot DuplicateMsg.CycleReadDnONS

ONS

EDC Gateway Cycle Read Message CycleReadMsg.ER <EQ0006_CycleReadMsg.ER>

Duplicate Message Detection Control Duplicate Cycle Read Done 1-Shot DuplicateMsg.CycleReadErONS

ONS

S:FS

/

Duplicate Message Detection Control Duplicate Cycle Read Count

ADD
Source A DuplicateMsg.CycleReadCount
0
Source B 1
Dest DuplicateMsg.CycleReadCount
0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 735
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Duplicate Message Detection Control Duplicate Message Found DuplicateMsg.FD

L

EDC Gateway Cycle Write Message CycleWriteMsg.EN <EQ0006_CycleWriteMsg.EN>

Duplicate Message Detection Control Duplicate Cycle Write Enable 1-Shot DuplicateMsg.CycleWriteEnONS

ONS

EDC Gateway Cycle Write Message CycleWriteMsg.DN <EQ0006_CycleWriteMsg.DN>

Duplicate Message Detection Control Duplicate Cycle Write Done 1-Shot DuplicateMsg.CycleWriteDnONS

ONS

EDC Gateway Cycle Write Message CycleWriteMsg.ER <EQ0006_CycleWriteMsg.ER>

Duplicate Message Detection Control Duplicate Cycle Write Done 1-Shot DuplicateMsg.CycleWriteErONS

ONS

S:FS

/

Duplicate Message Detection Control Duplicate Cycle Write Count

ADD
Source A DuplicateMsg.CycleWriteCount
0
Source B 1
Dest DuplicateMsg.CycleWriteCount
0

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 745
Input Par DuplicateMsg.DelayTMR.ACC
Input Par 0

Listen for Duplicate Message Tag activity before enabling Driver Messages

Disable/Inhibit Program if Duplicate Message Tag detected

9

Duplicate Message Detection Control Duplicate Message Found
DuplicateMsg.FD

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	905
Input Par	DuplicateMsg.DelayTMR.ACC
Input Par	0

Duplicate Message Detection Control Program Disable Flag for SSV Command

MOVE	
Source	-1
Dest	DuplicateMsg.DisableFlag
	0

Duplicate Message Detection Control Program Disable Flag for SSV Command

SSV	
Class Name	Program
Instance Name	THIS
Attribute Name	DisableFlag
Source	DuplicateMsg.DisableFlag
	0

(End)

Equipment Driver User Configuration

Equipment Data Collection
User Configuration

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 90
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Driver Enable

Driver Interface - choose Toyopuc, RemMsg or Logix
TpucDriverEN - FEC driver with Toyopuc data via I/O Module
RemMsgEN - Remote Message data from CompactLogix, PLC-5, etc.
LogixDriverEN - driver embedded in Logix Controller or FEC driver with Remote Messages

2

Equipment Data Collection - User Config Toyopuc Driver Enable
UserCfg.TpucDriverEN
U

Equipment Data Collection - User Config Remote Driver Enable
UserCfg.RemDriverEN
U

Equipment Data Collection - User Config Logix Driver Enable
UserCfg.LogixDriverEN
L

TimeSyncEN - enables Time Synchronization of EqDriver from EDC Gateway wallclock

3

Equipment Data Collection - User Config Time Synchronization Message Enable
UserCfg.TimeSyncEN
U

Equipment Driver User Configuration

EDC Gateway Logix Path

Configure Message Path for the EDC Gateway Logix Controller
(This Configuration Message is not used - it serves as a placeholder for the Path used in Driver Messages.)

Use Gateway EN2T in Slot 5 (or future 6), Logix in Slot 3
Path = Local path to Ethernet Network, Gateway Slot 5 IP Address, 0, 3

4

AFI

EDC Gateway Message
Path Configuration

MSG
Message Control UserCfgMsg ...
<EQ0006_UserCfgMsg>
EN
DN
ER

Equipment Driver User Configuration

v1.3 - Dec2016 - Multi-packet support for >8 DINT Equipment-Defined Faults
Packet Enable - Latch bit for each 8-DINT Packet used below
Word Count - Total number of FaultMsg_EqDef Data Words used in each Packet
(unused Words are ignored by Server to reduce processing of raw data)

Toggle the UserCfg.UpdateEQID if Fault Packet Enables are added/removed from the configuration.
v1.5 - moved Packet EN and Counts from FaultMsgCtl to UserCfg to fix Reset, Config, Initialize sequence

5

Equipment Data Collection - User Config Toggle ON if EQID Changed
UserCfg.UpdateEQID

NOP

Equipment Data Collection - User Config Equipment Defined Packet EN bit array
UserCfg.EqDefPacketEN.0 (L)
UserCfg.EqDefPacketEN.1 (L)
UserCfg.EqDefPacketEN.2 (L)
UserCfg.EqDefPacketEN.3 (U)
UserCfg.EqDefPacketEN.4 (U)
UserCfg.EqDefPacketEN.5 (U)
UserCfg.EqDefPacketEN.6 (U)
UserCfg.EqDefPacketEN.7 (U)

Equipment Data Collection - User Config Equipment Defined Packet Data Word Count

MOVE
Source 8
Dest UserCfg.EqDefDataCount[0] 8

MOVE
Source 8
Dest UserCfg.EqDefDataCount[1] 8

MOVE
Source 8
Dest UserCfg.EqDefDataCount[2] 8

MOVE
Source 8
Dest UserCfg.EqDefDataCount[3] 8

MOVE
Source 8
Dest UserCfg.EqDefDataCount[4] 8

MOVE
Source 8
Dest UserCfg.EqDefDataCount[5] 8

MOVE
Source 8
Dest UserCfg.EqDefDataCount[6] 8

MOVE
Source 8
Dest UserCfg.EqDefDataCount[7] 8

(End)

Equipment Data Collection
User Configuration for Logix and Remote Drivers
(Toyopuc Drivers configured programatically in Toyopuc routines)

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 92
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Driver Options Enable

Latch EN for each Driver function used:
DriverEN - enables Live Callout Msg
FaultMsgEN - enables Buffered Fault/Status Msg
QualityMsgEN - enables Buffered Quality/Health Msg
TrendMsgEN - enables Buffered Trend Msg
CycleMsgEN - enables Buffered Cycle Msg
VersionEN - enables Version Msg
TimeSyncEN - enables Time Synchronization from EDC Gateway wallclock

2

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN (L)

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN (L)

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN (L)

Equipment Data Collection - User Config Trend Message Enable
UserCfg.TrendMsgEN (U)

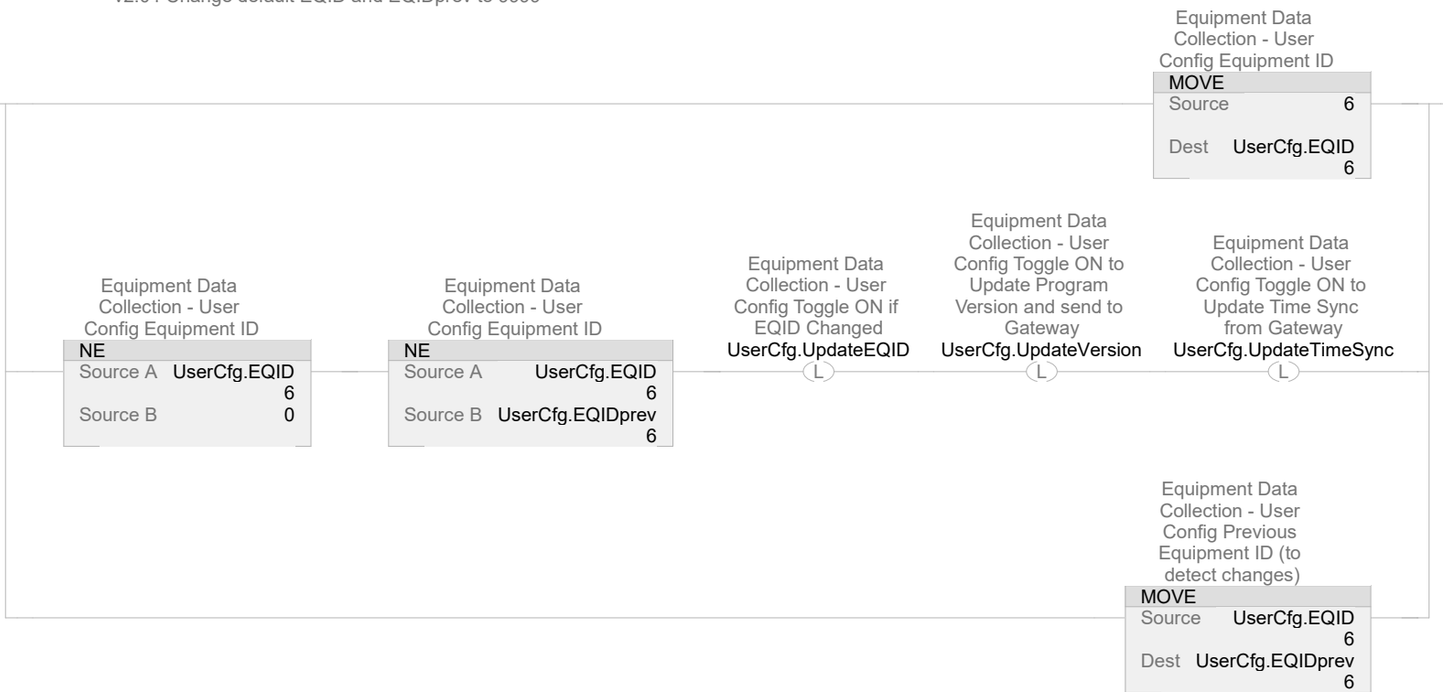
Equipment Data Collection - User Config Cycle Message Enable
UserCfg.CycleMsgEN (U)

Equipment Data Collection - User Config Version Message Enable
UserCfg.VersionEN (L)

Equipment ID

Load unique ID assigned to this Equipment Driver
(This must match the EQID configured on the EDC Server.)
v1.4 Add UpdateVersion and UpdateTimeSync
v2.04 Change default EQID and EQIDprev to 9999

3



(End)

EqDriver embedded in Logix Controller

EqDriver embedded in Logix Controller

This routine is only executed when the driver is configured for Logix

Replace with User Logic
xxUserCfgFaultBOOL
U

0

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	93
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

1

EqDriver embedded in Logix Controller

v1.6 Optional Message to get Overall Start from Gateway
Configure Message Source Element Array Index for desired EQID and remove AFI to Enable Rung
EQID PreDefined Fault word is loaded into RemPreDef

i.e. set Message Source to EDC_EqCallout[1,1].PreDef for EQID 101 -- loads RemotePreDef.OverallStartOn, ConvRunning and FIxedStopPos

2

AFI

EDC Gateway Overall Start Message Timer

MOVE
Source 1000
Dest OverallStartMsgTMR.PRE 250

EDC Gateway Overall Start Message Timer

TON
Timer OverallStartMsgTMR
Preset 250
Accum 0
EN
DN

EDC Gateway Overall Start Message Timer
OverallStartMsgTMR.DN

EDC Gateway Overall Start Read Message
OverallStartReadMsg.EN
<EQ0006_OverallStartReadMsg.EN>

EDC Gateway Overall Start Read Message

MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest OverallStartReadMsg.UnconnectedTimeout <EQ0006_OverallStartReadMsg.UnconnectedTimeout> 4000000

EDC Gateway Overall Start Read Message

MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest OverallStartReadMsg.ConnectionRate <EQ0006_OverallStartReadMsg.ConnectionRate> 1000000

EDC Gateway Overall Start Read Message

MOVE
Source CFG.CLx.ConnectionMult 0
Dest OverallStartReadMsg.TimeoutMultiplier <EQ0006_OverallStartReadMsg.TimeoutMultiplier> 0

EDC Gateway Overall Start Read Message
OverallStartReadMsg.EN_CC
<EQ0006_OverallStartReadMsg.EN_CC>
U

EDC Gateway Overall Start Read Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest OverallStartReadMsg.Path <EQ0006_OverallStartReadMsg.Path>
Length 1

EDC Gateway Overall Start Read Message

MSG
Message Control OverallStartReadMsg ... <EQ0006_OverallStartReadMsg>
EN
DN
ER

EDC Gateway Overall Start Message Timer
OverallStartMsgTMR
RES

EDC Gateway Overall Start Read Message
OverallStartReadMsg.ER
<EQ0006_OverallStartReadMsg.ER>

Remote PreDefined Faults from selected EQID

FLL
Source 0
Dest RemotePreDef <EQ0006_RemotePreDef>
Length 1

Gateway Overall Start Message ER
OverallStartReadMsgER
L

EqDriver embedded in Logix Controller

EDC Gateway Overall Start Read Message
OverallStartReadMsg.DN
<EQ0006_OverallStartReadMsg.DN>

Remote PreDefined Faults from selected EQID

COP	
Source	RemotePreDefDINT <EQ0006_RemotePreDefDINT>
Dest	RemotePreDef <EQ0006_RemotePreDef>
Length	1

Gateway Overall Start Message ER
OverallStartReadMsgER
U

Overall Start On
used by EDC to determine Equipment Downtime and OA%

Signal comes from Andon Consumed or EDC Gateway Read Message

3

Remote PreDefined Faults from selected EQID Line Production Work Time
RemotePreDef._00
<EQ0006_RemotePreDef._00>

Equipment Data Fault/Status Message Line Production Work Time
FaultMsg.PreDef._00

Conveyor Running
used by EDC to determine Equipment Downtime and OA%

Signal comes from Andon Consumed or EDC Gateway Read Message

4

Remote PreDefined Faults from selected EQID Line Conveyor Running
RemotePreDef._01
<EQ0006_RemotePreDef._01>
AFI

Equipment Data Fault/Status Message Line Conveyor Running
FaultMsg.PreDef._01

Fixed Stop Position
used by EDC to determine Equipment Downtime and OA%

Signal comes from Andon Consumed or EDC Gateway Read Message

5

Remote PreDefined Faults from selected EQID Line Conveyor Stopped at Fixed Stop Position (L10)
RemotePreDef._02
<EQ0006_RemotePreDef._02>
AFI

Equipment Data Fault/Status Message Line Conveyor Stopped at Fixed Stop Position (L10)
FaultMsg.PreDef._02

EqDriver embedded in Logix Controller

Fixed Stop Request
used by EDC to determine Equipment Downtime and OA%

Equipment Data
Fault/Status Message
EQ Fixed Stop
Request
FaultMsg.PreDef._03

6 AFI

Temp Stop Request
used by EDC to determine Equipment Downtime and OA%

Equipment Data
Fault/Status Message
Temp Stop Request
FaultMsg.PreDef._04

7 AFI

v2.0 Productive Stop Request, Fault_On, Alarm_On generated by EqDriver in _100_PreDefFaultData based on Event Mask configuration

8 NOP

Master On is OFF

Equipment Data
Fault/Status Message
Master On is OFF
FaultMsg.PreDef._08

9 AFI

Continuous is OFF

Equipment Data
Fault/Status Message
Continuous is OFF
FaultMsg.PreDef._09

10 AFI

Manual

Equipment Data
Fault/Status Message
Manual is ON
FaultMsg.PreDef._10

11 AFI

EqDriver embedded in Logix Controller

Cycle Time Over

12 AFI

Equipment Data
Fault/Status Message
Cycle Time is Over
FaultMsg.PreDef._11

Overload On

13 AFI

Equipment Data
Fault/Status Message
Overload is ON
FaultMsg.PreDef._12

v1.3 - Maintenance Call

14 AFI

Equipment Data
Fault/Status Message
Maintenance Call
FaultMsg.PreDef._13

v1.4 - Interlock Off - Equipment Bypassed

15 AFI

Equipment Data
Fault/Status Message
Interlock Off -
Equipment Bypassed
FaultMsg.PreDef._14

v2.1 Fault Memory generated by EqDriver in _100_PreDefFaultData

16 NOP

v2.0 - Don't send the HB as an Event (toggles continuously)
HB is monitored and reported as an Equipment Data Error if it stops

17 AFI

Equipment Data
Fault/Status Message
Inch Mode
FaultMsg.PreDef._16

v2.0 Alert_On generated by EqDriver in _100_PreDefFaultData based on Event Mask configuration

18 NOP

EqDriver embedded in Logix Controller

Equipment Data Fault/Status Message Line Down Callout Escalation
FaultMsg.PreDef._18

19 AFI

Health Alarm and Health Fault generated by Gateway from Server status download
Invalid Wall Clock and EqDataER generated by EqDriver

20 NOP

v2.1 Body Weld signal - Auto Run without Transfer

Equipment Data Fault/Status Message Continuous Off2 (BW Auto Run without Transfer)
FaultMsg.PreDef._23

21 AFI

v2.1 Delay (production hold - charge OR% but not OA%)

Equipment Data Fault/Status Message Delay (Prodution Hold)
FaultMsg.PreDef._24

22 AFI

v2.1 Short (charge OR% but not OA%)

Equipment Data Fault/Status Message Short or SubShort
FaultMsg.PreDef._25

23 AFI

v2.1 Full (charge OR% but not OA%)

Equipment Data Fault/Status Message Full
FaultMsg.PreDef._26

24 AFI

EqDriver embedded in Logix Controller

v2.1 In Demand Off (charge OR% but not OA%)
Equipment is not In Demand -- something else is holding line

Equipment Data
Fault/Status Message
In Demand Off (don't charge OA)
FaultMsg.PreDef._27

25 AFI

v2.1 Spare

Equipment Data
Fault/Status Message
Tool Change Mode
FaultMsg.PreDef._28

26 AFI

v2.1 Spare

Equipment Data
Fault/Status Message
Spare
FaultMsg.PreDef._29

27 AFI

RingBuffer FIlling and Message Error generated by EqDriver

28 NOP

Equipment Defined Fault/Status
8 DINTs - 256 signals unique to this Equipment

v1.3 Packet 0 - DINTs 0-7

Shop Status for Communications (Callout, Fault, Quality, Trend)
Bit 18 - Callout Sequence Error Blocked (normal occurrence for short duration events)

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM	
Source	0
Mask	16#0002_0fff
Dest	EqDefHold[0].Word[0] 16#0000_0000

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM	
Source	EDC.Perf.Sum.Shop.StateVal 16#0000_0a7e
Mask	16#fffd_f000
Dest	EqDefHold[0].Word[0] 16#0000_0000

29

EqDriver embedded in Logix Controller

Fault Ring Buffer

30

Equipment Data Collection Read Buffer Alarm EDC.RB_Fault01.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault02.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault03.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault04.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault05.BufferAlarm / Equipment Data Collection Read Buffer Alarm EDC.RB_Fault06.BufferAlarm / NoFaultBufferAlarm

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].0

NoFaultBufferAlarm /

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault01.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault02.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault03.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault04.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault05.WritePtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault06.WritePtrErr / NoFaultWritePtrErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].1

NoFaultWritePtrErr /

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault01.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault02.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault03.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault04.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault05.ReadPtrErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault06.ReadPtrErr / NoFaultReadPtrErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].2

NoFaultReadPtrErr /

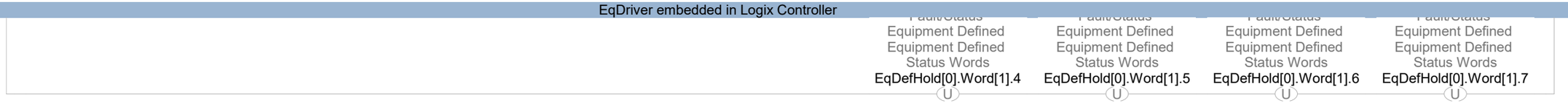
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault01.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault02.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault03.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault04.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault05.WriteBlockErr / Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Fault06.WriteBlockErr / NoFaultWriteBlockErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].3

NoFaultWriteBlockErr /

Equipment Data Fault/Status Equipment Data Fault/Status Equipment Data Fault/Status Equipment Data Fault/Status

EqDriver embedded in Logix Controller



EqDriver embedded in Logix Controller

Quality Ring Buffer
DN005 Health Status Alarm Callout

31

Equipment Data Collection Read Buffer Alarm EDC.RB_Quality01.BufferAlarm
Equipment Data Collection Read Buffer Alarm EDC.RB_Quality02.BufferAlarm
Equipment Data Collection Read Buffer Alarm EDC.RB_Quality03.BufferAlarm
Equipment Data Collection Read Buffer Alarm EDC.RB_Quality04.BufferAlarm
NoQualityBufferAlarm

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].8
NoQualityBufferAlarm

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality01.WritePtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality02.WritePtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality03.WritePtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality04.WritePtrErr
NoQualityWritePtrErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].9
NoQualityWritePtrErr

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality01.ReadPtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality02.ReadPtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality03.ReadPtrErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality04.ReadPtrErr
NoQualityReadPtrErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].10
NoQualityReadPtrErr

Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality01.WriteBlockErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality02.WriteBlockErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality03.WriteBlockErr
Equipment Data Collection Diagnostic - Manual Reset EDC.RB_Quality04.WriteBlockErr
NoQualityWriteBlockErr

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[1].11
NoQualityWriteBlockErr

Equipment Data

EqDriver embedded in Logix Controller

Equipment Data Collection Response has CIS Error - Send Callout Bit
EDC.ServerStat.Ctl.CisErrorSendCallout

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[0].Word[1].12

Equipment Data Collection Response has Health Status Error - Send Callout Bit
EDC.ServerStat.Ctl.HealthStatusSendCallout

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[0].Word[1].13

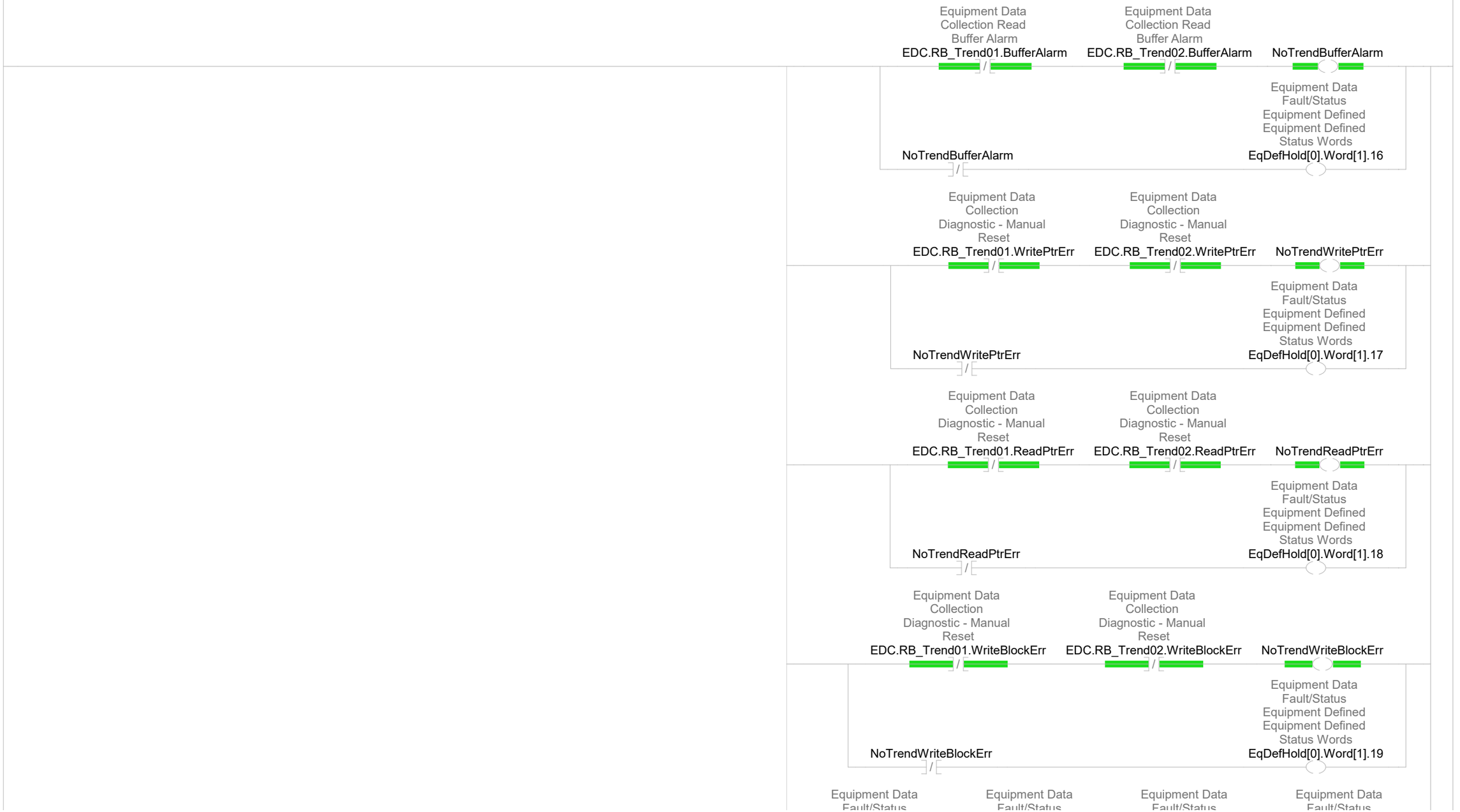
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[0].Word[1].14
U

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[0].Word[1].15
U

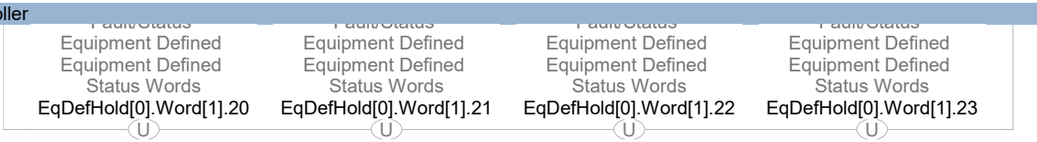
EqDriver embedded in Logix Controller

Trend Ring Buffer

32



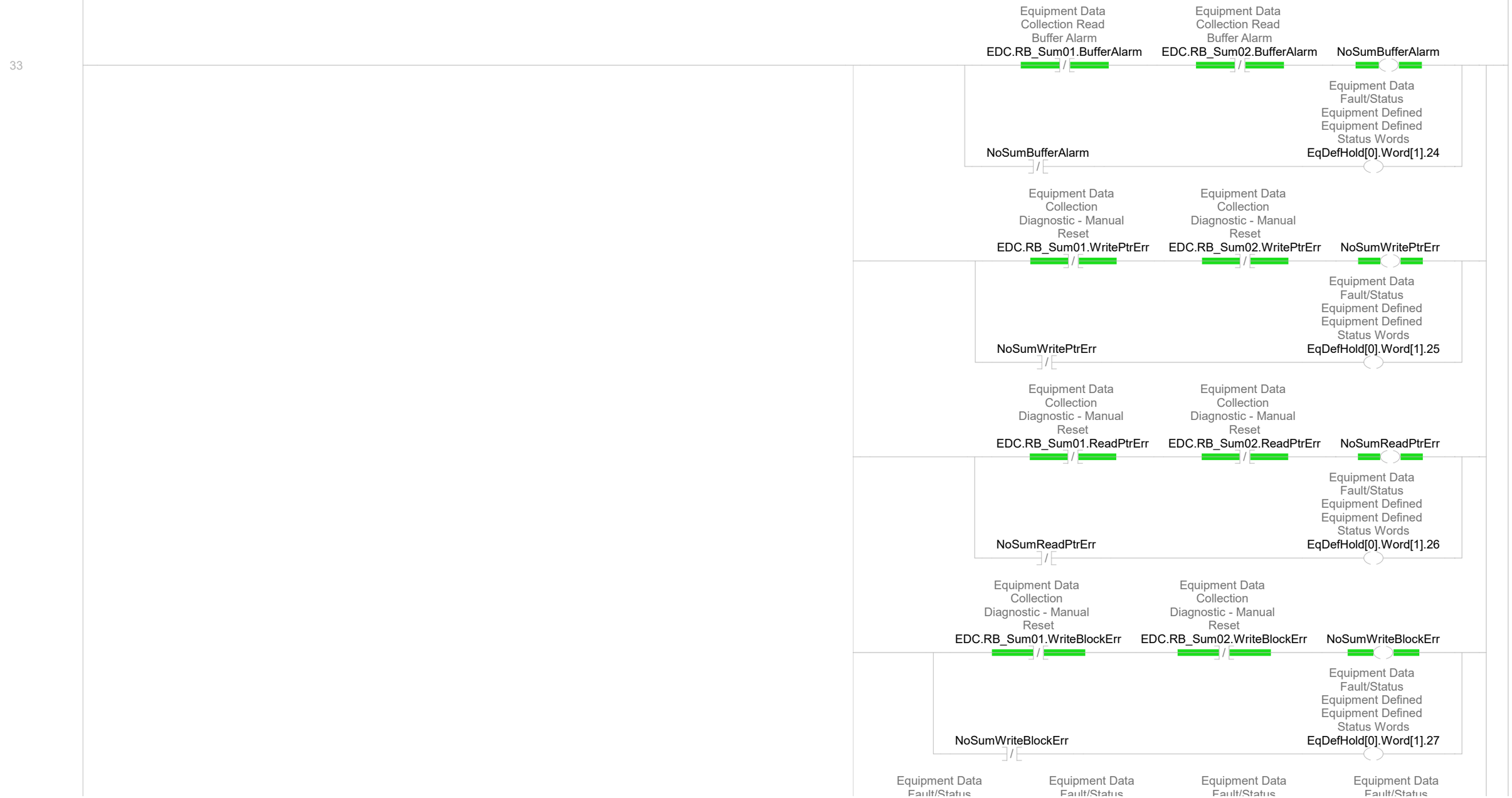
EqDriver embedded in Logix Controller



EqDriver embedded in Logix Controller

Summary Ring Buffer

33



EqDriver embedded in Logix Controller

Equipment Defined
Equipment Defined
Status Words
EqDefHold[0].Word[1].28
U

Equipment Defined
Equipment Defined
Status Words
EqDefHold[0].Word[1].29
U

Equipment Defined
Equipment Defined
Status Words
EqDefHold[0].Word[1].30
U

Equipment Defined
Equipment Defined
Status Words
EqDefHold[0].Word[1].31
U

Server Config Download

34

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source 0
Mask 16#0000FFFF
Dest EqDefHold[0].Word[2]
16#0000_0000

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source EDC_Status01.ServerCfg.StateVal
16#0000_0004
Mask 16#FFFF0000
Dest EqDefHold[0].Word[2]
16#0000_0000

Server Status Interface

35

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source 0
Mask 16#0000FFFF
Dest EqDefHold[0].Word[3]
16#0000_0000

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source EDC_Status01.ServerStat.StateVal
16#0000_0004
Mask 16#FFFF0000
Dest EqDefHold[0].Word[3]
16#0000_0000

EqDriver embedded in Logix Controller

DHF013 - Nov 27, 2017 -- Add Server Config Download All Event 4.0

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Equipment Data Collection Response Download All Started EDC.ServerCfg.Ctl.DownloadAllStarted	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].0
PdM Config Tool Configuration Array is Full PdM_Cfg.ToolArrayFull	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].1
PdM Config Trend Configuration Array is Full PdM_Cfg.TrendArrayFull	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].2
PdM Config Tool Processing Error - Invalid EQID PdM_Cfg.ToolInvalidEQID	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].3
PdM Config Tool Processing Error - Invalid Data Index PdM_Cfg.ToolInvalidDataIndex	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].4
PdM Config Tool Processing Error - Invalid Sample Period PdM_Cfg.ToolInvalidSamplePeriod	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].5
PdM Config Tool Processing Error - Invalid Network Data PdM_Cfg.ToolInvalidNetworkData	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].6
PdM Config Trend Processing Error - Invalid EQID PdM_Cfg.TrendInvalidEQID	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].7

EqDriver embedded in Logix Controller

PdM Config Trend Processing Error - Invalid Sample Period PdM_Cfg.TrendInvalidSamplePeriod	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].8
PdM Config Trend Processing Error - Invalid Sample Rate PdM_Cfg.TrendInvalidSampleRate	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].9
PdM Config Trend Processing Error - Invalid Network Data PdM_Cfg.TrendInvalidNetworkData	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].10
Equipment Data Collection Response Duplicate Error EDC.ServerCfg.Ctl.RspDupER	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].11
Equipment Data Collection Response Duplicate Error EDC.ServerPdm.Ctl.RspDupER	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].12
Equipment Data Collection Response Duplicate Error EDC.ServerMask.Ctl.RspDupER	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].13
Line Status Transaction Error LDD_LineStateSendER	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[0].Word[4].14
	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words MVM Source 0

EqDriver embedded in Logix Controller

		MVM Source Mask 16#FFFF8000 Dest EqDefHold[0].Word[4] 16#0000_0000
	Server Health Interface	
37	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words MVM Source 0 Mask 16#0000FFFF Dest EqDefHold[0].Word[5] 16#0000_0000	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words MVM Source EDC_Status01.ServerHealth.StateVal 16#0000_0004 Mask 16#FFFF0000 Dest EqDefHold[0].Word[5] 16#0000_0000
	Server Pdm Interface	
38	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words MVM Source 0 Mask 16#0000FFFF Dest EqDefHold[0].Word[6] 16#0000_0000	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words MVM Source EDC_Status01.ServerPdm.StateVal 16#0000_0004 Mask 16#FFFF0000 Dest EqDefHold[0].Word[6] 16#0000_0000

EqDriver embedded in Logix Controller

Server Mask Interface

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Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM	
Source	0
Mask	16#0000FFFF
Dest	EqDefHold[0].Word[7] 16#0000_0000

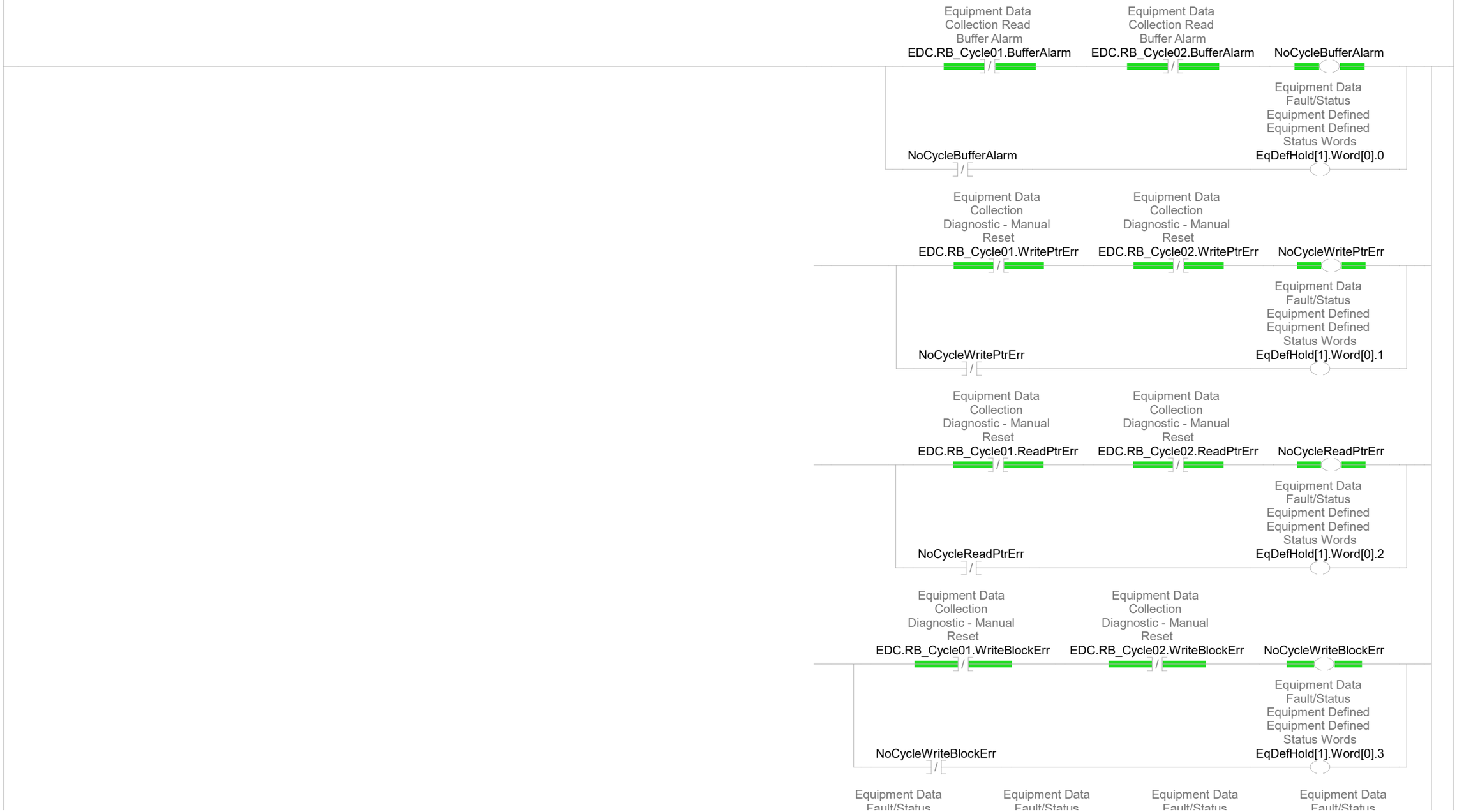
Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

MVM	
Source	EDC_Status01.ServerMask.StateVal 16#0000_0004
Mask	16#FFFF0000
Dest	EqDefHold[0].Word[7] 16#0000_0000

EqDriver embedded in Logix Controller

Cycle Ring Buffer

40



EqDriver embedded in Logix Controller

Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[0].4
U

Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[0].5
U

Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[0].6
U

Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[0].7
U

Packet 1 Spare

41

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
MVM
Source 0
Mask 16#FFFFFF00
Dest EqDefHold[1].Word[0]
16#0000_0000

Event Text Storage

42

CPT
Dest TextQtyLimit
175500
Expression EDC_HMI.CFG.EvTxt01.n *6 *0.9

Event
Controls Event Text
Qty
GT
Source A EDC_Event.CTL.TextQty
45032
Source B TextQtyLimit
175500

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].0

Server Interface
Status State = Red -
Equip Event Array
Full ER
EDC_Status01.ServerCfg.Update.EquipEventArrayFull

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].1

EqDriver embedded in Logix Controller

Ethernet 00 Status

43

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet00.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].2

Ethernet Module Status Last Scan
GE
Source A EDC_Enet00.CPU_Util.Last
580
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].4

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].4

Ethernet Module Status Last Scan
GE
Source A EDC_Enet00.CPU_Util.Last
580
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].4
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet00.CPU_Util.Last
580
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet00.CPU_Util.Last
580
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].3

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].3

Ethernet Module Status Last Scan
GE
Source A EDC_Enet00.CPU_Util.Last
580
Source B 675

EqDriver embedded in Logix Controller

Ethernet 01 Status

44

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet01.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].5

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].7

Ethernet Module Status Last Scan
GE
Source A EDC_Enet01.CPU_Util.Last 440
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].7

Ethernet Module Status Last Scan
GE
Source A EDC_Enet01.CPU_Util.Last 440
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].7
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet01.CPU_Util.Last 440
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet01.CPU_Util.Last 440
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].6

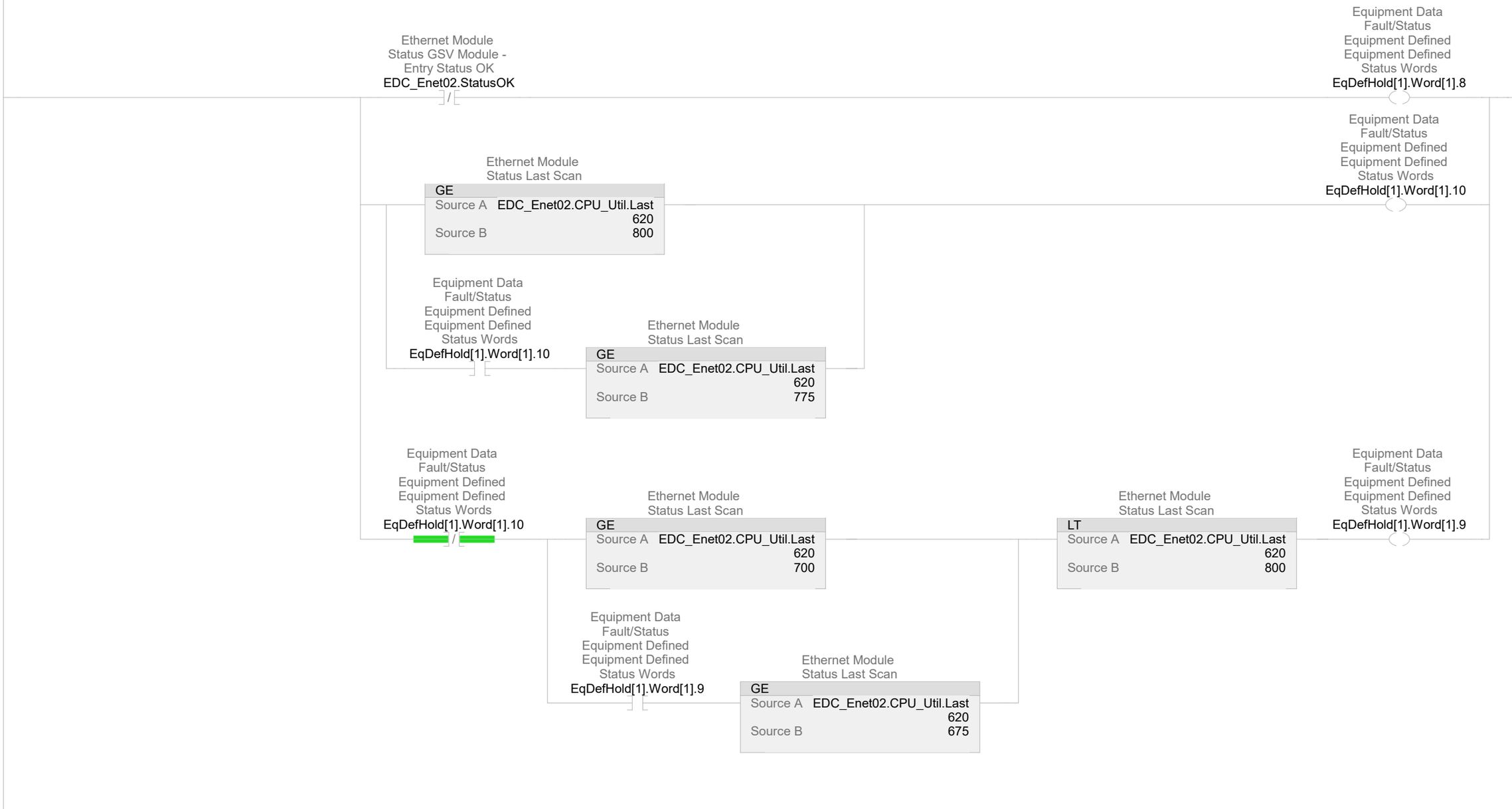
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].6

Ethernet Module Status Last Scan
GE
Source A EDC_Enet01.CPU_Util.Last 440
Source B 675

EqDriver embedded in Logix Controller

Ethernet 02 Status

45



EqDriver embedded in Logix Controller

Ethernet 03 Status

46

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet03.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].11

Ethernet Module Status Last Scan
GE
Source A EDC_Enet03.CPU_Util.Last 350
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].13

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].13

Ethernet Module Status Last Scan
GE
Source A EDC_Enet03.CPU_Util.Last 350
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].13
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet03.CPU_Util.Last 350
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet03.CPU_Util.Last 350
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].12

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].12

Ethernet Module Status Last Scan
GE
Source A EDC_Enet03.CPU_Util.Last 350
Source B 675

EqDriver embedded in Logix Controller

Ethernet 04 Status

47

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet04.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].14

Ethernet Module Status Last Scan
GE
Source A EDC_Enet04.CPU_Util.Last 130
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].16

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].16

Ethernet Module Status Last Scan
GE
Source A EDC_Enet04.CPU_Util.Last 130
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].16
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet04.CPU_Util.Last 130
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet04.CPU_Util.Last 130
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].15

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].15

Ethernet Module Status Last Scan
GE
Source A EDC_Enet04.CPU_Util.Last 130
Source B 675

EqDriver embedded in Logix Controller

Ethernet 05 Status

48

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet05.StatusOK

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].17

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].19

Ethernet Module Status Last Scan
GE
Source A EDC_Enet05.CPU_Util.Last 0
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].19

Ethernet Module Status Last Scan
GE
Source A EDC_Enet05.CPU_Util.Last 0
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].19

Ethernet Module Status Last Scan
GE
Source A EDC_Enet05.CPU_Util.Last 0
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet05.CPU_Util.Last 0
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].18

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].18

Ethernet Module Status Last Scan
GE
Source A EDC_Enet05.CPU_Util.Last 0
Source B 675

EqDriver embedded in Logix Controller

Ethernet 06 Status

49

Ethernet Module Status GSV Module - Entry Status OK
EDC_Enet06.StatusOK
/

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].20

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].22

Ethernet Module Status Last Scan
GE
Source A EDC_Enet06.CPU_Util.Last 0
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].22

Ethernet Module Status Last Scan
GE
Source A EDC_Enet06.CPU_Util.Last 0
Source B 775

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].22
/

Ethernet Module Status Last Scan
GE
Source A EDC_Enet06.CPU_Util.Last 0
Source B 700

Ethernet Module Status Last Scan
LT
Source A EDC_Enet06.CPU_Util.Last 0
Source B 800

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].21

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[1].21

Ethernet Module Status Last Scan
GE
Source A EDC_Enet06.CPU_Util.Last 0
Source B 675

EqDriver embedded in Logix Controller

Ethernet 07 Status

50 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].23

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].25

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].24

Ethernet 08 Status

51 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].26

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].28

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].27

EqDriver embedded in Logix Controller

Ethernet 09 Status

52 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].29

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].31

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[1].30

Ethernet 10 Status

53 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].0

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].2

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].1

EqDriver embedded in Logix Controller

Ethernet 11 Status

54 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].3

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].5

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].4

Ethernet 12 Status

55 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].6

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].8

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].7

EqDriver embedded in Logix Controller

Ethernet 13 Status

56 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].9

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].11

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].10

Ethernet 14 Status

57 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].12

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].14

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].13

EqDriver embedded in Logix Controller

Ethernet 15 Status

58 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].15

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].17

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].16

Ethernet 16 Status

59 AFI

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].18

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].20

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words
EqDefHold[1].Word[2].19

EqDriver embedded in Logix Controller

Unused
(future)

60

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[2].21	EqDefHold[1].Word[2].22	EqDefHold[1].Word[2].23	EqDefHold[1].Word[2].24	EqDefHold[1].Word[2].25	EqDefHold[1].Word[2].26	EqDefHold[1].Word[2].27	EqDefHold[1].Word[2].28	EqDefHold[1].Word[2].29
U	U	U	U	U	U	U	U	U

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words	Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[2].30	EqDefHold[1].Word[2].31
U	U

EqDriver embedded in Logix Controller

CLx Time Sync Errors

61

Logix Wall Clock Invalid Year (force time sync) SNTP_InvalidYear → Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].0

SNTP Query and Clock Sync Instruction There was an error retrieving the time. SNTP_AOI_TAG.ER → Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].1

SNTP Query and Clock Sync Instruction SNTP_AOI_TAG.Enable_Instruction (/) → Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].2

SNTP Query and Clock Sync Instruction SNTP_AOI_TAG.Sync_Controller_Clock (/) → Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].3

System time master CST_CurrentStatus.2 (/) → Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].4

Wall Clock Time Time Zone matches Config WCT_TimeZoneOK (/) → Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].5

Wall Clock Time DST matches Config WCT_DSTok (/) → Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].6

EqDriver embedded in Logix Controller

CLx High Task Scan Time > 0.5 sec
(500,000 microseconds)

62

CLxTaskScanHigh

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].7

63

AFI

Unused (future)

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].8 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].9 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].10 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].11 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].12 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].13 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].14 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].15 (U)

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].16 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].17 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].18 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].19 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].20 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].21 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].22 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].23 (U)

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].24 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].25 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].26 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].27 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].28 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].29 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].30 (U)
Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words EqDefHold[1].Word[3].31 (U)

EqDriver embedded in Logix Controller

64

Packet 1 Word 3-7
Fault Word from FEC1-FEC4 located in Slots 3-7
(FEC Faults are Messaged from FECs to Gateway using Slot# as Array Index)

If any FEC sends 'first scan' signal, refresh all Masks
(Gateway doesn't know which EqDrv are in which FEC -- so need to use the BIG hammer)

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source FEC_EqDrvFault[3] 16#0000_0000
Dest EqDefHold[1].Word[4] 16#0000_0000

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source FEC_EqDrvFault[4] 16#0000_0000
Dest EqDefHold[1].Word[5] 16#0000_0000

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source FEC_EqDrvFault[5] 16#0000_0000
Dest EqDefHold[1].Word[6] 16#0000_0000

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source FEC_EqDrvFault[6] 16#0000_0000
Dest EqDefHold[1].Word[7] 16#0000_0000

FEC1 Faults
FEC_EqDrvFault[3].4

FEC2 Faults
FEC_EqDrvFault[4].4

FEC3 Faults
FEC_EqDrvFault[5].4

FEC4 Faults
FEC_EqDrvFault[6].4

FEC_EqDrvFault_FirstScanONS
ONS

Any FEC First Scan
FEC_EqDrvFault_FirstScan

CLx High Program Scan Time

65

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
MOVE
Source CLxProgScanHigh 0
Dest EqDefHold[2].Word[0] 16#0000_0000

EqDriver embedded in Logix Controller

Spare Packets 2-7

66

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[2].Word[1]
Length 7

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[3].Word[0]
Length 8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[4].Word[0]
Length 8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[5].Word[0]
Length 8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[6].Word[0]
Length 8

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

FLL
Source 0
Dest EqDefHold[7].Word[0]
Length 8

generate a bunch of data very fast
to test the 10-record batch Fault Event driver

67

AFI

TON
Timer xxTestDataTMR
Preset 10000
Accum 0
EN
DN

xxTestDataTMR.DN

ADD
Source A xxTestCount
0
Source B 1
Dest xxTestCount
0

MOD
Source A xxTestCount
0
Source B 2
Dest xxTestResult
0

EQ
Source A xxTestResult
0
Source B 0

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].8
U

EQ
Source A xxTestResult
0
Source B 1

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words
EqDefHold[1].Word[3].8
L

GE
Source A xxTestCount
0
Source B 12

CLR
Dest xxTestCount
0

xxTestDataTMR
RES

(End)

Equipment Quality/Health Data Mapping

Equipment Quality/Health Data
User Data Mapping

Load Job Data
Load Job Data ONS
Send Tool Data
Send Tool Data ONS
Cfg Fault Code Clear
Cfg Fault Code Trigger

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 94
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

2

Tool02_Enable

MOVE
Source 20000
Dest Tool02_TMR.PRE 20000

TON
Timer Tool02_TMR
Preset 20000
Accum 0
EN
DN

Tool02_TMR.DN

Tool02_Start (L)
Tool02_Send (L)

ADD
Source A Tool02_Count 0
Source B 1
Dest Tool02_Count 0

GE
Source A Tool02_Count 0
Source B 10

Tool02_TMR (RES)

CLR
Dest Tool02_Count 0

Equipment Quality/Health Data Mapping

Load Tool Vehicle Data

3

ShiftReset

ToolVehONS[1]
ONS

Year
NE
Source A WCT_DstTool[0]
0
Source B 0

Tool01_Start

Tool01_Send

Tool02_Start

Tool02_Send

Equipment Data Quality/Health Message Program Number from SLCP
MOVE
Source 0
Dest QualityMsg.Data.AviPrgNum
0

Equipment Data Quality/Health Message Program Number Executed
MOVE
Source 0
Dest QualityMsg.Data.ActPrgNum
0

Equipment Data Quality/Health Message Assembly Body Number
MOVE
Source 0
Dest QualityMsg.Data.BodyNum
0

Equipment Data Quality/Health Message Assembly Sequence Number
MOVE
Source 0
Dest QualityMsg.Data.AssyNum
0

Replace with User Logic
FLL
Source 0
Dest QualityMsg.Data.VinNum[0]
Length 18

Equipment Data Quality/Health Message VIN Number
CLR
Dest QualityMsg.Data.VinNum[18]
'$00'

Equipment Data Quality/Health Message VIN Number
CLR
Dest QualityMsg.Data.VinNum[19]
'$00'

Equipment Data Quality/Health Message Employee Badge Number for Job Release
FLL
Source 0
Dest QualityMsg.Data.BadgeNum[0]
Length 12

Equipment Quality/Health Data Mapping

4

Tool 1
PdM Trigger
(Tool 1 Config has been changed to PdM Sample)

Tool01_Start
Tool01_Start (U)
Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Tool01_Send
Tool01_Send (U)
Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Tool Number
MOVE
Source 1
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 1
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Tool 2
PdM Trigger

tool to demo ability to reset Trigger data by resending Start signal
each Start Signal performs Pdm Reset/Init on Data array

5

Tool02_Start
Tool02_Start
U
Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Tool02_Send
Tool02_Send
U
Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Tool Number
MOVE
Source 2
Dest QualityMsg.Data.ToolNum 100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 4
Dest QualityMsg.Data.DataCount 10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus 1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode 0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source Tool02_Count 0
Dest QualityMsg.Data.CycleTime 0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source Tool02_Count 0
Dest QualityMsg.Data.CycleCount 0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Wallclock TimeZone and DST
Tool Number 100

6

Year

NE
Source A WCT_DstTool[0] 0
Source B 0

Equipment Data Quality/Health Message Tool Number

MOVE
Source 100
Dest QualityMsg.Data.ToolNum 100

Equipment Data Quality/Health Message Data Values Used (1-5)

MOVE
Source 10
Dest QualityMsg.Data.DataCount 10

Equipment Data Quality/Health Message Job Status

MOVE
Source 1
Dest QualityMsg.Data.JobStatus 1

Equipment Data Quality/Health Message Fault Code

MOVE
Source 0
Dest QualityMsg.Data.FaultCode 0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)

MOVE
Source 0
Dest QualityMsg.Data.CycleTime 0

Equipment Data Quality/Health Message Cycle Count

MOVE
Source 0
Dest QualityMsg.Data.CycleCount 0

Time Zone Udpated

EQ
Source A WCT_DstTool[7] 0
Source B 1

Equipment Data Quality/Health Message Fault Code

MOVE
Source 1
Dest QualityMsg.Data.FaultCode 0

DST Udpated

EQ
Source A WCT_DstTool[8] 0
Source B 1

Equipment Data Quality/Health Message Fault Code

MOVE
Source 2
Dest QualityMsg.Data.FaultCode 0

Time Zone Udpated

EQ
Source A WCT_DstTool[7] 0
Source B 1

DST Udpated

EQ
Source A WCT_DstTool[8] 0
Source B 1

Equipment Data Quality/Health Message Fault Code

MOVE
Source 3
Dest QualityMsg.Data.FaultCode 0

Equipment Data Quality/Health Message Quality Data 1

MOVE
Source WCT_DstTool[0] 0
Dest QualityMsg.Data.Data1 2026

Equipment Data Quality/Health Message Quality Data 2

MOVE
Source WCT_DstTool[1] 10
Dest QualityMsg.Data.Data2 10

Equipment Data Quality/Health Message Quality Data 3

MOVE
Source WCT_DstTool[2] 30
Dest QualityMsg.Data.Data3 30

Equipment Data Quality/Health Message Quality Data 4

MOVE
Source WCT_DstTool[3] 15
Dest QualityMsg.Data.Data4 15

Equipment Data Quality/Health Message Quality Data 5

MOVE
Source WCT_DstTool[4] 0
Dest QualityMsg.Data.Data5 0

Equipment Quality/Health Data Mapping

Equipment Data Quality/Health Message Quality Data 6

MOVE
Source WCT_DstTool[5] 0
Dest QualityMsg.Data.Data6 0

Equipment Data Quality/Health Message Quality Data 7

MOVE
Source WCT_DstTool[9] 2
Dest QualityMsg.Data.Data7 2

Equipment Data Quality/Health Message Quality Data 8

MOVE
Source WCT_DstTool[11] 1
Dest QualityMsg.Data.Data8 1

Equipment Data Quality/Health Message Quality Data 9

CLR
Dest QualityMsg.Data.Data9 0

Equipment Data Quality/Health Message Quality Data 10

CLR
Dest QualityMsg.Data.Data10 0

Code Rule for DST
0=Disable
1=Disable Dec-Feb
2=Enable Apr-Oct
3=Enable Mar
4=Enable Nov
5=Disable Mar/Nov

GT
Source A WCT_DstTool[9] 2
Source B 2

Equipment Data Quality/Health Message Quality Data 9

MOVE
Source WCT_DstTool[14] 3
Dest QualityMsg.Data.Data9 0

Equipment Data Quality/Health Message Quality Data 10

MOVE
Source WCT_DstTool[15] 29
Dest QualityMsg.Data.Data10 0

Year

CLR
Dest WCT_DstTool[0] 0

Load Quality Data record to Ring Buffer

JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Fault/Event RB Usage by Shift
Tool Number 101

7 ShiftReset ToolSendONS[1] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 101
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 6
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Fault01.PrevShiftCnt
85089
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Fault02.PrevShiftCnt
25487
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source EDC.RB_Fault03.PrevShiftCnt
172357
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source EDC.RB_Fault04.PrevShiftCnt
357752
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
MOVE
Source EDC.RB_Fault05.PrevShiftCnt
161029
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
MOVE
Source EDC.RB_Fault06.PrevShiftCnt
401978
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health

Equipment Quality/Health Data Mapping

Message Quality Data 10

CLR	
Dest	QualityMsg.Data.Data100

Load Quality Data record to Ring Buffer

JSR	
Routine Name	_132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Quality/Tool RB Usage by Shift
Tool Number 102

8

ShiftReset ToolSendONS[2] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 102
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 4
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Quality01.PrevShiftCnt
101079
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Quality02.PrevShiftCnt
102048
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
MOVE
Source EDC.RB_Quality03.PrevShiftCnt
125974
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
MOVE
Source EDC.RB_Quality04.PrevShiftCnt
121316
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
CLR
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
CLR
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health Message Quality Data 10
CLR
Dest QualityMsg.Data.Data10
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Equipment Quality/Health Data Mapping

Trend RB Usage by Shift
Tool Number 103

9 ShiftReset ToolSendONS[3] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 103
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 2
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Trend01.PrevShiftCnt
2553
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Trend02.PrevShiftCnt
3106
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
CLR
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
CLR
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
CLR
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
CLR
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health Message Quality Data 10
CLR
Dest QualityMsg.Data.Data10
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Equipment Quality/Health Data Mapping

Summary RB Usage by Shift
Tool Number 104

10

ShiftReset ToolSendONS[4] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 104
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 2
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Sum01.PrevShiftCnt
132552
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Sum02.PrevShiftCnt
132844
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
CLR
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
CLR
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
CLR
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
CLR
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health Message Quality Data 10
CLR
Dest QualityMsg.Data.Data10
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

Equipment Quality/Health Data Mapping

Cycle RB Usage by Shift
Tool Number 105

11 ShiftReset ToolSendONS[5] ONS

Equipment Data Quality/Health Message Tool Number
MOVE
Source 105
Dest QualityMsg.Data.ToolNum
100

Equipment Data Quality/Health Message Data Values Used (1-5)
MOVE
Source 2
Dest QualityMsg.Data.DataCount
10

Equipment Data Quality/Health Message Job Status
MOVE
Source 1
Dest QualityMsg.Data.JobStatus
1

Equipment Data Quality/Health Message Fault Code
MOVE
Source 0
Dest QualityMsg.Data.FaultCode
0

Equipment Data Quality/Health Message Cycle Time (xx.x sec)
MOVE
Source 0
Dest QualityMsg.Data.CycleTime
0

Equipment Data Quality/Health Message Cycle Count
MOVE
Source 0
Dest QualityMsg.Data.CycleCount
0

Equipment Data Quality/Health Message Quality Data 1
MOVE
Source EDC.RB_Cycle01.PrevShiftCnt
8603
Dest QualityMsg.Data.Data1
2026

Equipment Data Quality/Health Message Quality Data 2
MOVE
Source EDC.RB_Cycle02.PrevShiftCnt
8544
Dest QualityMsg.Data.Data2
10

Equipment Data Quality/Health Message Quality Data 3
CLR
Dest QualityMsg.Data.Data3
30

Equipment Data Quality/Health Message Quality Data 4
CLR
Dest QualityMsg.Data.Data4
15

Equipment Data Quality/Health Message Quality Data 5
CLR
Dest QualityMsg.Data.Data5
0

Equipment Data Quality/Health Message Quality Data 6
CLR
Dest QualityMsg.Data.Data6
0

Equipment Data Quality/Health Message Quality Data 7
CLR
Dest QualityMsg.Data.Data7
2

Equipment Data Quality/Health Message Quality Data 8
CLR
Dest QualityMsg.Data.Data8
1

Equipment Data Quality/Health Message Quality Data 9
CLR
Dest QualityMsg.Data.Data9
0

Equipment Data Quality/Health Message Quality Data 10
CLR
Dest QualityMsg.Data.Data10
0

Load Quality Data record to Ring Buffer
JSR
Routine Name _132_QualityDataLoadRB

Equipment Quality/Health Data Mapping

(End)

Equipment Trend Data Mapping

Equipment Trend Data
User Data Mapping

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 95
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

(End)

Equipment Cycle Data Mapping

Equipment Cycle Data
User Data Mapping

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 96
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

(End)

Pre-Defined Fault/Status Data Mapping

Predefined Fault/Status Data
v2.0 - Aug2022 - PreDefined Events defined by EqDriver

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 100
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

v1.13 EDC EQ Driver - Wallclock Time Invalid

2

CLx Module Status
CLx Transaction Time Invalid
CLX.TransTimeInvalid

Equipment Data Fault/Status Message Invalid Wallclock Time (from EqDriver)
FaultMsg.PreDef._21

Pre-Defined Fault/Status Data Mapping

v1.5 EDC EQ Driver - Equipment Data Error
FEC EQ Driver not Polling Fault, Quality and/or Trend Data
v1.6 Optional Message to get Overall Start from Gateway

3

Gateway Overall Start Message ER
OverallStartReadMsgER

Equipment Data Collection - User Config Toyopuc Driver Enable
UserCfg.TpucDriverEN

TpucDriverER

Equipment Data Collection - User Config Remote Driver Enable
UserCfg.RemDriverEN

RemoteDriverER

Equipment Data Fault/Status Message Equipment Data Error (from EqDriver)
FaultMsg.PreDef._22

EDC EQ Driver - Ring Buffers Not Empty after 5 minutes
v2.1 - combined all RB Filling to a single Event
v2.09 Change RBCtl.NotEmptyTMR.DN to RBCtl.BufferAlarm to include Not Empty or Full condition

4

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN

EDC Ring Buffer Control Read Buffer Alarm
FaultRBCtl.BufferAlarm

Diagnostic - manual reset
RingBufferFIlling_Fault

Equipment Data Fault/Status Message RingBuffer Filling (from EqDriver)
FaultMsg.PreDef._30

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN

EDC Ring Buffer Control Read Buffer Alarm
QualityRBCtl.BufferAlarm

Diagnostic - manual reset
RingBufferFIlling_Quality

Equipment Data Collection - User Config Trend Message Enable
UserCfg.TrendMsgEN

EDC Ring Buffer Control Read Buffer Alarm
TrendRBCtl.BufferAlarm

Diagnostic - manual reset
RingBufferFIlling_Trend

Equipment Data Collection - User Config Cycle Message Enable
UserCfg.CycleMsgEN

EDC Ring Buffer Control Read Buffer Alarm
CycleRBCtl.BufferAlarm

Diagnostic - manual reset
RingBufferFIlling_Cycle

Pre-Defined Fault/Status Data Mapping

EDC EQ Driver - Message Error
v2.1 combined Callout, Fault, Quality, Trend, Time Sync, Version and Cycle to a single Event
(Note: If Callout Error is ON, Callout Msg with this status is not delivered to Gateway -- this Error is just a placeholder -- the EDC Gateway will detect Callout Msg Timeout and annunciate Error)

5

Equipment Data Collection - User Config Driver Enable
UserCfg.DriverEN

Equipment Data Collection - Line Equipment Status State = Red - Callout Tx ER
Status.State.CalloutTxER

Diagnostic - manual reset
MsgER_Callout
L

Equipment Data Fault/Status Message Gateway Message Error (from EqDriver)
FaultMsg.PreDef._31

EDC Gateway Callout Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

EDC Gateway Callout Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN

Equipment Data Collection - Line Equipment Status State = Red - Fault Tx ER
Status.State.FaultTxER

Diagnostic - manual reset
MsgER_Fault
L

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Equipment Data Collection - User Config Quality Message Enable
UserCfg.QualityMsgEN

Equipment Data Collection - Line Equipment Status State = Red - Quality Tx ER
Status.State.QualityTxER

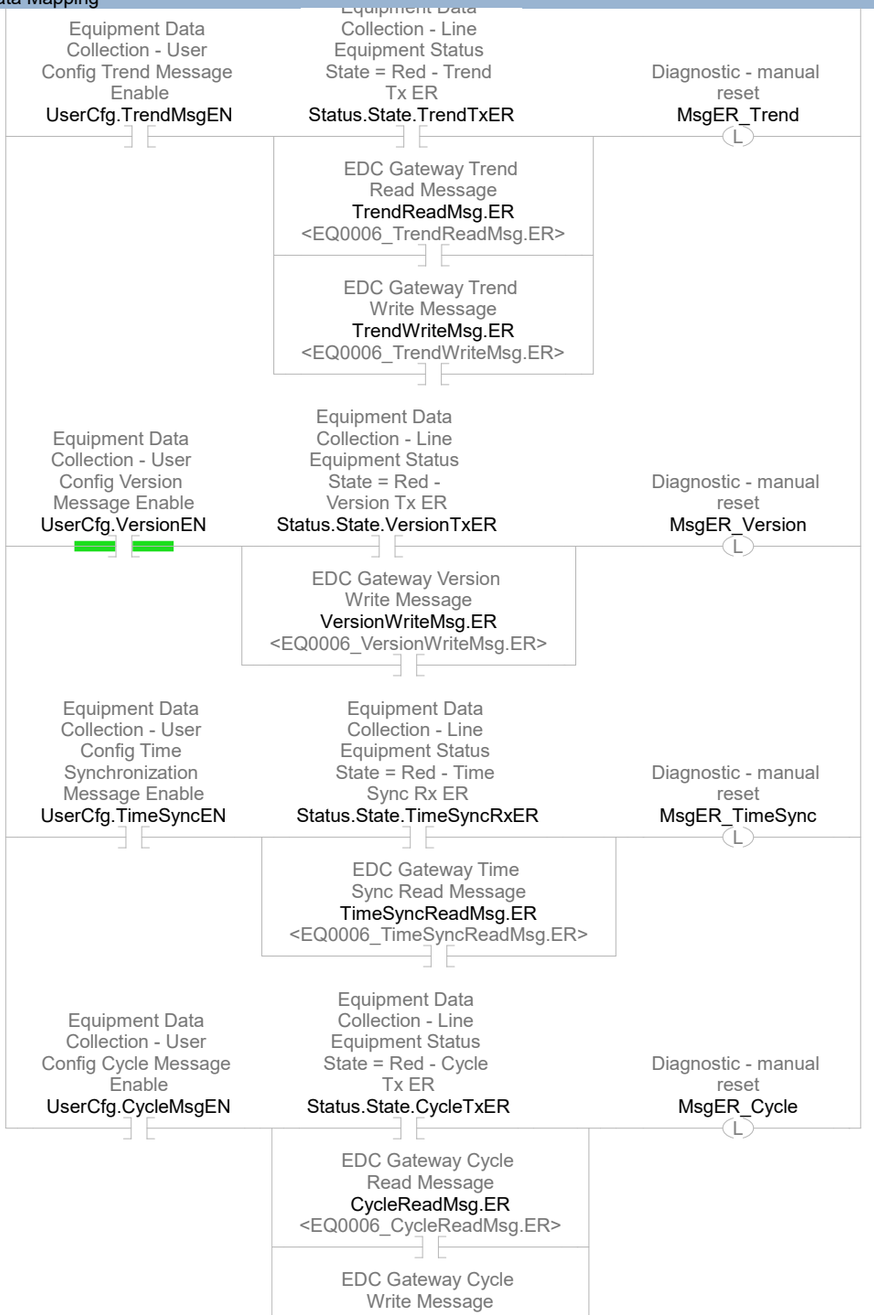
Diagnostic - manual reset
MsgER_Quality
L

EDC Gateway Quality Read Message
QualityReadMsg.ER
<EQ0006_QualityReadMsg.ER>

EDC Gateway Quality Write Message
QualityWriteMsg.ER
<EQ0006_QualityWriteMsg.ER>

Equipment Data

Pre-Defined Fault/Status Data Mapping



Pre-Defined Fault/Status Data Mapping

CycleWriteMsg.ER
<EQ0006_CycleWriteMsg.ER>

Move PreDef bits to DINT for Masking
Load hard-coded Mask for Events used to determine Fault, Alarm, Alert, Maint
(PreDefMaskUsed - ignore PreDefined signals that are generated as a result of Fault, Alarm, Alert)
EDC2 Sep 30, 2022 - remove Pre.15 Fault Memory from PreDefMaskUsed

6

COP	
Source	FaultMsg.PreDef
Dest	PreDefVal
Length	1

Events used to determine Fault, Alarm, Alert, Maint

MOVE	
Source	16#fffd3f00
Dest	PreDefMaskUsed 16#fffd_3f00

Equipment Fault - EqDef and PreDef
used by EDC to determine Equipment Downtime and OA%
(Red Status on Overview)
v2.00 - Fault Mask downloaded from EDC Server

7

Checks EqDef Events against the selected Mask

aoiEDCv2_00_EqDefMaskCheck	
aoiEDCv2_00_EqDefMas...	aoiEqDefFaultCheck
EqDef	FaultMsgCtl.EqDef
EqDefMask	EqDefFaultMask
UserCfg	UserCfg
EventOn	EqDefFaultOn 0

AND	
Source A	PreDefVal 16#0000_0021
Source B	PreDefMaskUsed 16#fffd_3f00
Dest	PreDefFault 16#0000_0000

AND	
Source A	PreDefFault 16#0000_0000
Source B	PreDefMaskFault 16#2035_cb50
Dest	PreDefFault 16#0000_0000

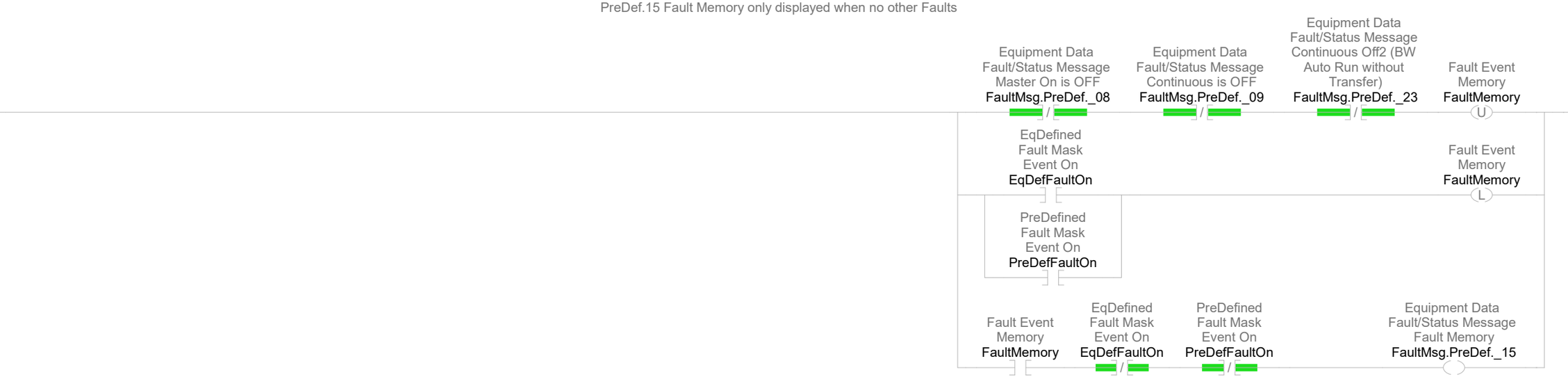
NE	
Source A	PreDefFault 16#0000_0000
Source B	0

PreDefined Fault Mask Event On
PreDefFaultOn

Pre-Defined Fault/Status Data Mapping

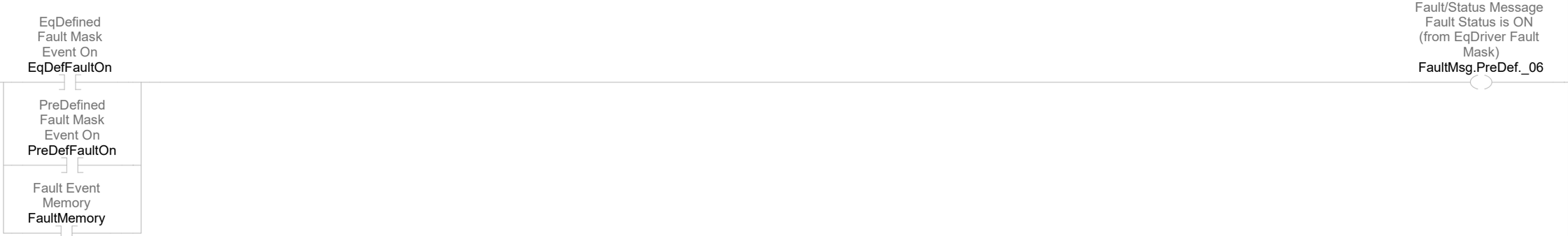
Equipment Fault - Fault Memory
On with any Fault
Off when ready to run
PreDef.15 Fault Memory only displayed when no other Faults

8



Equipment Fault - EqDef, PreDef or PreDefMemory

9



Pre-Defined Fault/Status Data Mapping

Equipment Alarm
(Yellow Status on Overview)
v2.00 - Alarm Mask downloaded from EDC Server

10

Checks EqDef Events against the selected Mask

aoiEDCv2_00_EqDefMaskCheck

aoiEDCv2_00_EqDefMas...	aoiEqDefAlarmCheck
EqDef	FaultMsgCtl.EqDef
EqDefMask	EqDefAlarmMask
UserCfg	UserCfg
EventOn	EqDefAlarmOn
	0

AND

Source A	PreDefVal 16#0000_0021
Source B	PreDefMaskUsed 16#fffd_3f00
Dest	PreDefAlarm 16#0000_0000

AND

Source A	PreDefAlarm 16#0000_0000
Source B	PreDefMaskAlarm 16#c0c8_2088
Dest	PreDefAlarm 16#0000_0000

NE

Source A	PreDefAlarm 16#0000_0000
Source B	0

PreDefined Alarm Mask Event On
PreDefAlarmOn

EqDefined Alarm Mask Event On
EqDefAlarmOn

PreDefined Alarm Mask Event On
PreDefAlarmOn

Equipment Data Fault/Status Message Alarm Status is ON (from EqDriver Alarm Mask)
FaultMsg.PreDef._07

Pre-Defined Fault/Status Data Mapping

v1.17 - Equipment Alert
(Lt Blue Status on Overview)
v2.00 - Alert Mask downloaded from EDC Server

11

Checks EqDef Events against the selected Mask

aoiEDCv2_00_EqDefMaskCheck

aoiEDCv2_00_EqDefMas...	aoiEqDefAlertCheck
EqDef	FaultMsgCtl.EqDef
EqDefMask	EqDefAlertMask
UserCfg	UserCfg
EventOn	EqDefAlertOn
	0

AND

Source A	PreDefVal 16#0000_0021
Source B	PreDefMaskUsed 16#fffd_3f00
Dest	PreDefAlert 16#0000_0000

AND

Source A	PreDefAlert 16#0000_0000
Source B	PreDefMaskAlert 16#1002_0020
Dest	PreDefAlert 16#0000_0000

NE

Source A	PreDefAlert 16#0000_0000
Source B	0

PreDefined Alert Mask Event On
PreDefAlertOn

EqDefined Alert Mask Event On
EqDefAlertOn

PreDefined Alert Mask Event On
PreDefAlertOn

Equipment Data Fault/Status Message Alert Status is ON (from EqDriver Alert Mask)
FaultMsg.PreDef._17

Pre-Defined Fault/Status Data Mapping

Stop Request caused by Production
used by EDC to determine Equipment Downtime and OA%
v2.00 - Maintenance Mask downloaded from EDC Server

12

Checks EqDef Events against the selected Mask

aoiEDCv2_00_EqDefMaskCheck

aoiEDCv2_00_EqDefMas...	aoiEqDefMaintCheck ...
EqDef	FaultMsgCtl.EqDef
EqDefMask	EqDefMaintMask
UserCfg	UserCfg
EventOn	EqDefMaintOn
	0

AND

Source A	PreDefVal 16#0000_0021
Source B	PreDefMaskUsed 16#fffd_3f00
Dest	PreDefMaint 16#0000_0000

AND

Source A	PreDefMaint 16#0000_0000
Source B	PreDefMaskMaint 16#2035_cb40
Dest	PreDefMaint 16#0000_0000

NE

Source A	PreDefMaint 16#0000_0000
Source B	0

PreDefined Maint Mask Event On
PreDefMaintOn

EqDefined Maint Mask Event On
EqDefMaintOn

PreDefined Maint Mask Event On
PreDefMaintOn

MaintActive

MaintActive
/

Equipment Data Fault/Status Message Production Stop Request (from EqDriver Mask Prod/Maint Mask)
FaultMsg.PreDef._05

(End)

Callout Data Load for Message

Equipment Fault/Status Data
Callout Data Load
v1.3 - Dec2016 - Multi-packet support for >8 DINT Equipment-Defined Faults

PKTi - Packet Index for multiplexed Fault Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 112
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

1

Exit if This Packet not Enabled

Fault Ring Buffer Message Control for multiplexed packets Data Change
FaultMsgCtl.Change.[PKTi]

2

lbl_Exit
JMP

Send This Packet

Fault Ring Buffer Message Control for multiplexed packets Data Change
FaultMsgCtl.Change.[PKTi]
U

Message Control Send Data
CalloutMsgCtl.Send
L

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK
U

Message Control Write Command Done
CalloutMsgCtl.WriteOK
U

3

Callout Data Load for Message

Increment Transaction Number

4

Message Control TM Transaction Trigger

ADD
Source A CalloutMsgCtl.TransNum 9993
Source B 1
Dest CalloutMsgCtl.TransNum 9993

Message Control TM Transaction Trigger

LT
Source A CalloutMsgCtl.TransNum 9993
Source B 1

Message Control TM Transaction Trigger

MOVE
Source 1
Dest CalloutMsgCtl.TransNum 9993

Message Control TM Transaction Trigger

GT
Source A CalloutMsgCtl.TransNum 9993
Source B CFG.MsgTransNumRollover 9999

Callout Data Load for Message

Load Message Header, Timestamp, Data and Footer
v1.3 - Added Header.PacketID for multiplexed data packets
v2.00 Footer.Spare = 2 indicates v2 with Fault Event Mask in Callout Read Packets

5

Equipment Data Callout/Status Message Header - Transaction Number (must match Footer)

MOVE
Source CalloutMsgCtl.TransNum 9993
Dest CalloutMsg.Header.TransNum 9993

Equipment Data Callout/Status Message Header - Command Number

MOVE
Source 0
Dest CalloutMsg.Header.CmdNum 0

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source UserCfg.EQID 6
Dest CalloutMsg.Header.EQID 6

Equipment Data Callout/Status Message Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE
Source PKTi -1
Dest CalloutMsg.Header.PacketID 1

Equipment Data Callout/Status Message Transaction Time Stamp

COP
Source CLX.TransTime
Dest CalloutMsg.TransTime
Length 1

Equipment Data Callout/Status Message Data - Predefined Status

COP
Source FaultMsg.PreDef
Dest CalloutMsg.PreDef
Length 1

Equipment Data Callout/Status Message Data - Equipment Defined Status

COP
Source FaultMsgCtl.EqDef[PKTi]
Dest CalloutMsg.EqDef
Length 1

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)

MOVE
Source 2
Dest CalloutMsg.Footer.Spare 2

Equipment Data Callout/Status Message Footer - Transaction Number (must match Footer)

MOVE
Source CalloutMsgCtl.TransNum 9993
Dest CalloutMsg.Footer.TransNum 9993

Callout Data Load for Message

Load Message Header Config Bits

6

Equipment Data Collection - User Config Driver Enable UserCfg.DriverEN	Equipment Data Callout/Status Message Driver Enabled CalloutMsg.Header.DriverEN
Equipment Data Collection - User Config Fault Message Enable UserCfg.FaultMsgEN	Equipment Data Callout/Status Message Fault Message Enabled CalloutMsg.Header.FaultMsgEN
Equipment Data Collection - User Config Quality Message Enable UserCfg.QualityMsgEN	Equipment Data Callout/Status Message Quality Message Enabled CalloutMsg.Header.QualityMsgEN
Equipment Data Collection - User Config Trend Message Enable UserCfg.TrendMsgEN	Equipment Data Callout/Status Message Trend Message Enabled CalloutMsg.Header.TrendMsgEN
Equipment Data Collection - User Config Version Message Enable UserCfg.VersionEN	Equipment Data Callout/Status Message Version Message Enabled CalloutMsg.Header.VersionMsgEN
Equipment Data Collection - User Config Time Synchronization Message Enable UserCfg.TimeSyncEN	Equipment Data Callout/Status Message Time Synchronization Message Enabled CalloutMsg.Header.TimeSyncMsgEN
Equipment Data Collection - User Config Cycle Message Enable UserCfg.CycleMsgEN	Equipment Data Callout/Status Message Cycle Message Enabled CalloutMsg.Header.CycleMsgEN
AFI	Equipment Data Callout/Status Message Spare CalloutMsg.Header.Spare7

Callout Data Load for Message

Break out of the FOR Loop when a Packet is found for Callout Message

7

BRK

Exit Routine

lbl_Exit

8

LBL

NOP

(End)

Callout Data Message

Equipment Fault/Status Data
Callout Live Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 114
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Message Control Message Transaction Timer

MOVE
Source 999999999
Dest CalloutMsgCtl.Trans.TMR.PRE 999999999

Message Control Message Transaction Timer

TON
Timer CalloutMsgCtl.Trans.TMR
Preset 999999999
Accum 24725
EN
DN

Message Delay Timer
prevent Message from sending too quickly for Gateway
Gateway processes one Line per scan -- Equipment Message could take 500-1000 msec to process
any attempt to send faster will generate extra Read Message traffic until previous message has been processed

3

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

Message Control Message Delay Timer

MOVE
Source CFG.MsgDelayTime 250
Dest CalloutMsgCtl.DelayTMR.PRE 250

Message Control Message Delay Timer

TON
Timer CalloutMsgCtl.DelayTMR
Preset 250
Accum 253
EN
DN

Callout Data Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

Message Control Send Data
CalloutMsgCtl.Send

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

Message Control Message Resend Timer
MOVE
Source CFG.LiveMsgResendTime
31000
Dest CalloutMsgCtl.ResendTMR.PRE
31000

Message Control Message Resend Timer
TON
Timer CalloutMsgCtl.ResendTMR
Preset 31000
Accum 24698
EN
DN

Message Control Message Resend Timer
CalloutMsgCtl.ResendTMR.DN

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend
L

Message Control Message Resend Timer
CalloutMsgCtl.ResendTMR
RES

Message Control Read Command
CalloutMsgCtl.Read
U

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK
U

Message Control Write Data
CalloutMsgCtl.Write
U

Message Control Write Command Done
CalloutMsgCtl.WriteOK
U

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source CalloutMsg.Header.PacketID
1
Dest Trans.Ctl.Opt[1]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 122
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

Send Data Initialize

5

Message Control Send Data
CalloutMsgCtl.Send

Message Control Send Data 1-shot
CalloutMsgCtl.SendONS
ONS

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Message Control Message Transaction Timer
CalloutMsgCtl.Trans.TMR
RES

Message Control Transaction Elapsed Time for Last Event (used to calculate differential)
CLR
Dest CalloutMsgCtl.Trans.Last
15

Callout Data Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

v2.00 - Gateway also sends the Fault, Alarm, Alert, Ignore and Maintenance Masks in the 'Empty Record'
v3.50 jumper CalloutMsg Delay TMR - need fast updates for accurate timing and fast PdM Triggers

6

Message Control Send Data
CalloutMsgCtl.Send

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK

Message Control Message Delay Timer
CalloutMsgCtl.DelayTMR.DN

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

EDC Gateway Callout Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

Message Control Message Retry Timer
CalloutMsgCtl.RetryTMR.DN

Message Control Read Command
CalloutMsgCtl.Read
(L)

Message Control Read Command 1-Shot
CalloutMsgCtl.ReadONS

EDC Gateway Callout Read Message

COP
Source UserCfgMsg.Path
<EQ0006_UserCfgMsg.Path>
Dest CalloutReadMsg.Path
<EQ0006_CalloutReadMsg.Path>
Length 1

EDC Gateway Callout Read Message

COP
Source CFG.MsgDest.CalloutDest
Dest CalloutReadMsg.RemoteElement
<EQ0006_CalloutReadMsg.RemoteElement>
Length 1

EDC Gateway Callout Read Message

MOVE
Source 14
Dest CalloutReadMsg.REQ_LEN
<EQ0006_CalloutReadMsg.REQ_LEN>
14

EDC Gateway Callout Read Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest CalloutReadMsg.UnconnectedTimeout
<EQ0006_CalloutReadMsg.UnconnectedTimeout>
4000000

EDC Gateway Callout Read Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest CalloutReadMsg.ConnectionRate
<EQ0006_CalloutReadMsg.ConnectionRate>
1000000

EDC Gateway Callout Read Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest CalloutReadMsg.TimeoutMultiplier
<EQ0006_CalloutReadMsg.TimeoutMultiplier>
0

EDC Gateway Callout Read Message
CalloutReadMsg.EN_CC
<EQ0006_CalloutReadMsg.EN_CC>
(U)

EDC Gateway Callout Read Message

MSG
Message Control CalloutReadMsg ...
<EQ0006_CalloutReadMsg>
EN
DN
ER

Callout Data Message

Read Message EN Performance & Diagnostics

7

Message Control Read
Command 1-Shot
CalloutMsgCtl.ReadONS

Performance
Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CalloutMsgCtl.Trans
PerfTrans	Status.CalloutTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	130
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Callout Data Message

Read Message OK
Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record and loads Mask data)

MEQ used to get first and last INTs holding Header and Footer TransNum
Callout Rx Message contains non-zero Mask data in the adjacent INTs

8

Message Control Read Command
CalloutMsgCtl.Read

EDC Gateway Callout Read Message
CalloutReadMsg.DN
<EQ0006_CalloutReadMsg.DN>

Message Control Read Command
CalloutMsgCtl.Read
(U)

EDC Message Rx Data Callout/Status Receive Message

MEQ
Source EqRxData.Callout[0]
<EQ0006_RxData.Callout[0]> -65536
Mask 16#0000_ffff
Compare 0

EDC Message Rx Data Callout/Status Receive Message

MEQ
Source EqRxData.Callout[CFG.CalloutMSG.ei]
<EQ0006_RxData.Callout[CFG.CalloutMSG.ei]> 248
Mask 16#ffff_0000
Compare 0

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK
(L)

Store Event Mask in Callout Read Message

JSR
Routine Name _115_CalloutDataMask

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans CalloutMsgCtl.Trans
PerfTrans Status.CalloutTxRead.DN

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans CalloutMsgCtl.Trans
PerfTrans Status.CalloutTxRead.KA

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EqRxData.Callout[0]
<EQ0006_RxData.Callout[0]> -65536
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EqRxData.Callout[CFG.CalloutMSG.ei]
<EQ0006_RxData.Callout[CFG.CalloutMSG.ei]> 248
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 131
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

Callout Data Message

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Callout Read Message CalloutReadMsg.EN <EQ0006_CalloutReadMsg.EN>

Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest CalloutMsgCtl.ReadTimeoutTMR.PRE
5000

Message Control Message Timeout Timer

TON
Timer CalloutMsgCtl.ReadTimeoutTMR
Preset 5000
Accum 0
EN
DN

Message Control Message Timeout Timer CalloutMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Callout Read Message CalloutReadMsg.TO <EQ0006_CalloutReadMsg.TO>

L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 134
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

Read Message ER Performance & Diagnostics

10

EDC Gateway Callout Read Message CalloutReadMsg.ER <EQ0006_CalloutReadMsg.ER>

Message Control Read Error 1-shot CalloutMsgCtl.ReadErONS

ONS

Equipment Data Collection - Line Equipment Status State = Red - Callout Tx ER Status.Update.CalloutTxER

L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans CalloutMsgCtl.Trans
PerfTrans Status.CalloutTxRead.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 133
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

Callout Data Message

Write Message

v3.50 remove CalloutMsg Delay TMR - need fast updates for accurate timing and fast PdM Triggers

11

Message Control Send Data
CalloutMsgCtl.Send

Message Control Resend Data (Keep Alive)
CalloutMsgCtl.Resend

Message Control Read Buffer Ready
CalloutMsgCtl.ReadOK

Message Control Write Command Done
CalloutMsgCtl.WriteOK

EDC Gateway Callout Read Message
CalloutReadMsg.EN
<EQ0006_CalloutReadMsg.EN>

EDC Gateway Callout Write Message
CalloutWriteMsg.EN
<EQ0006_CalloutWriteMsg.EN>

EDC Gateway Callout Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Message Control Message Retry Timer
CalloutMsgCtl.RetryTMR.DN

Message Control Write Data
CalloutMsgCtl.Write
L

Message Control Message Delay Timer
CalloutMsgCtl.DelayTMR
RES

EDC Gateway Callout Write Message

COP
Source UserCfgMsg.Path
<EQ0006_UserCfgMsg.Path>
Dest CalloutWriteMsg.Path
<EQ0006_CalloutWriteMsg.Path>
Length 1

EDC Gateway Callout Write Message

COP
Source CFG.MsgDest.CalloutDest
Dest CalloutWriteMsg.RemoteElement
<EQ0006_CalloutWriteMsg.RemoteElement>
Length 1

EDC Gateway Callout Write Message

MOVE
Source CFG.CalloutMSG.e
14
Dest CalloutWriteMsg.REQ_LEN
<EQ0006_CalloutWriteMsg.REQ_LEN>
14

EDC Gateway Callout Write Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest CalloutWriteMsg.UnconnectedTimeout
<EQ0006_CalloutWriteMsg.UnconnectedTimeout>
4000000

EDC Gateway Callout Write Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest CalloutWriteMsg.ConnectionRate
<EQ0006_CalloutWriteMsg.ConnectionRate>
1000000

EDC Gateway Callout Write Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest CalloutWriteMsg.TimeoutMultiplier
<EQ0006_CalloutWriteMsg.TimeoutMultiplier>
0

EDC Gateway Callout Write Message
CalloutWriteMsg.EN_CC
<EQ0006_CalloutWriteMsg.EN_CC>
U

EDC Message Tx Data Callout/Status Receive Message

COP
Source CalloutMsg
Dest EqTxData.Callout[0]
<EQ0006_TxData.Callout[0]>
Length CFG.CalloutMSG.n

EDC Gateway Callout Write Message

MSG
Message Control CalloutWriteMsg ...
<EQ0006_CalloutWriteMsg>
EN
DN
ER

Callout Data Message

Message EN Performance & Diagnostics

12

Message Control Write Data: CalloutMsgCtl.Write

Message Control Write Data 1-shot: CalloutMsgCtl.WriteONS

ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans

aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	CalloutMsgCtl.Trans
PerfTrans	Status.CalloutTxWrite.EN

Transaction Log Control SBR External Load -- Optional Parameters

MOVE

Source	CalloutMsg.Header.PacketID
	1
Dest	Trans.Ctl.Opt[1]
	-1

Log Transactions

JSR

Routine Name	_902_TransLogSBR
Input Par	140
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Callout Data Message

13

Message Control Write Data CalloutMsgCtl.Write

EDC Gateway Callout Write Message Done CalloutWriteMsg.DN <EQ0006_CalloutWriteMsg.DN>

Message Control Write Data CalloutMsgCtl.Write (U)

Message Control Write Command Done CalloutMsgCtl.WriteOK (L)

Message Control Send Data CalloutMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Callout Tx DN Status.Update.CalloutTxOK (L)

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Message Control Resend Data (Keep Alive) CalloutMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Callout Keep Alive Tx DN Status.Update.CalloutKaTxOK (L)

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CalloutMsgCtl.Trans
PerfTrans	Status.CalloutTxWrite.DN

Message Control Resend Data (Keep Alive) CalloutMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CalloutMsgCtl.Trans
PerfTrans	Status.CalloutTxWrite.KA

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.PacketID 1
Dest	Trans.Ctl.Opt[1] -1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	141
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Message Control Send Data CalloutMsgCtl.Send (U)

Message Control Resend Data (Keep Alive) CalloutMsgCtl.Resend (U)

Callout Data Message

Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Callout Write Message CalloutWriteMsg.EN <EQ0006_CalloutWriteMsg.EN>

Message Control Message Timeout Timer

MOVE	
Source	CFG.CLx.MsgTimeout
	5000
Dest	CalloutMsgCtl.WriteTimeoutTMR.PRE
	5000

Message Control Message Timeout Timer

TON		
Timer	CalloutMsgCtl.WriteTimeoutTMR	EN
Preset	5000	DN
Accum	0	

Message Control Message Timeout Timer CalloutMsgCtl.WriteTimeoutTMR.DN

EDC Gateway Callout Write Message CalloutWriteMsg.TO <EQ0006_CalloutWriteMsg.TO> (L)

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.PacketID
	1
Dest	Trans.Ctl.Opt[1]
	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	144
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Message ER Performance & Diagnostics

15

EDC Gateway Callout Write Message CalloutWriteMsg.ER <EQ0006_CalloutWriteMsg.ER>

Message Control Write Error 1-shot CalloutMsgCtl.WriteErONS (ONS)

Equipment Data Collection - Line Equipment Status State = Red - Callout Tx ER Status.Update.CalloutTxER (L)

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate		
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate	...
Status	Status	

Performance Transaction

aoiEDCv1_5_EqPerfTrans		
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans	...
PerfMsgTrans	CalloutMsgCtl.Trans	
PerfTrans	Status.CalloutTxWrite.ER	

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	CalloutMsg.Header.PacketID
	1
Dest	Trans.Ctl.Opt[1]
	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	143
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

Callout Data Message

Message Error Retry

16

EDC Gateway Callout Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

EDC Gateway Callout Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Message Control Message Retry Timer
MOVE
Source CFG.MsgRetryTime
5000
Dest CalloutMsgCtl.RetryTMR.PRE
0

Message Control Message Retry Timer
TON
Timer CalloutMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

Message Control Message Retry Timer
CalloutMsgCtl.RetryTMR.DN

Message Control Retry 1-shot
CalloutMsgCtl.RetryONS
ONS

EDC Gateway Callout Read Message
CalloutReadMsg.ER
<EQ0006_CalloutReadMsg.ER>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source CalloutMsg.Header.PacketID
1
Dest Trans.Ctl.Opt[1]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 132
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

EDC Gateway Callout Write Message
CalloutWriteMsg.ER
<EQ0006_CalloutWriteMsg.ER>

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source CalloutMsg.Header.PacketID
1
Dest Trans.Ctl.Opt[1]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 142
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

(End)

Store Event Mask in Callout Read Message

Equipment Fault/Status Data
Callout Data Mask Store from Read Message

Callout Read Message contains Mask for one Packet (8 words) for Fault, Alarm, Alert, Ignore or Maint
or a Request to Resend Data for one Packet

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 115
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Store Event Mask in Callout Read Message

2

Check for Valid EQID
Exit if Invalid

Equipment Data Callout/Status Message

COP
Source EqRxData.Callout[0] <EQ0006_RxData.Callout[0]>
Dest CalloutMsgMask
Length 1

Equipment Data Callout/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

NE
Source A CalloutMsgMask.Header.EQID 6
Source B UserCfg.EQID 6

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.EQID 6
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.PacketID 1
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.CmdNum -1
Dest Trans.Ctl.Opt[3] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 136
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Check for Valid PacketID
Exit if Invalid

3

Equipment Data
Callout/Status
Message Packet ID -
generates offset for
8-DINT mulitplexed
data packet

LT	
Source A	CalloutMsgMask.Header.PacketID 1
Source B	0

Equipment Data
Callout/Status
Message Packet ID -
generates offset for
8-DINT mulitplexed
data packet

GT	
Source A	CalloutMsgMask.Header.PacketID 1
Source B	CFG.FaultEq.ei 2

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	CalloutMsgMask.Header.EQID 6
Dest	Trans.Ctl.Opt[1] -1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	CalloutMsgMask.Header.PacketID 1
Dest	Trans.Ctl.Opt[2] -1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	CalloutMsgMask.Header.CmdNum -1
Dest	Trans.Ctl.Opt[3] -1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	137
Input Par	CalloutMsgCtl.Trans.TMR.ACC
Input Par	CalloutMsgCtl.TransNum

lbl_Exit
JMP

Load Valid PacketID to PKi index

4

Mask Storage
Packet Index

MOVE	
Source	CalloutMsgMask.Header.PacketID 1
Dest	PKi 1

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Maintenance Mask

5

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -1

Data Refresh
By Packet
DataRefreshPacket.[PKi]
/

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef
16#2035_cb40
Source B PreDefMaskMaint
16#2035_cb40

PreDef Mask
Change
RefreshPreDef

PreDef Mask
Change
RefreshPreDef
/

EqDef Mask
Change
MaskFSC
RES

EqDef Mask
Change
MOVE
Source CFG.FaultEq.n
8
Dest MaskFSC.LEN
8

EqDef Mask
Change
CLR
Dest MaskFSC.POS
7

EqDef Mask
Change
MaskFSC.FD

EqDef Mask
Change
FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefMaintMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskMaint
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefMaintMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Ignore Mask

6

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -2

Data Refresh
By Packet
DataRefreshPacket.[PKi]

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef
16#2035_cb40
Source B PreDefMaskIgnore
16#ffff_ffff

PreDef Mask
Change
RefreshPreDef

PreDef Mask
Change
RefreshPreDef

EqDef Mask
Change
MaskFSC
RES

EqDef Mask
Change
MOVE
Source CFG.FaultEq.n
8
Dest MaskFSC.LEN
8

EqDef Mask
Change
CLR
Dest MaskFSC.POS
7

EqDef Mask
Change
MaskFSC.FD

EqDef Mask
Change
FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefIgnoreMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskIgnore
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefIgnoreMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Fault Mask

7

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -3

Data Refresh
By Packet
DataRefreshPacket.[PKi]

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef
16#2035_cb40
Source B PreDefMaskFault
16#2035_cb50

PreDef Mask
Change
RefreshPreDef

PreDef Mask
Change
RefreshPreDef

EqDef Mask
Change
MaskFSC
RES

EqDef Mask
Change

MOVE
Source CFG.FaultEq.n
8
Dest MaskFSC.LEN
8

EqDef Mask
Change

CLR
Dest MaskFSC.POS
7

EqDef Mask
Change
MaskFSC.FD

EqDef Mask
Change

FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefFaultMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskFault
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefFaultMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Alarm Mask

8

Equipment Data Callout/Status Message Header - Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum -1
Source B -4

Data Refresh By Packet
DataRefreshPacket.[PKi]

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef 16#2035_cb40
Source B PreDefMaskAlarm 16#c0c8_2088

PreDef Mask Change
RefreshPreDef

PreDef Mask Change
RefreshPreDef

EqDef Mask Change
MaskFSC
RES

EqDef Mask Change
MOVE
Source CFG.FaultEq.n 8
Dest MaskFSC.LEN 8

EqDef Mask Change
CLR
Dest MaskFSC.POS 7

EqDef Mask Change
MaskFSC.FD

EqDef Mask Change
FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefAlarmMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh By Packet
DataRefreshPacket.[PKi]
L

Equipment Data Fault/Status Equipment Defined Equipment Defined Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskAlarm
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefAlarmMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Store Event Mask in Callout Read Message

Refresh Packet if Mask Changed
Save Alert Mask

9

Equipment Data
Callout/Status
Message Header -
Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum
-1
Source B -5

Data Refresh
By Packet
DataRefreshPacket.[PKi]

COP
Source CalloutMsgMask.PreDef
Dest PreDef
Length 1

NE
Source A PreDef
16#2035_cb40
Source B PreDefMaskAlert
16#1002_0020

PreDef Mask
Change
RefreshPreDef

PreDef Mask
Change
RefreshPreDef

EqDef Mask
Change
MaskFSC
RES

EqDef Mask
Change

MOVE
Source CFG.FaultEq.n
8
Dest MaskFSC.LEN
8

EqDef Mask
Change

CLR
Dest MaskFSC.POS
7

EqDef Mask
Change
MaskFSC.FD

EqDef Mask
Change

FSC
Control MaskFSC
Length 8
Position 7
Mode ALL
Expression CalloutMsgMask.EqDef.Word[MaskFSC.POS]<>EqDefAlertMask[PKi].Word[MaskFSC.POS]
EN
DN
ER

Data Refresh
By Packet
DataRefreshPacket.[PKi]
L

Equipment Data
Fault/Status
Equipment Defined
Equipment Defined
Status Words

COP
Source CalloutMsgMask.PreDef
Dest PreDefMaskAlert
Length 1

COP
Source CalloutMsgMask.EqDef.Word[0]
Dest EqDefAlertMask[PKi].Word[0]
Length CFG.FaultEq.n

lbl_Exit
JMP

Store Event Mask in Callout Read Message

Data Refresh Request by Packet
for Gateway first scan or Gateway Status Display Refresh PB

10

Equipment Data Callout/Status Message Header - Command Number

EQ
Source A CalloutMsgMask.Header.CmdNum -1
Source B -10

Data Refresh By Packet
DataRefreshPacket.[PKi]
L

lbl_Exit
JMP

Invalid Command Number

11

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.EQID 6
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.PacketID 1
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CalloutMsgMask.Header.CmdNum -1
Dest Trans.Ctl.Opt[3] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 138
Input Par CalloutMsgCtl.Trans.TMR.ACC
Input Par CalloutMsgCtl.TransNum

lbl_Exit
JMP

Exit Routine

12

lbl_Exit
LBL

NOP

(End)

Collect Fault/Status Data

Equipment Fault/Status Data
Collect Data for Callout Live Message and Fault Buffered Message
v1.3 - Dec2016 - Multi-packet support for >8 DINT Equipment-Defined Faults

PKTi - Packet Index for multiplexed Fault Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 121
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Exit if this Packet not Enabled

2

Equipment Data Collection - User Config Equipment Defined Packet EN bit array
UserCfg.EqDefPacketEN.[PKTi]

lbl_Exit
JMP

Collect Fault/Status Data

Filter Equipment Defined Fault/Status Packet with Ignore Mask downloaded by Server Config
Sep 22, 2022 - Replaced MVM with AND (MVM doesn't update masked bits)

3

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[0] ??
Source B EqDefIgnoreMask[PKTi].Word[0] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[0] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[1] ??
Source B EqDefIgnoreMask[PKTi].Word[1] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[1] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[2] ??
Source B EqDefIgnoreMask[PKTi].Word[2] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[2] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[3] ??
Source B EqDefIgnoreMask[PKTi].Word[3] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[3] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[4] ??
Source B EqDefIgnoreMask[PKTi].Word[4] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[4] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[5] ??
Source B EqDefIgnoreMask[PKTi].Word[5] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[5] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[6] ??
Source B EqDefIgnoreMask[PKTi].Word[6] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[6] ??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

AND
Source A EqDefHold[PKTi].Word[7] ??
Source B EqDefIgnoreMask[PKTi].Word[7] ??
Dest FaultMsgCtl.EqDef[PKTi].Word[7] ??

EqDriver Internal PreDefined Faults
(RingBuffer Filling, Message Faults, Maint/Fault/Alarm/Alert Masks etc.)

v2.0 Update PreDefined when everything else is complete to provide correct EQ Status to Gateway

4

Fault EqDefined Packet Index

EQ
Source A PKTi -1
Source B 0

Simultation for Fault Data Enable Simulation

SimFault.EN

Pre-Defined Fault/Status Data Mapping

JSR
Routine Name _100_PreDefFaultData

Collect Fault/Status Data

Check for Data Change in this Packet
Predefined must be copied to DINT for EQU

PreDefined changes are always sent in Packet 0
(This sometimes causes additional data packet, but simplifies logic and server data processing)

5

Fault EqDefined Packet Index

EQ
Source A: PKTi
-1
Source B: 0

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults DINT

COP
Source: FaultMsg.PreDef
Dest: FaultMsgCtl.PreDefVal
Length: 1

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults Last DINT

COP
Source: FaultMsgCtl.PreDefLast
Dest: FaultMsgCtl.PreDefLastVal
Length: 1

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults DINT

MEQ
Source: FaultMsgCtl.PreDefVal
16#0000_0021
Mask: 16#ffff_ffff
Compare: FaultMsgCtl.PreDefLastVal
16#0000_0021

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A: FaultMsgCtl.EqDef[PKTi].Word[0]
??
Source B: FaultMsgCtl.EqDefLast[PKTi].Word[0]
??

Fault EqDefined Packet Index

NE
Source A: PKTi
-1
Source B: 0

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A: FaultMsgCtl.EqDef[PKTi].Word[1]
??
Source B: FaultMsgCtl.EqDefLast[PKTi].Word[1]
??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A: FaultMsgCtl.EqDef[PKTi].Word[2]
??
Source B: FaultMsgCtl.EqDefLast[PKTi].Word[2]
??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A: FaultMsgCtl.EqDef[PKTi].Word[3]
??
Source B: FaultMsgCtl.EqDefLast[PKTi].Word[3]
??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A: FaultMsgCtl.EqDef[PKTi].Word[4]
??
Source B: FaultMsgCtl.EqDefLast[PKTi].Word[4]
??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A: FaultMsgCtl.EqDef[PKTi].Word[5]
??
Source B: FaultMsgCtl.EqDefLast[PKTi].Word[5]
??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A: FaultMsgCtl.EqDef[PKTi].Word[6]
??
Source B: FaultMsgCtl.EqDefLast[PKTi].Word[6]
??

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

EQ
Source A: FaultMsgCtl.EqDef[PKTi].Word[7]
??
Source B: FaultMsgCtl.EqDefLast[PKTi].Word[7]
??

Fault Ring Buffer Message Control for multiplexed packets No Data Change
FaultMsgCtl.NoChange.[PKTi]

Collect Fault/Status Data

Data Change in this Packet
Update Last Values
Load Fault/Status Ring Buffer
Special Case to generate a Callout Message when Gateway Mask Download Required

PreDefined changes are always sent in Packet 0
(This sometimes causes additional data packet, but simplifies logic and server data processing)

v2.09 Data Refresh signal (daily, pre-shift)
v3.02 Data Refresh Packet signal (first scan, mask updates and Gateway request); block Packet refresh until all Mask Downloads are complete

6

Fault Ring Buffer Message Control for multiplexed packets No Data Change
FaultMsgCtl.NoChange.[PKTi]

Load Fault RingBuffer
FaultMsgChange

Data Refresh 1-shot
DataRefresh

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)
CalloutMsgMask.Footer.Spare.[PKTi]

Data Refresh By Packet
DataRefreshPacket.[PKTi]

Data Refresh By Packet
DataRefreshPacket.[PKTi]

Equipment Data Callout/Status Message Footer - Spare (fills unused portion of DINT)
CalloutMsgMask.Footer.Spare.[PKTi]

Fault Ring Buffer Message Control for multiplexed packets Data Change

EQ
Source A FaultMsgCtl.Change
0
Source B 0

Generate Callout Message for Mask
MaskDownloadRequired

Fault Ring Buffer Message Control for multiplexed packets Data Change
FaultMsgCtl.Change.[PKTi]

Load Fault RingBuffer
FaultMsgChange

Fault EqDefined Packet Index

EQ
Source A PKTi
-1
Source B 0

Fault Ring Buffer Message Control for multiplexed packets Predefined Faults Last

COP
Source FaultMsg.PreDef
Dest FaultMsgCtl.PreDefLast
Length 1

Collect Fault/Status Data

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined

COP
Source FaultMsgCtl.EqDef[PKTi]
Dest FaultMsgCtl.EqDefLast[PKTi]
Length 1

Equipment Data Collection - User Config Fault Message Enable
UserCfg.FaultMsgEN

CLx Module Status CLx Transaction Time Invalid
CLX.TransTimeInvalid

Load Fault Data record to Ring Buffer

JSR
Routine Name _122_FaultDataLoadRB

Exit Routine

7

lbl_Exit
LBL

NOP

(End)

Load Fault Data record to Ring Buffer

Equipment Fault/Status Data
Ring Buffer Load
v1.3 - Dec2016 - Multi-packet support for >8 DINT Equipment-Defined Faults

PKTi - Packet Index for multiplexed Fault Data

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 122
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

EDC Ring Buffer Control Transaction Number

ADD
Source A FaultRBCtl.TransactionNumber 3759
Source B 1
Dest FaultRBCtl.TransactionNumber 3759

EDC Ring Buffer Control Transaction Number

LT
Source A FaultRBCtl.TransactionNumber 3759
Source B 1

EDC Ring Buffer Control Transaction Number

GT
Source A FaultRBCtl.TransactionNumber 3759
Source B CFG.MsgTransNumRollover 9999

EDC Ring Buffer Control Transaction Number

MOVE
Source 1
Dest FaultRBCtl.TransactionNumber 3759

Load Fault Data record to Ring Buffer

Load Message Header, Timestamp and Footer Data
v1.3 - Added Header.PacketID for multiplexed data packets
Copy This Packet to FaultMsg.EqDef

3

Equipment Data Fault/Status Message Header - Transaction Number (must match Footer)

MOVE	
Source	FaultRBCtl.TransactionNumber 3759
Dest	FaultMsg.Header.TransNum 3759

Equipment Data Fault/Status Message Header - Command Number

MOVE	
Source	0
Dest	FaultMsg.Header.CmdNum 0

Equipment Data Fault/Status Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE	
Source	UserCfg.EQID 6
Dest	FaultMsg.Header.EQID 6

Equipment Data Fault/Status Message Header - Data Count - Total Qty of Words used in all Packets

MOVE	
Source	UserCfg.EqDefDataCount[PKTi] ??
Dest	FaultMsg.Header.DataCount 8

Equipment Data Fault/Status Message Header - Packet ID - generates offset for 8-DINT mulitplexed data packet

MOVE	
Source	PKTi -1
Dest	FaultMsg.Header.PacketID 0

Equipment Data Fault/Status Message Transaction Time Stamp

COP	
Source	CLX.TransTime
Dest	FaultMsg.TransTime
Length	1

Equipment Data Fault/Status Message Data - Predefined Status

COP	
Source	FaultMsg.PreDef
Dest	FaultMsg.PreDef
Length	1

Equipment Data Fault/Status Message Data - Equipment Defined Status

COP	
Source	FaultMsgCtl.EqDef[PKTi]
Dest	FaultMsg.EqDef
Length	1

Equipment Data Fault/Status Message Footer - Spare (fills unused portion of DINT)

MOVE	
Source	0
Dest	FaultMsg.Footer.Spare 0

Equipment Data Fault/Status Message Footer - Transaction Number (must match Footer)

MOVE	
Source	FaultRBCtl.TransactionNumber 3759
Dest	FaultMsg.Footer.TransNum 3759

Load Fault Data record to Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

4

EDC Ring Buffer Control Next Write Record

LT
Source A FaultRBCtl.WritePtr 141
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A FaultRBCtl.WritePtr 141
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Diagnostic - Manual Reset

FaultRBCtl.WritePtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source FaultRBCtl.WritePtr 141
Dest FaultRBCtl.WritePtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Header.PacketID 0
Dest Trans.Ctl.Opt[3] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 213
Input Par 0
Input Par FaultMsg.Header.TransNum

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest FaultRBCtl.WritePtr 141

Load Fault Data record to Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

5

EDC Ring Buffer Control Next Write Record

EQ
Source A FaultRBCtl.WritePtr 141
Source B FaultRBCtl.ReadPtr 140

EDC Ring Buffer Control Buffer Full - Block Writes

FaultRBCtl.FullBlockWrite

EDC Ring Buffer Control Diagnostic - Manual Reset

FaultRBCtl.WriteBlockErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A FaultRBCtl.WriteBlockErrCnt 1557127
Source B 1
Dest FaultRBCtl.WriteBlockErrCnt 1557127

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultMsg.Header.PacketID 0
Dest Trans.Ctl.Opt[3] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 214
Input Par 0
Input Par FaultMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans (...)
PerfMsgTrans TempMsgTrans
PerfTrans Status.FaultRx.ER

lbl_Exit

JMP

Load Fault Data record to Ring Buffer

RB Write -- Load Message into Ring Buffer

6

Equipment Data Fault/Status Message
COP
Source FaultMsg
Dest FaultRB[FaultRBCtl.WritePtr]
Length 1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source FaultMsg.Header.PacketID 0
Dest Trans.Ctl.Opt[3] -1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 211
Input Par 0
Input Par FaultMsg.Header.TransNum

Message Control Transaction Timing
FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TempMsgTrans
PerfTrans Status.FaultRx.DN

RB Write -- Increment Write Pointer

7

EDC Ring Buffer Control Next Write Record
ADD
Source A FaultRBCtl.WritePtr 141
Source B 1
Dest FaultRBCtl.WritePtr 141

EDC Ring Buffer Control Next Write Record
LT
Source A FaultRBCtl.WritePtr 141
Source B 0

EDC Ring Buffer Control Next Write Record
GT
Source A FaultRBCtl.WritePtr 141
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Next Write Record
MOVE
Source 0
Dest FaultRBCtl.WritePtr 141

Exit Routine

8

lbl_Exit
LBL

NOP

Load Fault Data record to Ring Buffer

(End)

Load Read Records from RingBuffer to Gateway Message

v3.5 Equipment Bulk Fault/Status Data
Ring Buffer Unload to Gateway Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 123
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Load Read Records from RingBuffer to Gateway Message

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Next Read Record

LT
Source A FaultRBCtl.ReadPtr 140
Source B 0

EDC Ring Buffer Control Next Read Record

GT
Source A FaultRBCtl.ReadPtr 140
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Diagnostic - Manual Reset

FaultRBCtl.ReadPtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source FaultRBCtl.ReadPtr 140
Dest FaultRBCtl.ReadPtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 223
Input Par 0
Input Par FaultRBCtl.TransactionNumber

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest FaultRBCtl.WritePtr 141

EDC Ring Buffer Control Next Read Record

MOVE
Source CFG.FaultRB.mi 249
Dest FaultRBCtl.ReadPtr 140

Load Read Records from RingBuffer to Gateway Message

Next Read Location
ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

ADD
Source A FaultRBCtl.ReadPtr 140
Source B 1
Dest FaultRBCtl.ReadNextPtr 141

EDC Ring Buffer Control Next Read Record

GT
Source A FaultRBCtl.ReadNextPtr 141
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest FaultRBCtl.ReadNextPtr 141

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Next Read Record

EQ
Source A FaultRBCtl.ReadPtr 140
Source B FaultRBCtl.WritePtr 141

EDC Ring Buffer Control Buffer Full
FaultRBCtl.Full

EDC Ring Buffer Control Read Buffer Record in Progress
FaultRBCtl.ReadInProgress
/

EDC Ring Buffer Control Next Read Record

EQ
Source A FaultRBCtl.ReadNextPtr 141
Source B FaultRBCtl.WritePtr 141

EDC Ring Buffer Control Buffer Empty
FaultRBCtl.Empty

Load Read Records from RingBuffer to Gateway Message Buffer Count

5

EDC Ring Buffer Control Read Buffer Record in Progress
FaultRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
FaultRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record
GT
Source A FaultRBCtl.WritePtr 141
Source B FaultRBCtl.ReadPtr 140

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest FaultRBCtl.RecordCount 0
Expression FaultRBCtl.WritePtr-FaultRBCtl.ReadPtr

EDC Ring Buffer Control Next Write Record
LT
Source A FaultRBCtl.WritePtr 141
Source B FaultRBCtl.ReadPtr 140

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest FaultRBCtl.RecordCount 0
Expression CFG.FaultRB.n-(FaultRBCtl.ReadPtr-FaultRBCtl.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
FaultRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
FaultRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record
GE
Source A FaultRBCtl.WritePtr 141
Source B FaultRBCtl.ReadNextPtr 141

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest FaultRBCtl.RecordCount 0
Expression FaultRBCtl.WritePtr-FaultRBCtl.ReadNextPtr

EDC Ring Buffer Control Next Write Record
LT
Source A FaultRBCtl.WritePtr 141
Source B FaultRBCtl.ReadNextPtr 141

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest FaultRBCtl.RecordCount 0
Expression CFG.FaultRB.n-(FaultRBCtl.ReadNextPtr-FaultRBCtl.WritePtr)

EDC Ring Buffer Control Ring Buffer Record Count
GT
Source A FaultRBCtl.RecordCount 0
Source B FaultRBCtl.RecordCountMax 249

EDC Ring Buffer Control Ring Buffer Record Count Max
MOVE
Source FaultRBCtl.RecordCount 0
Dest FaultRBCtl.RecordCountMax 249

Load Read Records from RingBuffer to Gateway Message

Read Next
Ready for next data to Gateway Message

6

EDC Ring Buffer Control Buffer Empty FaultRBCtl.Empty /
EDC Ring Buffer Control Read Buffer Record in Progress FaultRBCtl.ReadInProgress /
Fault Ring Buffer Message Control for multiplexed packets Send Data FaultMsgCtl.Send /
Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive) FaultMsgCtl.Resend /
EDC Ring Buffer Control Read Next Buffer Record 1-shot FaultRBCtl.ReadNextONS [ONS]
EDC Ring Buffer Control Read Next Buffer Record FaultRBCtl.ReadNext (OTE)
EDC Ring Buffer Control Read Buffer Record in Progress FaultRBCtl.ReadInProgress (L)
EDC Ring Buffer Control Read Buffer Record Complete FaultRBCtl.ReadComplete (U)

Skip ReadNext code if not loading new records
(faster execution)

7

EDC Ring Buffer Control Read Next Buffer Record FaultRBCtl.ReadNext /
lbl_SkipReadNext (JMP)

Next Message
Initialize Bulk Fault Message Loop
use RecordCount to load up to 10 messages

8

EDC Ring Buffer Control Read Next Buffer Record FaultRBCtl.ReadNext

EDC Message Tx Data Bulk Fault/Status Receive Message
FLL
Source 0
Dest EqTxData.Fault[0]
<EQ0006_TxData.Fault[0]>
Length CFG.FaultMSG.e

Record Index
CLR
Dest RECi 2

Record Max Index
SUB
Source A FaultRBCtl.RecordCount 0
Source B 1
Dest RECmi 1

Record Max Index
LE
Source A RECmi 1
Source B 0

Record Max Index
MOVE
Source 0
Dest RECmi 1

Record Max Index
GT
Source A RECmi 1
Source B 9

Record Max Index
MOVE
Source 9
Dest RECmi 1

diagnostic record count
ADD
Source A xxRECmi[RECmi] 5794
Source B 1
Dest xxRECmi[RECmi] 5794

Load Read Records from RingBuffer to Gateway Message

Next Message
Bulk Fault Message Loop

9

lbl_MsgLoop
LBL

NOP

Read Next
Increment Read Pointer

10

EDC Ring Buffer Control Read Next Buffer Record
FaultRBCtl.ReadNext

EDC Ring Buffer Control Next Read Record
ADD
Source A FaultRBCtl.ReadPtr 140
Source B 1
Dest FaultRBCtl.ReadPtr 140

EDC Ring Buffer Control Next Read Record
LT
Source A FaultRBCtl.ReadPtr 140
Source B 0

EDC Ring Buffer Control Next Read Record
GT
Source A FaultRBCtl.ReadPtr 140
Source B CFG.FaultRB.mi 249

EDC Ring Buffer Control Next Read Record
MOVE
Source 0
Dest FaultRBCtl.ReadPtr 140

Load Read Records from RingBuffer to Gateway Message

Read Next
Load Read Record

calculate offset for each RECi
14 DINT/record + 2 DINT Header

11

EDC Ring Buffer
Control Read Next
Buffer Record
FaultRBCtl.ReadNext

Fault Msg
Record Index

CPT	
Dest	FLTi
	16
Expression	(CFG.FaultRec.e * RECi) + 2

EDC Message Tx Data
Bulk Fault/Status
Receive Message

COP	
Source	FaultRB[FaultRBCtl.ReadPtr]
Dest	EqTxData.Fault[FLTi]
	<EQ0006_TxData.Fault[FLTi]>
Length	CFG.FaultRec.e

Fault Ring Buffer
Message Control
for multiplexed
packets Transaction
Number

MOVE	
Source	FaultRB[FaultRBCtl.ReadPtr].Header.TransNum
	3759
Dest	FaultMsgCtl.TransNum
	3759

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	FaultRBCtl.WritePtr
	141
Dest	Trans.Ctl.Opt[1]
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	FaultRBCtl.ReadPtr
	140
Dest	Trans.Ctl.Opt[2]
	-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE	
Source	FaultRB[FaultRBCtl.ReadPtr].Header.PacketID
	0
Dest	Trans.Ctl.Opt[3]
	-1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	221
Input Par	0
Input Par	FaultMsgCtl.TransNum

Load Read Records from RingBuffer to Gateway Message

load Bulk Message Header and Footer Transaction Number from first record
overwrite DataCount/PacketID with Bulk Message RecordCount

the footer location and Write Message length are based on record count
(most messages will have a single record - no point in sending 10x the data when not needed)
14 DINT/record + 2 DINT Header

12

EDC Ring Buffer Control Read Next Buffer Record: FaultRBCtl.ReadNext

Record Index

EQ
- Source A: RECi = 2
- Source B: 0

Equipment Data Message Header

COP
- Source: FaultRB[FaultRBCtl.ReadPtr].Header
- Dest: BulkHeader
- Length: 1

Equipment Data Message Header Header - Data Count - Total Qty of Words used in all Packets

ADD
- Source A: RECmi = 1
- Source B: 1
- Dest: BulkHeader.DataCount = 2

Equipment Data Message Header Header - Packet ID - generates offset for 8-DINT mulitplexed data packet

CLR
- Dest: BulkHeader.PacketID = 0

EDC Message Tx Data Bulk Fault/Status Receive Message

COP
- Source: BulkHeader
- Dest: EqTxData.Fault[0] <EQ0006_TxData.Fault[0]>
- Length: 2

CPT
- Dest: FLTei = 30
- Expression: (CFG.FaultRec.e * (RECmi+1)) + 2

EDC Message Tx Data Bulk Fault/Status Receive Message

COP
- Source: FaultRB[FaultRBCtl.ReadPtr].Footer
- Dest: EqTxData.Fault[FLTei] <EQ0006_TxData.Fault[FLTei]>
- Length: 1

Increment Loop Pointer and Jump until all Records loaded

13

EDC Ring Buffer Control Read Next Buffer Record: FaultRBCtl.ReadNext

Record Index

ADD
- Source A: RECi = 2
- Source B: 1
- Dest: RECi = 2

Record Index

LE
- Source A: RECi = 2
- Source B: RECmi = 1

lbl_MsgLoop
JMP

Load Read Records from RingBuffer to Gateway Message

Read Next
Set Data Ready Handshake

14

EDC Ring Buffer Control Read Next Buffer Record
FaultRBCtl.ReadNext

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send
(L)

Fault Ring Buffer Message Control for multiplexed packets Read Command
FaultMsgCtl.Read
(U)

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready
FaultMsgCtl.ReadOK
(U)

Fault Ring Buffer Message Control for multiplexed packets Write Command
FaultMsgCtl.Write
(U)

Fault Ring Buffer Message Control for multiplexed packets Write Command Done
FaultMsgCtl.WriteOK
(U)

Skip ReadNext code if not loading new records
(faster execution)

15

lbl_SkipReadNext
LBL

NOP

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

16

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send

Fault Ring Buffer Message Control for multiplexed packets Write Command Done
FaultMsgCtl.WriteOK

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send
(U)

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend
(U)

EDC Ring Buffer Control Read Buffer Record Complete
FaultRBCtl.ReadComplete
(L)

Load Read Records from RingBuffer to Gateway Message

Read Complete Delay
Allow Gateway processing time before attempting next record send.

17

EDC Ring Buffer Control Read Buffer Record Complete
FaultRBCtl.ReadComplete

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source 100
Dest FaultRBCtl.ReadCompleteDelayTMR.PRE
100

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer FaultRBCtl.ReadCompleteDelayTMR
Preset 100
Accum 107
EN
DN

EDC Ring Buffer Control Read Complete Delay Timer
FaultRBCtl.ReadCompleteDelayTMR.DN

EDC Ring Buffer Control Read Buffer Record Complete Delay
FaultRBCtl.ReadCompleteDelay

EDC Ring Buffer Control Read Buffer Record in Progress
FaultRBCtl.ReadInProgress
U

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

18

EDC Ring Buffer Control Buffer Empty
FaultRBCtl.Empty
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 180000
Dest FaultRBCtl.NotEmptyTMR.PRE
180000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer FaultRBCtl.NotEmptyTMR
Preset 180000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A FaultRBCtl.NotEmptyTMR.ACC
0
Source B FaultRBCtl.NotEmptyMax
180003

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source FaultRBCtl.NotEmptyTMR.ACC
0
Dest FaultRBCtl.NotEmptyMax
180003

Load Read Records from RingBuffer to Gateway Message

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

19

EDC Ring Buffer Control Buffer Full
FaultRBCtl.Full

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
FaultRBCtl.NotEmptyTMR.DN

EDC Ring Buffer Control Read Buffer Alarm
FaultRBCtl.BufferAlarm

(End)

Fault Data Ring Buffer and Message

Equipment Fault/Status Data
Buffered Fault Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	124
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Fault Ring Buffer Message Control for multiplexed packets Message Transaction Timer

MOVE	
Source	999999999
Dest	FaultMsgCtl.Trans.TMR.PRE 999999999

Fault Ring Buffer Message Control for multiplexed packets Message Transaction Timer

TON		
Timer	FaultMsgCtl.Trans.TMR	EN
Preset	999999999	DN
Accum	211470	

Fault Data Ring Buffer and Message

Read Message Delay Timer
If driver sends data too fast, it will generate extra Read Message traffic until previous message has been processed
Optimization uses the ReadOk and ReadNotOk counts to tune the Delay Time to the Gateway response

3

EDC Gateway Fault Write Message
FaultWriteMsg.EN
<EQ0006_FaultWriteMsg.EN>

EDC Gateway Fault Read Message
FaultReadMsg.EN
<EQ0006_FaultReadMsg.EN>

Fault Ring Buffer Message Control for multiplexed packets Read Message Ok Count

GE
Source A FaultMsgCtl.ReadOkCount 4
Source B CfgRB_ReadOkLimit <CfgRB.ReadOkLimit> 5

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

SUB
Source A FaultMsgCtl.DelayTMR.PRE 100
Source B CfgRB_ReadOkSub <CfgRB.ReadOkSub> 10
Dest FaultMsgCtl.DelayTMR.PRE 100

Fault Ring Buffer Message Control for multiplexed packets Read Message Not Ok Count

GE
Source A FaultMsgCtl.ReadNotOkCount 0
Source B CfgRB_ReadNotOkLimit <CfgRB.ReadNotOkLimit> 2

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

ADD
Source A FaultMsgCtl.DelayTMR.PRE 100
Source B CfgRB_ReadNotOkAdd <CfgRB.ReadNotOkAdd> 20
Dest FaultMsgCtl.DelayTMR.PRE 100

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

LE
Source A FaultMsgCtl.DelayTMR.PRE 100
Source B CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

MOVE
Source CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100
Dest FaultMsgCtl.DelayTMR.PRE 100

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

GT
Source A FaultMsgCtl.DelayTMR.PRE 100
Source B CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500

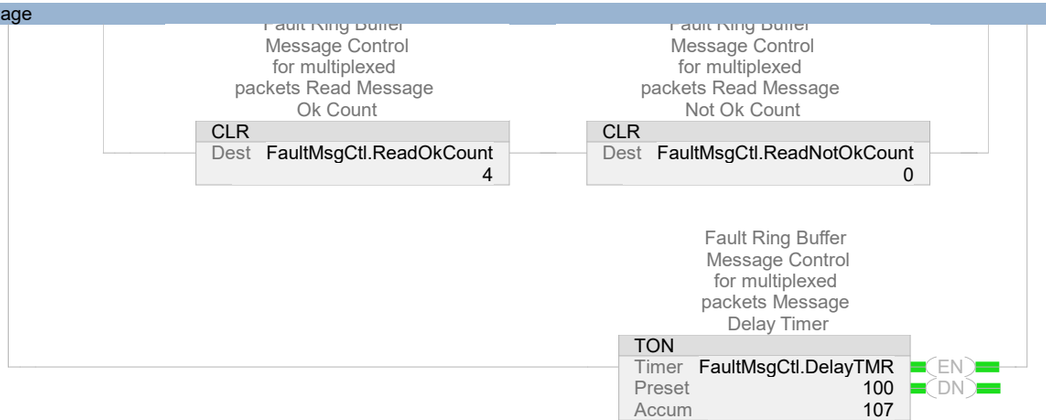
Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer

MOVE
Source CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500
Dest FaultMsgCtl.DelayTMR.PRE 100

Fault Ring Buffer

Fault Ring Buffer

Fault Data Ring Buffer and Message



Fault Data Ring Buffer and Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

Fault Ring Buffer Message Control for multiplexed packets Send Data FaultMsgCtl.Send

EDC Gateway Fault Write Message FaultWriteMsg.EN <EQ0006_FaultWriteMsg.EN>

EDC Gateway Fault Read Message FaultReadMsg.EN <EQ0006_FaultReadMsg.EN>

Fault Ring Buffer Message Control for multiplexed packets Message Resend Timer

MOVE
Source CFG.BuffMsgResendTime 301000
Dest FaultMsgCtl.ResendTMR.PRE 301000

Fault Ring Buffer Message Control for multiplexed packets Message Resend Timer

TON
Timer FaultMsgCtl.ResendTMR
Preset 301000
Accum 211442
EN
DN

Fault Ring Buffer Message Control for multiplexed packets Message Resend Timer FaultMsgCtl.ResendTMR.DN

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive) FaultMsgCtl.Resend

L

Fault Ring Buffer Message Control for multiplexed packets Message Resend Timer FaultMsgCtl.ResendTMR

RES

Fault Ring Buffer Message Control for multiplexed packets Read Command FaultMsgCtl.Read

U

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready FaultMsgCtl.ReadOK

U

Fault Ring Buffer Message Control for multiplexed packets Write Command FaultMsgCtl.Write

U

Fault Ring Buffer Message Control for multiplexed packets Write Command Done FaultMsgCtl.WriteOK

U

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.WritePtr 141
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRBCtl.ReadPtr 140
Dest Trans.Ctl.Opt[2] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source FaultRB[FaultRBCtl.ReadPtr].Header.PacketID 0
Dest Trans.Ctl.Opt[3] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 222
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message
Send Data Initialize

5

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Send Data 1-Shot
FaultMsgCtl.SendONS
ONS

Fault Ring Buffer Message Control for multiplexed packets Message Transaction Timer
FaultMsgCtl.Trans.TMR
RES

Fault Ring Buffer Message Control for multiplexed packets Transaction Elapsed Time for Last Event (used to calculate differential)
CLR
Dest FaultMsgCtl.Trans.Last
14

Fault Data Ring Buffer and Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

variable Read Msg length for variable record count - read back what was sent with last WriteMsg

6

Fault Ring Buffer Message Control for multiplexed packets Send Data FaultMsgCtl.Send

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready FaultMsgCtl.ReadOK

Fault Ring Buffer Message Control for multiplexed packets Message Delay Timer FaultMsgCtl.DelayTMR.DN

EDC Gateway Fault Read Message FaultReadMsg.EN <EQ0006_FaultReadMsg.EN>

EDC Gateway Fault Write Message FaultWriteMsg.EN <EQ0006_FaultWriteMsg.EN>

EDC Gateway Fault Read Message FaultReadMsg.ER <EQ0006_FaultReadMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive) FaultMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer FaultMsgCtl.RetryTMR.DN

Fault Ring Buffer Message Control for multiplexed packets Read Command FaultMsgCtl.Read (L)

Fault Ring Buffer Message Control for multiplexed packets Read Command 1-Shot FaultMsgCtl.ReadONS

EDC Gateway Fault Read Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest FaultReadMsg.Path <EQ0006_FaultReadMsg.Path>
Length 1

EDC Gateway Fault Read Message

COP
Source CFG.MsgDest.FaultDest
Dest FaultReadMsg.RemoteElement <EQ0006_FaultReadMsg.RemoteElement>
Length 1

EDC Gateway Fault Read Message

MOVE
Source FaultWriteMsg.REQ_LEN <EQ0006_FaultWriteMsg.REQ_LEN>
31
Dest FaultReadMsg.REQ_LEN <EQ0006_FaultReadMsg.REQ_LEN>
31

EDC Gateway Fault Read Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest FaultReadMsg.UnconnectedTimeout <EQ0006_FaultReadMsg.UnconnectedTimeout>
4000000

EDC Gateway Fault Read Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest FaultReadMsg.ConnectionRate <EQ0006_FaultReadMsg.ConnectionRate>
1000000

EDC Gateway Fault Read Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest FaultReadMsg.TimeoutMultiplier <EQ0006_FaultReadMsg.TimeoutMultiplier>
0

EDC Gateway Fault Read Message FaultReadMsg.EN_CC <EQ0006_FaultReadMsg.EN_CC> (U)

EDC Gateway Fault Read Message

MSG
Message Control FaultReadMsg <EQ0006_FaultReadMsg>
EN
DN
ER

Fault Data Ring Buffer and Message

Read Message EN Performance & Diagnostics

7

Fault Ring Buffer Message Control for multiplexed packets Read Command 1-Shot
FaultMsgCtl.ReadONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	230
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Read Message OK - Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record)

Increment ReadOk and ReadNotOk counts
only increment ReadOk count when multiple records in Buffer (single records are always ready to send and don't test Gateway servicing speed)

variable Read Msg length for variable record count - check TransNum=0 in first & last word sent with last WriteMsg

8

Fault Ring Buffer Message Control for multiplexed packets Read Command
FaultMsgCtl.Read

EDC Gateway Fault Read Message
FaultReadMsg.DN
<EQ0006_FaultReadMsg.DN>

Fault Ring Buffer Message Control for multiplexed packets Read Command
FaultMsgCtl.Read
U

EDC Message Rx Data Bulk Fault/Status Receive Message

EQ
Source A EqRxData.Fault[0]
<EQ0006_RxData.Fault[0]>
0
Source B 0
0

EDC Message Rx Data Bulk Fault/Status Receive Message

EQ
Source A EqRxData.Fault[FaultReadMsg.REQ_LEN-1]
<EQ0006_RxData.Fault[FaultReadMsg.REQ_LEN-1]>
0
Source B 0
0

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready
FaultMsgCtl.ReadOK
L

EDC Ring Buffer Control Ring Buffer Record Count

GE
Source A FaultRBCtl.RecordCount
0
Source B 2

Fault Ring Buffer Message Control for multiplexed packets Read Message Ok Count

ADD
Source A FaultMsgCtl.ReadOkCount
4
Source B 1
Dest FaultMsgCtl.ReadOkCount
4

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready
FaultMsgCtl.ReadOK
/

Fault Ring Buffer Message Control for multiplexed packets Read Message Not Ok Count

ADD
Source A FaultMsgCtl.ReadNotOkCount
0
Source B 1
Dest FaultMsgCtl.ReadNotOkCount
0

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans FaultMsgCtl.Trans
PerfTrans Status.FaultTxRead.DN

Fault Ring Buffer Message Control for multiplexed packets Resend Data

Performance

Fault Data Ring Buffer and Message

(Keep Alive)
FaultMsgCtl.Resend

Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans FaultMsgCtl.Trans
PerfTrans Status.FaultTxRead.KA

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source EqRxData.Fault[0]
<EQ0006_RxData.Fault[0]> 0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters
MOVE
Source EqRxData.Fault[FaultReadMsg.REQ_LEN-1]
<EQ0006_RxData.Fault[FaultReadMsg.REQ_LEN-1]> 0
Dest Trans.Ctl.Opt[2]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 231
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Fault
Read Message
FaultReadMsg.EN
<EQ0006_FaultReadMsg.EN>

Fault Ring Buffer
Message Control
for multiplexed
packets Message
Timeout Timer
MOVE
Source CFG.CLx.MsgTimeout
5000
Dest FaultMsgCtl.ReadTimeoutTMR.PRE
5000

Fault Ring Buffer
Message Control
for multiplexed
packets Message
Timeout Timer
TON
Timer FaultMsgCtl.ReadTimeoutTMR
Preset 5000
Accum 0
EN
DN

Fault Ring Buffer
Message Control
for multiplexed
packets Message
Timeout Timer
FaultMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Fault
Read Message
FaultReadMsg.TO
<EQ0006_FaultReadMsg.TO>
L

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 234
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Read Message ER Performance & Diagnostics

10

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Read Error 1-shot
FaultMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Fault Tx ER
Status.Update.FaultTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxRead.ER

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	233
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Write Message
Write Data is loaded by Ring Buffer routine

variable Write Msg length for variable record count - only send what is needed (most messages will only have 1 record)
14 DINT/record + 3 DINT Header/Footer

11

Fault Ring Buffer Message Control for multiplexed packets Send Data FaultMsgCtl.Send

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive) FaultMsgCtl.Resend

Fault Ring Buffer Message Control for multiplexed packets Read Buffer Ready FaultMsgCtl.ReadOK

Fault Ring Buffer Message Control for multiplexed packets Write Command Done FaultMsgCtl.WriteOK

EDC Gateway Fault Read Message FaultReadMsg.EN <EQ0006_FaultReadMsg.EN>

EDC Gateway Fault Write Message FaultWriteMsg.EN <EQ0006_FaultWriteMsg.EN>

EDC Gateway Fault Write Message FaultWriteMsg.ER <EQ0006_FaultWriteMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer FaultMsgCtl.RetryTMR.DN

Fault Ring Buffer Message Control for multiplexed packets Write Command FaultMsgCtl.Write (L)

EDC Gateway Fault Write Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest FaultWriteMsg.Path <EQ0006_FaultWriteMsg.Path>
Length 1

EDC Gateway Fault Write Message

COP
Source CFG.MsgDest.FaultDest
Dest FaultWriteMsg.RemoteElement <EQ0006_FaultWriteMsg.RemoteElement>
Length 1

Equipment Data Message Header

COP
Source EqTxData.Fault[0] <EQ0006_TxData.Fault[0]>
Dest BulkHeader
Length 2

EDC Gateway Fault Write Message

CPT
Dest FaultWriteMsg.REQ_LEN <EQ0006_FaultWriteMsg.REQ_LEN>
31
Expression (BulkHeader.DataCount * CFG.FaultRec.e) + 3

EDC Gateway Fault Write Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest FaultWriteMsg.UnconnectedTimeout <EQ0006_FaultWriteMsg.UnconnectedTimeout>
4000000

EDC Gateway Fault Write Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest FaultWriteMsg.ConnectionRate <EQ0006_FaultWriteMsg.ConnectionRate>
1000000

EDC Gateway Fault Write Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest FaultWriteMsg.TimeoutMultiplier <EQ0006_FaultWriteMsg.TimeoutMultiplier>
0

EDC Gateway Fault Write Message FaultWriteMsg.EN_CC <EQ0006_FaultWriteMsg.EN_CC> (U)

EDC Gateway Fault Write Message

MSG
Message Control FaultWriteMsg <EQ0006_FaultWriteMsg>
(EN) (DN) (ER)

Fault Data Ring Buffer and Message
Write Message EN Performance & Diagnostics

12

Fault Ring Buffer Message Control for multiplexed packets Write Command
FaultMsgCtl.Write

Fault Ring Buffer Message Control for multiplexed packets Write Command 1-Shot
FaultMsgCtl.WriteONS
ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxWrite.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	240
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

13

Write Message Done

Fault Ring Buffer Message Control for multiplexed packets Write Command
FaultMsgCtl.Write

EDC Gateway Fault Write Message
FaultWriteMsg.DN
<EQ0006_FaultWriteMsg.DN>

Fault Ring Buffer Message Control for multiplexed packets Write Command
FaultMsgCtl.Write
U

Fault Ring Buffer Message Control for multiplexed packets Write Command Done
FaultMsgCtl.WriteOK
L

Fault Ring Buffer Message Control for multiplexed packets Send Data
FaultMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Fault Tx DN
Status.Update.FaultTxOK
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Fault Keep Alive Tx DN
Status.Update.FaultKaTxOK
L

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxWrite.DN

Fault Ring Buffer Message Control for multiplexed packets Resend Data (Keep Alive)
FaultMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	FaultMsgCtl.Trans
PerfTrans	Status.FaultTxWrite.KA

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	241
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Write Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Fault Write Message FaultWriteMsg.EN <EQ0006_FaultWriteMsg.EN>

Fault Ring Buffer Message Control for multiplexed packets Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout 5000
Dest FaultMsgCtl.WriteTImeoutTMR.PRE 5000

Fault Ring Buffer Message Control for multiplexed packets Message Timeout Timer

TON
Timer FaultMsgCtl.WriteTImeoutTMR
Preset 5000
Accum 0
EN
DN

Fault Ring Buffer Message Control for multiplexed packets Message Timeout Timer FaultMsgCtl.WriteTImeoutTMR.DN

EDC Gateway Fault Write Message FaultWriteMsg.TO <EQ0006_FaultWriteMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 244
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Write Message ER Performance & Diagnostics

15

EDC Gateway Fault Write Message FaultWriteMsg.ER <EQ0006_FaultWriteMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Write Error 1-shot FaultMsgCtl.WriteErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Fault Tx ER Status.Update.FaultTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans FaultMsgCtl.Trans
PerfTrans Status.FaultTxWrite.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 243
Input Par FaultMsgCtl.Trans.TMR.ACC
Input Par FaultMsgCtl.TransNum

Fault Data Ring Buffer and Message

Read/Write Message Error Retry

16

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer
FaultMsgCtl.RetryTMR.DN

Fault Ring Buffer Message Control for multiplexed packets Retry 1-Shot
FaultMsgCtl.RetryONS
ONS

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer

MOVE	
Source	CFG.MsgRetryTime 5000
Dest	FaultMsgCtl.RetryTMR.PRE 5000

Fault Ring Buffer Message Control for multiplexed packets Message Retry Timer

TON		
Timer	FaultMsgCtl.RetryTMR	EN
Preset	5000	DN
Accum	0	

EDC Gateway Fault Read Message
FaultReadMsg.ER
<EQ0006_FaultReadMsg.ER>

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	232
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

EDC Gateway Fault Write Message
FaultWriteMsg.ER
<EQ0006_FaultWriteMsg.ER>

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	242
Input Par	FaultMsgCtl.Trans.TMR.ACC
Input Par	FaultMsgCtl.TransNum

(End)

Load Quality Data record to Ring Buffer

Equipment Quality/Health Data
Ring Buffer Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 132
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

EDC Ring Buffer Control Transaction Number

ADD
Source A QualityRBCtl.TransactionNumber 4157
Source B 1
Dest QualityRBCtl.TransactionNumber 4157

EDC Ring Buffer Control Transaction Number

LT
Source A QualityRBCtl.TransactionNumber 4157
Source B 1

EDC Ring Buffer Control Transaction Number

GT
Source A QualityRBCtl.TransactionNumber 4157
Source B CFG.MsgTransNumRollover 9999

EDC Ring Buffer Control Transaction Number

MOVE
Source 1
Dest QualityRBCtl.TransactionNumber 4157

Load Quality Data record to Ring Buffer

Load Message Header, Timestamp and Footer Data
v1.6 - Packet Type added (0=5 data, 1=10 data)
v1.16 DHF024 Allow Quality Targets to pass -1, -2 Commands to CmdNum using Spare1
v2.00 Remove Targets from CmdNum; Add PdM Triggers to CmdNum

3

Equipment Data Quality/Health Message Header - Transaction Number (must match Footer)

MOVE
Source QualityRBCtl.TransactionNumber 4157
Dest QualityMsg.Header.TransNum 4157

Equipment Data Quality/Health Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source UserCfg.EQID 6
Dest QualityMsg.Header.EQID 6

Equipment Data Quality/Health Message Header - PacketType (0=5 Data, 1=10 Data)

MOVE
Source 1
Dest QualityMsg.Header.PacketType 1

Equipment Data Quality/Health Message Header - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest QualityMsg.Header.Spare 0

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source 0
Dest QualityMsg.Header.CmdNum 0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source -11
Dest QualityMsg.Header.CmdNum 0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source -12
Dest QualityMsg.Header.CmdNum 0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.0

Equipment Data Quality/Health Message USER CmdNum bit 0 - Start bit 1 - Stop/Send
QualityMsg.Data.Spare1.1

Equipment Data Quality/Health Message Header - Command Number

MOVE
Source -13
Dest QualityMsg.Header.CmdNum 0

Equipment Data Quality/Health Message Transaction Time Stamp

COP
Source CLX.TransTime
Dest QualityMsg.TransTime
Length 1

Load Quality Data record to Ring Buffer

Equipment Data Quality/Health Message Footer - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest QualityMsg.Footer.Spare 0

Equipment Data Quality/Health Message Footer - Transaction Number (must match Footer)

MOVE
Source QualityRBCtl.TransactionNumber 4157
Dest QualityMsg.Footer.TransNum 4157

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

4

EDC Ring Buffer Control Next Write Record

LT
Source A QualityRBCtl.WritePtr 43
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A QualityRBCtl.WritePtr 43
Source B CFG.QualityRB.mi 99

EDC Ring Buffer Control Diagnostic - Manual Reset
QualityRBCtl.WritePtrErr
(L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source QualityRBCtl.WritePtr 43
Dest QualityRBCtl.WritePtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.WritePtr 43
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.ReadPtr 42
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 313
Input Par 0
Input Par QualityMsg.Header.TransNum

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest QualityRBCtl.WritePtr 43

Load Quality Data record to Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

5

EDC Ring Buffer Control Next Write Record

EQ
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadPtr 42

EDC Ring Buffer Control Buffer Full - Block Writes
QualityRBCtl.FullBlockWrite

EDC Ring Buffer Control Diagnostic - Manual Reset
QualityRBCtl.WriteBlockErr
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A QualityRBCtl.WriteBlockErrCnt 221
Source B 1
Dest QualityRBCtl.WriteBlockErrCnt 221

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.WritePtr 43
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.ReadPtr 42
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 314
Input Par 0
Input Par QualityMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TempMsgTrans
PerfTrans Status.QualityRx.ER

lbl_Exit
JMP

Load Quality Data record to Ring Buffer

RB Write -- Load Message into Ring Buffer

6

Equipment Data Quality/Health Message

COP	
Source	QualityMsg
Dest	QualityRB[QualityRBCtl.WritePtr]
Length	1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	QualityRBCtl.WritePtr 43
Dest	Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	QualityRBCtl.ReadPtr 42
Dest	Trans.Ctl.Opt[2] -1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	311
Input Par	0
Input Par	QualityMsg.Header.TransNum

Message Control Transaction Timing

FLL	
Source	0
Dest	TempMsgTrans
Length	1

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	TempMsgTrans
PerfTrans	Status.QualityRx.DN

RB Write -- Increment Write Pointer

7

EDC Ring Buffer Control Next Write Record

ADD	
Source A	QualityRBCtl.WritePtr 43
Source B	1
Dest	QualityRBCtl.WritePtr 43

EDC Ring Buffer Control Next Write Record

LT	
Source A	QualityRBCtl.WritePtr 43
Source B	0

EDC Ring Buffer Control Next Write Record

GT	
Source A	QualityRBCtl.WritePtr 43
Source B	CFG.QualityRB.mi 99

EDC Ring Buffer Control Next Write Record

MOVE	
Source	0
Dest	QualityRBCtl.WritePtr 43

Load Quality Data record to Ring Buffer

Exit Routine

8

lbl_Exit
LBL

NOP

(End)

Load Read Records from RingBuffer to Gateway Message

Equipment Quality/Health Data
Ring Buffer Unload to Gateway Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 133
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Load Read Records from RingBuffer to Gateway Message

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Next Read Record

LT
Source A QualityRBCtl.ReadPtr 42
Source B 0

EDC Ring Buffer Control Next Read Record

GT
Source A QualityRBCtl.ReadPtr 42
Source B CFG.QualityRB.mi 99

EDC Ring Buffer Control Diagnostic - Manual Reset

QualityRBCtl.ReadPtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source QualityRBCtl.ReadPtr 42
Dest QualityRBCtl.ReadPtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.WritePtr 43
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source QualityRBCtl.ReadPtr 42
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 323
Input Par 0
Input Par QualityRBCtl.TransactionNumber

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest QualityRBCtl.WritePtr 43

EDC Ring Buffer Control Next Read Record

MOVE
Source CFG.QualityRB.mi 99
Dest QualityRBCtl.ReadPtr 42

Load Read Records from RingBuffer to Gateway Message

Next Read Location
ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

ADD
Source A QualityRBCtl.ReadPtr 42
Source B 1
Dest QualityRBCtl.ReadNextPtr 43

EDC Ring Buffer Control Next Read Record

GT
Source A QualityRBCtl.ReadNextPtr 43
Source B CFG.QualityRB.mi 99

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest QualityRBCtl.ReadNextPtr 43

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Next Read Record

EQ
Source A QualityRBCtl.ReadPtr 42
Source B QualityRBCtl.WritePtr 43

EDC Ring Buffer Control Buffer Full
QualityRBCtl.Full

EDC Ring Buffer Control Read Buffer Record in Progress
QualityRBCtl.ReadInProgress
/

EDC Ring Buffer Control Next Read Record

EQ
Source A QualityRBCtl.ReadNextPtr 43
Source B QualityRBCtl.WritePtr 43

EDC Ring Buffer Control Buffer Empty
QualityRBCtl.Empty

Load Read Records from RingBuffer to Gateway Message

5

Buffer Count

EDC Ring Buffer Control Read Buffer Record in Progress
QualityRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
QualityRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GT
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadPtr 42

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest QualityRBCtl.RecordCount 0
Expression QualityRBCtl.WritePtr-QualityRBCtl.ReadPtr

EDC Ring Buffer Control Next Write Record

LT
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadPtr 42

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest QualityRBCtl.RecordCount 0
Expression CFG.QualityRB.n-(QualityRBCtl.ReadPtr-QualityRBCtl.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
QualityRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
QualityRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GE
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadNextPtr 43

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest QualityRBCtl.RecordCount 0
Expression QualityRBCtl.WritePtr-QualityRBCtl.ReadNextPtr

EDC Ring Buffer Control Next Write Record

LT
Source A QualityRBCtl.WritePtr 43
Source B QualityRBCtl.ReadNextPtr 43

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest QualityRBCtl.RecordCount 0
Expression CFG.QualityRB.n-(QualityRBCtl.ReadNextPtr-QualityRBCtl.WritePtr)

EDC Ring Buffer Control Ring Buffer Record Count

GT
Source A QualityRBCtl.RecordCount 0
Source B QualityRBCtl.RecordCountMax 99

EDC Ring Buffer Control Ring Buffer Record Count Max

MOVE
Source QualityRBCtl.RecordCount 0
Dest QualityRBCtl.RecordCountMax 99

Load Read Records from RingBuffer to Gateway Message

6

Read Next
Ready for next data to Gateway Message

EDC Ring Buffer Control Buffer Empty: QualityRBCtl.Empty [/]
EDC Ring Buffer Control Read Buffer Record in Progress: QualityRBCtl.ReadInProgress [/]
Ring Buffer Message Control Send Data: QualityMsgCtl.Send [/]
Ring Buffer Message Control Resend Data (Keep Alive): QualityMsgCtl.Resend [/]
EDC Ring Buffer Control Read Next Buffer Record 1-shot: QualityRBCtl.ReadNextONS [ONS]
EDC Ring Buffer Control Read Next Buffer Record: QualityRBCtl.ReadNext ( )
EDC Ring Buffer Control Read Buffer Record in Progress: QualityRBCtl.ReadInProgress (L)
EDC Ring Buffer Control Read Buffer Record Complete: QualityRBCtl.ReadComplete (U)

7

Read Next
Increment Read Pointer

EDC Ring Buffer Control Read Next Buffer Record: QualityRBCtl.ReadNext

EDC Ring Buffer Control Next Read Record — ADD
- Source A: QualityRBCtl.ReadPtr 42
- Source B: 1
- Dest: QualityRBCtl.ReadPtr 42

EDC Ring Buffer Control Next Read Record — LT
- Source A: QualityRBCtl.ReadPtr 42
- Source B: 0

EDC Ring Buffer Control Next Read Record — GT
- Source A: QualityRBCtl.ReadPtr 42
- Source B: CFG.QualityRB.mi 99

EDC Ring Buffer Control Next Read Record — MOVE
- Source: 0
- Dest: QualityRBCtl.ReadPtr 42

8

Read Next
Load Read Record

EDC Ring Buffer Control Read Next Buffer Record: QualityRBCtl.ReadNext

EDC Message Tx Data Quality/Health Receive Message — COP
- Source: QualityRB[QualityRBCtl.ReadPtr]
- Dest: EqTxData.Quality[0] <EQ0006_TxData.Quality[0]>
- Length: CFG.QualityMSG.e

Ring Buffer Message Control Transaction Number — MOVE
- Source: QualityRB[QualityRBCtl.ReadPtr].Header.TransNum 4157
- Dest: QualityMsgCtl.TransNum 4157

Transaction Log Control SBR External Load -- Optional Parameters — MOVE
- Source: QualityRBCtl.WritePtr 43
- Dest: Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters — MOVE
- Source: QualityRBCtl.ReadPtr 42
- Dest: Trans.Ctl.Opt[2] -1

Log Transactions — JSR
- Routine Name: _902_TransLogSBR
- Input Par: 321
- Input Par: 0
- Input Par: QualityMsgCtl.TransNum

Load Read Records from RingBuffer to Gateway Message

Read Next
Set Data Ready Handshake

9

EDC Ring Buffer Control Read Next Buffer Record
QualityRBCtl.ReadNext

Ring Buffer Message Control Send Data
QualityMsgCtl.Send (L)

Ring Buffer Message Control Read Command
QualityMsgCtl.Read (U)

Ring Buffer Message Control Read Buffer Ready
QualityMsgCtl.ReadOK (U)

Ring Buffer Message Control Write Command
QualityMsgCtl.Write (U)

Ring Buffer Message Control Write Command Done
QualityMsgCtl.WriteOK (U)

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

10

Ring Buffer Message Control Send Data
QualityMsgCtl.Send

Ring Buffer Message Control Write Command Done
QualityMsgCtl.WriteOK

Ring Buffer Message Control Resend Data (Keep Alive)
QualityMsgCtl.Resend

Ring Buffer Message Control Send Data
QualityMsgCtl.Send (U)

Ring Buffer Message Control Resend Data (Keep Alive)
QualityMsgCtl.Resend (U)

EDC Ring Buffer Control Read Buffer Record Complete
QualityRBCtl.ReadComplete (L)

Read Complete Delay
Allow Gateway processing time before attempting next record send.

11

EDC Ring Buffer Control Read Buffer Record Complete
QualityRBCtl.ReadComplete

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source 100
Dest QualityRBCtl.ReadCompleteDelayTMR.PRE
100

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer QualityRBCtl.ReadCompleteDelayTMR
Preset 100
Accum 113
EN
DN

EDC Ring Buffer Control Read Complete Delay Timer
QualityRBCtl.ReadCompleteDelayTMR.DN

EDC Ring Buffer Control Read Buffer Record Complete Delay
QualityRBCtl.ReadCompleteDelay

EDC Ring Buffer Control Read Buffer Record in Progress
QualityRBCtl.ReadInProgress (U)

Load Read Records from RingBuffer to Gateway Message

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

12

EDC Ring Buffer Control Buffer Empty
QualityRBCtl.Empty
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest QualityRBCtl.NotEmptyTMR.PRE 300000

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer QualityRBCtl.NotEmptyTMR
Preset 300000
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A QualityRBCtl.NotEmptyTMR.ACC 0
Source B QualityRBCtl.NotEmptyMax 300001

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source QualityRBCtl.NotEmptyTMR.ACC 0
Dest QualityRBCtl.NotEmptyMax 300001

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

13

EDC Ring Buffer Control Buffer Full
QualityRBCtl.Full

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
QualityRBCtl.NotEmptyTMR.DN

EDC Ring Buffer Control Read Buffer Alarm
QualityRBCtl.BufferAlarm

(End)

Quality Data Ring Buffer and Message

Equipment Quality Data
Buffered Fault Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 134
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Ring Buffer Message Control Message Transaction Timer

MOVE
Source 999999999
Dest QualityMsgCtl.Trans.TMR.PRE 999999999

Ring Buffer Message Control Message Transaction Timer

TON
Timer QualityMsgCtl.Trans.TMR
Preset 999999999
Accum 287979
EN
DN

Quality Data Ring Buffer and Message

Read Message Delay Timer
If driver sends data too fast, it will generate extra Read Message traffic until previous message has been processed
Optimization uses the ReadOk and ReadNotOk counts to tune the Delay Time to the Gateway response

3

EDC Gateway Quality Write Message
QualityWriteMsg.EN
<EQ0006_QualityWriteMsg.EN>

EDC Gateway Quality Read Message
QualityReadMsg.EN
<EQ0006_QualityReadMsg.EN>

Ring Buffer Message Control Read Message Ok Count

GE		
Source A	QualityMsgCtl.ReadOkCount	3
Source B	CfgRB_ReadOkLimit <CfgRB.ReadOkLimit>	5

Ring Buffer Message Control Message Delay Timer

SUB		
Source A	QualityMsgCtl.DelayTMR.PRE	100
Source B	CfgRB_ReadOkSub <CfgRB.ReadOkSub>	10
Dest	QualityMsgCtl.DelayTMR.PRE	100

Ring Buffer Message Control Message Delay Timer

LE		
Source A	QualityMsgCtl.DelayTMR.PRE	100
Source B	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin>	100

Ring Buffer Message Control Message Delay Timer

MOVE		
Source	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin>	100
Dest	QualityMsgCtl.DelayTMR.PRE	100

Ring Buffer Message Control Read Message Not Ok Count

GE		
Source A	QualityMsgCtl.ReadNotOkCount	0
Source B	CfgRB_ReadNotOkLimit <CfgRB.ReadNotOkLimit>	2

Ring Buffer Message Control Message Delay Timer

ADD		
Source A	QualityMsgCtl.DelayTMR.PRE	100
Source B	CfgRB_ReadNotOkAdd <CfgRB.ReadNotOkAdd>	20
Dest	QualityMsgCtl.DelayTMR.PRE	100

Ring Buffer Message Control Message Delay Timer

GT		
Source A	QualityMsgCtl.DelayTMR.PRE	100
Source B	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax>	500

Ring Buffer Message Control Message Delay Timer

MOVE		
Source	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax>	500
Dest	QualityMsgCtl.DelayTMR.PRE	100

Ring Buffer Message Control Read Message Ok Count

CLR		
Dest	QualityMsgCtl.ReadOkCount	3

Ring Buffer Message Control Read Message Not Ok Count

CLR		
Dest	QualityMsgCtl.ReadNotOkCount	0

Ring Buffer Message Control Message Delay Timer

TON			
Timer	QualityMsgCtl.DelayTMR		EN
Preset		100	DN
Accum		101	

Quality Data Ring Buffer and Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

Ring Buffer Message Control Send Data QualityMsgCtl.Send

EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>

EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>

Ring Buffer Message Control Message Resend Timer

MOVE
Source CFG.BuffMsgResendTime 301000
Dest QualityMsgCtl.ResendTMR.PRE 301000

Ring Buffer Message Control Message Resend Timer

TON
Timer QualityMsgCtl.ResendTMR
Preset 301000
Accum 287951
EN
DN

Ring Buffer Message Control Message Resend Timer QualityMsgCtl.ResendTMR.DN

Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend (L)

Ring Buffer Message Control Message Resend Timer QualityMsgCtl.ResendTMR (RES)

Ring Buffer Message Control Read Command QualityMsgCtl.Read (U)

Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK (U)

Ring Buffer Message Control Write Command QualityMsgCtl.Write (U)

Ring Buffer Message Control Write Command Done QualityMsgCtl.WriteOK (U)

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 322
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Send Data Initialize

5

Ring Buffer Message Control Send Data QualityMsgCtl.Send

Ring Buffer Message Control Send Data 1-Shot QualityMsgCtl.SendONS (ONS)

Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend

Ring Buffer Message Control Message Transaction Timer QualityMsgCtl.Trans.TMR (RES)

Ring Buffer Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest QualityMsgCtl.Trans.Last 13

Quality Data Ring Buffer and Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

6

Ring Buffer Message Control Send Data QualityMsgCtl.Send
Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK
Ring Buffer Message Control Message Delay Timer QualityMsgCtl.DelayTMR.DN
EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>
EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>
EDC Gateway Quality Read Message QualityReadMsg.ER <EQ0006_QualityReadMsg.ER>
Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend
Ring Buffer Message Control Message Retry Timer QualityMsgCtl.RetryTMR.DN

Ring Buffer Message Control Read Command QualityMsgCtl.Read (L)
Ring Buffer Message Control Read Command 1-Shot QualityMsgCtl.ReadONS

EDC Gateway Quality Read Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest QualityReadMsg.Path <EQ0006_QualityReadMsg.Path>
Length 1

EDC Gateway Quality Read Message

COP
Source CFG.MsgDest.QualityDest
Dest QualityReadMsg.RemoteElement <EQ0006_QualityReadMsg.RemoteElement>
Length 1

EDC Gateway Quality Read Message

MOVE
Source CFG.QualityMSG.e 39
Dest QualityReadMsg.REQ_LEN <EQ0006_QualityReadMsg.REQ_LEN> 39

EDC Gateway Quality Read Message

MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest QualityReadMsg.UnconnectedTimeout <EQ0006_QualityReadMsg.UnconnectedTimeout> 4000000

EDC Gateway Quality Read Message

MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest QualityReadMsg.ConnectionRate <EQ0006_QualityReadMsg.ConnectionRate> 1000000

EDC Gateway Quality Read Message

MOVE
Source CFG.CLx.ConnectionMult 0
Dest QualityReadMsg.TimeoutMultiplier <EQ0006_QualityReadMsg.TimeoutMultiplier> 0

EDC Gateway Quality Read Message QualityReadMsg.EN_CC <EQ0006_QualityReadMsg.EN_CC> (U)

EDC Gateway Quality Read Message

MSG
Message Control QualityReadMsg <EQ0006_QualityReadMsg> ...
EN
DN
ER

Quality Data Ring Buffer and Message

Read Message EN Performance & Diagnostics

7

Ring Buffer Message Control Read Command 1-Shot

QualityMsgCtl.ReadONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans [...]
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	330
Input Par	QualityMsgCtl.Trans.TMR.ACC
Input Par	QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

Read Message OK - Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record)

Increment ReadOk and ReadNotOk counts
only increment ReadOk count when multiple records in Buffer (single records are always ready to send and don't test Gateway servicing speed)

8

Ring Buffer Message Control Read Command QualityMsgCtl.Read

EDC Gateway Quality Read Message QualityReadMsg.DN <EQ0006_QualityReadMsg.DN>

Ring Buffer Message Control Read Command QualityMsgCtl.Read (U)

EDC Message Rx Data Quality/Health Receive Message
EQ
Source A EqRxData.Quality[0] <EQ0006_RxData.Quality[0]> 0
Source B 0

EDC Message Rx Data Quality/Health Receive Message
EQ
Source A EqRxData.Quality[CFG.QualityMSG.ei] <EQ0006_RxData.Quality[CFG.QualityMSG.ei]> 0
Source B 0

Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK (L)

EDC Ring Buffer Control Ring Buffer Record Count
GE
Source A QualityRBCtl.RecordCount 0
Source B 2

Ring Buffer Message Control Read Message Ok Count
ADD
Source A QualityMsgCtl.ReadOkCount 3
Source B 1
Dest QualityMsgCtl.ReadOkCount 3

Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK (/)

Ring Buffer Message Control Read Message Not Ok Count
ADD
Source A QualityMsgCtl.ReadNotOkCount 0
Source B 1
Dest QualityMsgCtl.ReadNotOkCount 0

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans QualityMsgCtl.Trans
PerfTrans Status.QualityTxRead.DN

Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans QualityMsgCtl.Trans
PerfTrans Status.QualityTxRead.KA

Transaction Log Control SBR External Load -- Optional Parameters

Transaction Log Control SBR External Load -- Optional Parameters

Log Transactions

Quality Data Ring Buffer and Message

MOVE
Source EqRxData.Quality[0]
<EQ0006_RxData.Quality[0]>
0
Dest Trans.Ctl.Opt[1]
-1

MOVE
Source EqRxData.Quality[CFG.QualityMSG.ei]
<EQ0006_RxData.Quality[CFG.QualityMSG.ei]>
0
Dest Trans.Ctl.Opt[2]
-1

JSR
Routine Name _902_TransLogSBR
Input Par 331
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Quality
Read Message
QualityReadMsg.EN
<EQ0006_QualityReadMsg.EN>

Ring Buffer Message
Control Message
Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest QualityMsgCtl.ReadTimeoutTMR.PRE
5000

Ring Buffer Message
Control Message
Timeout Timer

TON
Timer QualityMsgCtl.ReadTimeoutTMR
Preset 5000
Accum 0
EN
DN

Ring Buffer Message
Control Message
Timeout Timer
QualityMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Quality
Read Message
QualityReadMsg.TO
<EQ0006_QualityReadMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 334
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

Read Message ER Performance & Diagnostics

10

EDC Gateway Quality Read Message
QualityReadMsg.ER
<EQ0006_QualityReadMsg.ER>

Ring Buffer Message Control Read Error 1-shot
QualityMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Quality Tx ER
Status.Update.QualityTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxRead.ER

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	333
Input Par	QualityMsgCtl.Trans.TMR.ACC
Input Par	QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

Write Message
Write Data is loaded by Ring Buffer routine

11

Ring Buffer Message Control Send Data QualityMsgCtl.Send

Ring Buffer Message Control Read Buffer Ready QualityMsgCtl.ReadOK

Ring Buffer Message Control Write Command Done QualityMsgCtl.WriteOK

EDC Gateway Quality Read Message QualityReadMsg.EN <EQ0006_QualityReadMsg.EN>

EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>

EDC Gateway Quality Write Message QualityWriteMsg.ER <EQ0006_QualityWriteMsg.ER>

Ring Buffer Message Control Resend Data (Keep Alive) QualityMsgCtl.Resend

Ring Buffer Message Control Message Retry Timer QualityMsgCtl.RetryTMR.DN

Ring Buffer Message Control Write Command QualityMsgCtl.Write (L)

EDC Gateway Quality Write Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest QualityWriteMsg.Path <EQ0006_QualityWriteMsg.Path>
Length 1

EDC Gateway Quality Write Message

COP
Source CFG.MsgDest.QualityDest
Dest QualityWriteMsg.RemoteElement <EQ0006_QualityWriteMsg.RemoteElement>
Length 1

EDC Gateway Quality Write Message

MOVE
Source CFG.QualityMSG.e 39
Dest QualityWriteMsg.REQ_LEN <EQ0006_QualityWriteMsg.REQ_LEN> 39

EDC Gateway Quality Write Message

MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest QualityWriteMsg.UnconnectedTimeout <EQ0006_QualityWriteMsg.UnconnectedTimeout> 4000000

EDC Gateway Quality Write Message

MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest QualityWriteMsg.ConnectionRate <EQ0006_QualityWriteMsg.ConnectionRate> 1000000

EDC Gateway Quality Write Message

MOVE
Source CFG.CLx.ConnectionMult 0
Dest QualityWriteMsg.TimeoutMultiplier <EQ0006_QualityWriteMsg.TimeoutMultiplier> 0

EDC Gateway Quality Write Message QualityWriteMsg.EN_CC <EQ0006_QualityWriteMsg.EN_CC> (U)

EDC Gateway Quality Write Message

MSG
Message Control QualityWriteMsg ... <EQ0006_QualityWriteMsg>
EN
DN
ER

Quality Data Ring Buffer and Message

Write Message EN Performance & Diagnostics

12

Ring Buffer Message Control Write Command
QualityMsgCtl.Write

Ring Buffer Message Control Write Command 1-Shot
QualityMsgCtl.WriteONS
ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxWrite.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	340
Input Par	QualityMsgCtl.Trans.TMR.ACC
Input Par	QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

13

Write Message Done

Ring Buffer Message Control Write Command
QualityMsgCtl.Write

EDC Gateway Quality Write Message
QualityWriteMsg.DN
<EQ0006_QualityWriteMsg.DN>

Ring Buffer Message Control Write Command
QualityMsgCtl.Write
U

Ring Buffer Message Control Write Command Done
QualityMsgCtl.WriteOK
L

Ring Buffer Message Control Send Data
QualityMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Quality Tx DN
Status.Update.QualityTxOK
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate ...
Status	Status

Ring Buffer Message Control Resend Data (Keep Alive)
QualityMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Quality Keep Alive Tx DN
Status.Update.QualityKaTxOK
L

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxWrite.DN

Ring Buffer Message Control Resend Data (Keep Alive)
QualityMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	QualityMsgCtl.Trans
PerfTrans	Status.QualityTxWrite.KA

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	341
Input Par	QualityMsgCtl.Trans.TMR.ACC
Input Par	QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message

Write Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Quality Write Message QualityWriteMsg.EN <EQ0006_QualityWriteMsg.EN>

Ring Buffer Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest QualityMsgCtl.WriteTImeoutTMR.PRE
5000

Ring Buffer Message Control Message Timeout Timer

TON
Timer QualityMsgCtl.WriteTImeoutTMR
Preset 5000
Accum 0
EN
DN

Ring Buffer Message Control Message Timeout Timer QualityMsgCtl.WriteTImeoutTMR.DN

EDC Gateway Quality Write Message QualityWriteMsg.TO <EQ0006_QualityWriteMsg.TO>

L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 344
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Write Message ER Performance & Diagnostics

15

EDC Gateway Quality Write Message QualityWriteMsg.ER <EQ0006_QualityWriteMsg.ER>

Ring Buffer Message Control Write Error 1-shot QualityMsgCtl.WriteErONS

ONS

Equipment Data Collection - Line Equipment Status State = Red - Quality Tx ER Status.Update.QualityTxER

L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans QualityMsgCtl.Trans
PerfTrans Status.QualityTxWrite.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 343
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

Quality Data Ring Buffer and Message
Read/Write Message Error Retry

16

EDC Gateway Quality
Read Message
QualityReadMsg.ER
<EQ0006_QualityReadMsg.ER>

EDC Gateway Quality
Write Message
QualityWriteMsg.ER
<EQ0006_QualityWriteMsg.ER>

Ring Buffer Message
Control Message
Retry Timer
QualityMsgCtl.RetryTMR.DN

Ring Buffer Message
Control Retry 1-Shot
QualityMsgCtl.RetryONS
ONS

Ring Buffer Message
Control Message
Retry Timer
MOVE
Source CFG.MsgRetryTime
5000
Dest QualityMsgCtl.RetryTMR.PRE
0

Ring Buffer Message
Control Message
Retry Timer
TON
Timer QualityMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Quality
Read Message
QualityReadMsg.ER
<EQ0006_QualityReadMsg.ER>

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 332
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

EDC Gateway Quality
Write Message
QualityWriteMsg.ER
<EQ0006_QualityWriteMsg.ER>

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 342
Input Par QualityMsgCtl.Trans.TMR.ACC
Input Par QualityMsgCtl.TransNum

(End)

Collect Trend Data

Equipment Trend Data
Collect Data

0

Trend Index

SBR	
Input Par	TRi

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	141
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Exit Routine if Index is out of range

2

Trend Index

LT	
Source A	TRi 4
Source B	0

Trend Index

GT	
Source A	TRi 4
Source B	CFG.Trend.mi 20

lbl_Exit
JMP

Copy Trend Arrays to local tags

3

Trend Collect Control

COP	
Source	TrendCfg[TRi].CTL
Dest	TrendCollectCtl
Length	1

Equipment Data Trend Message Data

COP	
Source	TrendCfg[TRi].Data
Dest	TrendCollect
Length	1

Collect Trend Data

Trend Initialize
Load user Sample Rate if within limits or default to 1 second
(Rate = Samples/Second)

4

Trend Collect Control Trend Start Sampling
TrendCollectCtl.Start

Trend Collect Control Trend Start 1-Shot
TrendCollectCtl.StartONS
ONS

Trend Collect Control Trend Init Sampling
TrendCollectCtl.Init

Trend Collect Control Trend Done Sampling
TrendCollectCtl.Done
U

Trend Collect Control Trend Save
TrendCollectCtl.Loaded
U

Equipment Data Trend Message Data
FLL
Source 0
Dest TrendCollect
Length 1

Trend Collect Control Trend Sample Rate Actual
MOVE
Source 1
Dest TrendCollectCtl.RateAct
0

Trend Collect Control Trend Sample Rate Requested (1-10 tenths of second)
LIMIT
Low Limit 1
Test TrendCollectCtl.Rate
0
High Limit 10

Trend Collect Control Trend Sample Rate Actual
MOVE
Source TrendCollectCtl.Rate
0
Dest TrendCollectCtl.RateAct
0

Trend Collect Control Trend Sample Time Preset
DIV
Source A 1000
Source B TrendCollectCtl.RateAct
0
Dest TrendCollectCtl.SampleTimePre
0

Trend Collect Control Trend Sample Last Time
CLR
Dest TrendCollectCtl.SampleTimeLast
0

Trend Collect Control Trend Array Index
CLR
Dest TrendCollectCtl.ARRi
0

Collect Trend Data

Trend Init
JobStatus and FaultCode are updates on Start and Send

5

Trend Collect
Control Trend Init
Sampling
TrendCollectCtl.Init

Equipment Data
Trend Message Data
Tool Number

MOVE	
Source	TRi 4
Dest	TrendCollect.ToolNum 0

Equipment Data
Trend Message Data
Data Values Used
(1-120)

MOVE	
Source	CFG.TrData.n 120
Dest	TrendCollect.DataCount 0

Equipment Data
Trend Message Data
Trend Rate - samples
per second (1-10)

MOVE	
Source	TrendCollectCtl.RateAct 0
Dest	TrendCollect.Rate 0

Equipment Data
Trend Message Data
Trend Scale

MOVE	
Source	TrendCollectCtl.Scale 0
Dest	TrendCollect.Scale 0

Equipment Data
Trend Message Data
Job Status

MOVE	
Source	TrendMsg.Data.JobStatus 0
Dest	TrendCollect.JobStatus 0

Equipment Data
Trend Message Data
Fault Code

MOVE	
Source	TrendMsg.Data.FaultCode 0
Dest	TrendCollect.FaultCode 0

Collect Trend Data

Trend Vehicle Data

6

Trend Collect
Control Trend Load
Vehicle Data
TrendCollectCtl.LoadVeh

Equipment Data
Trend Message Data
Program Number from
SLCP

MOVE
Source TrendMsg.Data.AviPrgNum 0
Dest TrendCollect.AviPrgNum 0

Equipment Data
Trend Message Data
Program Number
Executed

MOVE
Source TrendMsg.Data.ActPrgNum 0
Dest TrendCollect.ActPrgNum 0

Equipment Data
Trend Message Data
Assembly Body Number

MOVE
Source TrendMsg.Data.BodyNum 0
Dest TrendCollect.BodyNum 0

Equipment Data
Trend Message Data
Assembly Sequence
Number

MOVE
Source TrendMsg.Data.AssyNum 0
Dest TrendCollect.AssyNum 0

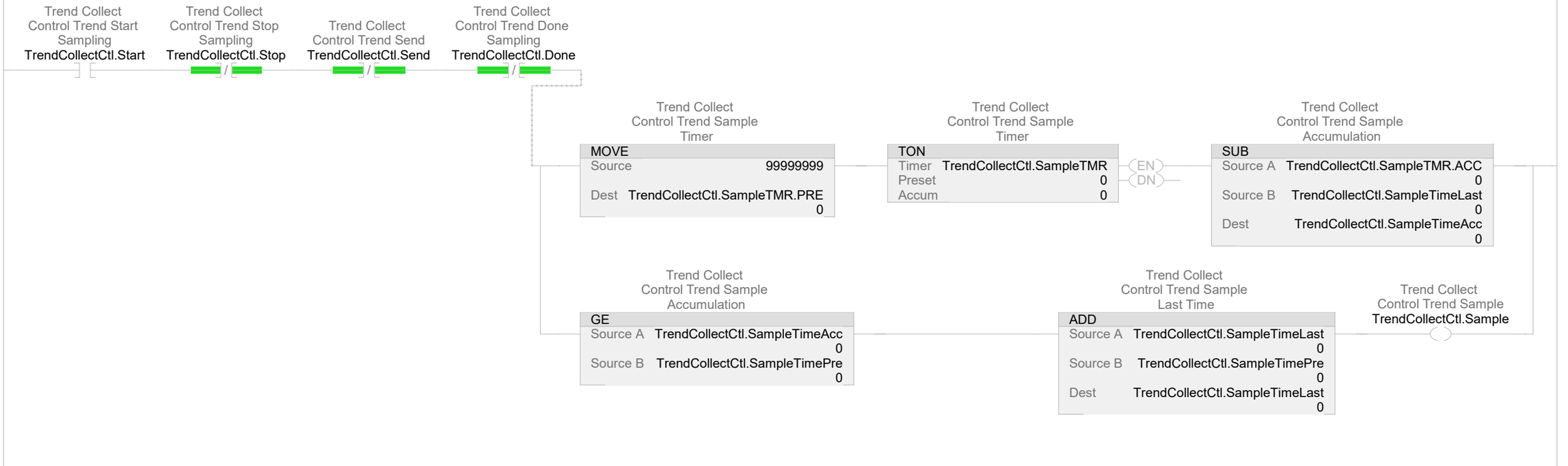
Equipment Data
Trend Message Data
Pdm Triggers: Start
(-11) and Send (-12)

MOVE
Source TrendMsg.Data.PdmTrigger 0
Dest TrendCollect.PdmTrigger 0

Collect Trend Data

Trend Sample TImer
Use free-running timer and samle interval to avoid accumulating Timer overruns throughout test.

7



Collect Trend Data

Trend Sample

8

Trend Collect
Control Trend Init
Sampling
TrendCollectCtl.Init

Trend Collect
Control Trend Sample
TrendCollectCtl.Sample

Equipment Data
Trend Message Data
Trend Data

MOVE
Source TrendCollectCtl.Data
0
Dest TrendCollect.Trend[TrendCollectCtl.ARRi]
0

Trend Collect
Control Trend Array
Index

ADD
Source A TrendCollectCtl.ARRi
0
Source B 1
Dest TrendCollectCtl.ARRi
0

Trend Collect
Control Trend Array
Index

LT
Source A TrendCollectCtl.ARRi
0
Source B 0

Trend Collect
Control Trend Array
Index

GT
Source A TrendCollectCtl.ARRi
0
Source B CFG.TrData.mi
119

Trend Collect
Control Trend Done
Sampling
TrendCollectCtl.Done
L

Collect Trend Data

Trend Samples Done
Update Status/Fault and Load RIng Buffer

9

Trend Collect Control Trend Send TrendCollectCtl.Send

Trend Collect Control Trend Save TrendCollectCtl.Loaded

Trend Collect Control Trend Send TrendCollectCtl.Send

Equipment Data Trend Message Data Job Status

MOVE
Source TrendMsg.Data.JobStatus 0
Dest TrendCollect.JobStatus 0

Equipment Data Trend Message Data Fault Code

MOVE
Source TrendMsg.Data.FaultCode 0
Dest TrendCollect.FaultCode 0

Equipment Data Trend Message Data Data Values Used (1-120)

MOVE
Source TrendCollectCtl.ARRi 0
Dest TrendCollect.DataCount 0

Load Trend Data record to Ring Buffer

JSR
Routine Name _142_TrendDataLoadRB

Trend Collect Control Trend Save TrendCollectCtl.Loaded (L)

Trend Collect Control Trend Start Sampling TrendCollectCtl.Start (U)

Trend Collect Control Trend Send TrendCollectCtl.Send (U)

Copy local tags to Trend Array

10

Trend Data Collection Trend Control

COP
Source TrendCollectCtl
Dest TrendCfg[TRi].CTL
Length 1

Trend Data Collection Trend Data

COP
Source TrendCollect
Dest TrendCfg[TRi].Data
Length 1

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Load Trend Data record to Ring Buffer

Equipment Trend Data
Ring Buffer Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 142
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

EDC Ring Buffer Control Transaction Number

ADD
Source A TrendRBCtl.TransactionNumber 0
Source B 1
Dest TrendRBCtl.TransactionNumber 0

EDC Ring Buffer Control Transaction Number

LT
Source A TrendRBCtl.TransactionNumber 0
Source B 1

EDC Ring Buffer Control Transaction Number

GT
Source A TrendRBCtl.TransactionNumber 0
Source B CFG.MsgTransNumRollover 9999

EDC Ring Buffer Control Transaction Number

MOVE
Source 1
Dest TrendRBCtl.TransactionNumber 0

Load Trend Data record to Ring Buffer

Load Message Header, Timestamp, Trend Data and Footer Data

3

Equipment Data Trend Message Header - Transaction Number (must match Footer)

MOVE	
Source	TrendRBCtl.TransactionNumber 0
Dest	TrendMsg.Header.TransNum 0

Equipment Data Trend Message Header - Command Number

MOVE	
Source	0
Dest	TrendMsg.Header.CmdNum 0

Equipment Data Trend Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE	
Source	UserCfg.EQID 6
Dest	TrendMsg.Header.EQID 0

Equipment Data Trend Message Header - Spare (fills unused portion of DINT)

MOVE	
Source	0
Dest	TrendMsg.Header.Spare 0

Equipment Data Trend Message Transaction Time Stamp

COP	
Source	CLX.TransTime
Dest	TrendMsg.TransTime
Length	1

Equipment Data Trend Message Data - Trend

COP	
Source	TrendCollect
Dest	TrendMsg.Data
Length	1

Equipment Data Trend Message Footer - Spare (fills unused portion of DINT)

MOVE	
Source	0
Dest	TrendMsg.Footer.Spare 0

Equipment Data Trend Message Footer - Transaction Number (must match Footer)

MOVE	
Source	TrendRBCtl.TransactionNumber 0
Dest	TrendMsg.Footer.TransNum 0

Load Trend Data record to Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

4

EDC Ring Buffer Control Next Write Record

LT
Source A: TrendRBCtl.WritePtr 0
Source B: 0

EDC Ring Buffer Control Next Write Record

GT
Source A: TrendRBCtl.WritePtr 0
Source B: CFG.TrendRB.mi 49

EDC Ring Buffer Control Diagnostic - Manual Reset

TrendRBCtl.WritePtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source: TrendRBCtl.WritePtr 0
Dest: TrendRBCtl.WritePtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: TrendRBCtl.WritePtr 0
Dest: Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: TrendRBCtl.ReadPtr 49
Dest: Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name: _902_TransLogSBR
Input Par: 413
Input Par: 0
Input Par: TrendMsg.Header.TransNum

EDC Ring Buffer Control Next Write Record

MOVE
Source: 0
Dest: TrendRBCtl.WritePtr 0

Load Trend Data record to Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

5

EDC Ring Buffer Control Next Write Record

EQ	
Source A	TrendRBCtl.WritePtr 0
Source B	TrendRBCtl.ReadPtr 49

EDC Ring Buffer Control Buffer Full - Block Writes
TrendRBCtl.FullBlockWrite

EDC Ring Buffer Control Diagnostic - Manual Reset
TrendRBCtl.WriteBlockErr
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD	
Source A	TrendRBCtl.WriteBlockErrCnt 0
Source B	1
Dest	TrendRBCtl.WriteBlockErrCnt 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	TrendRBCtl.WritePtr 0
Dest	Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE	
Source	TrendRBCtl.ReadPtr 49
Dest	Trans.Ctl.Opt[2] -1

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	414
Input Par	0
Input Par	TrendMsg.Header.TransNum

Message Control Transaction Timing

FLL	
Source	0
Dest	TempMsgTrans
Length	1

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	TempMsgTrans
PerfTrans	Status.TrendRx.ER

lbl_Exit
JMP

Load Trend Data record to Ring Buffer

RB Write -- Load Message into Ring Buffer

6

Equipment Data Trend Message

COP
Source TrendMsg
Dest TrendRB[TrendRBCtl.WritePtr]
Length 1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendRBCtl.WritePtr
0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendRBCtl.ReadPtr
49
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 411
Input Par 0
Input Par TrendMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TempMsgTrans
PerfTrans Status.TrendRx.DN

RB Write -- Increment Write Pointer

7

EDC Ring Buffer Control Next Write Record

ADD
Source A TrendRBCtl.WritePtr
0
Source B 1
Dest TrendRBCtl.WritePtr
0

EDC Ring Buffer Control Next Write Record

LT
Source A TrendRBCtl.WritePtr
0
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A TrendRBCtl.WritePtr
0
Source B CFG.TrendRB.mi
49

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest TrendRBCtl.WritePtr
0

Exit Routine

8

lbl_Exit
LBL

NOP

Load Trend Data record to Ring Buffer

(End)

Load Read Records from RingBuffer to Gateway Message

Equipment Trend Data
Ring Buffer Unload to Gateway Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 143
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Load Read Records from RingBuffer to Gateway Message

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Next Read Record

LT
Source A TrendRBCtl.ReadPtr 49
Source B 0

EDC Ring Buffer Control Next Read Record

GT
Source A TrendRBCtl.ReadPtr 49
Source B CFG.TrendRB.mi 49

EDC Ring Buffer Control Diagnostic - Manual Reset

TrendRBCtl.ReadPtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source TrendRBCtl.ReadPtr 49
Dest TrendRBCtl.ReadPtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendRBCtl.WritePtr 0
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source TrendRBCtl.ReadPtr 49
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 423
Input Par 0
Input Par TrendRBCtl.TransactionNumber

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest TrendRBCtl.WritePtr 0

EDC Ring Buffer Control Next Read Record

MOVE
Source CFG.TrendRB.mi 49
Dest TrendRBCtl.ReadPtr 49

Load Read Records from RingBuffer to Gateway Message

Next Read Location
ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

ADD
Source A TrendRBCtl.ReadPtr 49
Source B 1
Dest TrendRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Next Read Record

GT
Source A TrendRBCtl.ReadNextPtr 0
Source B CFG.TrendRB.mi 49

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest TrendRBCtl.ReadNextPtr 0

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Next Read Record

EQ
Source A TrendRBCtl.ReadPtr 49
Source B TrendRBCtl.WritePtr 0

EDC Ring Buffer Control Buffer Full
TrendRBCtl.Full

EDC Ring Buffer Control Read Buffer Record in Progress
TrendRBCtl.ReadInProgress
/

EDC Ring Buffer Control Next Read Record

EQ
Source A TrendRBCtl.ReadNextPtr 0
Source B TrendRBCtl.WritePtr 0

EDC Ring Buffer Control Buffer Empty
TrendRBCtl.Empty

Load Read Records from RingBuffer to Gateway Message

5

Buffer Count

EDC Ring Buffer Control Read Buffer Record in Progress: TrendRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete: TrendRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GT
Source A: TrendRBCtl.WritePtr 0
Source B: TrendRBCtl.ReadPtr 49

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest: TrendRBCtl.RecordCount 0
Expression: TrendRBCtl.WritePtr-TrendRBCtl.ReadPtr

EDC Ring Buffer Control Next Write Record

LT
Source A: TrendRBCtl.WritePtr 0
Source B: TrendRBCtl.ReadPtr 49

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest: TrendRBCtl.RecordCount 0
Expression: CFG.TrendRB.n-(TrendRBCtl.ReadPtr-TrendRBCtl.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress: TrendRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete: TrendRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record

GE
Source A: TrendRBCtl.WritePtr 0
Source B: TrendRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest: TrendRBCtl.RecordCount 0
Expression: TrendRBCtl.WritePtr-TrendRBCtl.ReadNextPtr

EDC Ring Buffer Control Next Write Record

LT
Source A: TrendRBCtl.WritePtr 0
Source B: TrendRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Ring Buffer Record Count

CPT
Dest: TrendRBCtl.RecordCount 0
Expression: CFG.TrendRB.n-(TrendRBCtl.ReadNextPtr-TrendRBCtl.WritePtr)

EDC Ring Buffer Control Ring Buffer Record Count

GT
Source A: TrendRBCtl.RecordCount 0
Source B: TrendRBCtl.RecordCountMax 0

EDC Ring Buffer Control Ring Buffer Record Count Max

MOVE
Source: TrendRBCtl.RecordCount 0
Dest: TrendRBCtl.RecordCountMax 0

Load Read Records from RingBuffer to Gateway Message

Read Next
Ready for next data to Gateway Message

6

EDC Ring Buffer Control Buffer Empty	EDC Ring Buffer Control Read Buffer Record in Progress	EDC Gateway Trend Message Control Send Data	EDC Gateway Trend Message Control Resend Data (Keep Alive)	EDC Ring Buffer Control Read Next Buffer Record 1-shot	EDC Ring Buffer Control Read Next Buffer Record	EDC Ring Buffer Control Read Buffer Record in Progress	EDC Ring Buffer Control Read Buffer Record Complete
TrendRBCtl.Empty	TrendRBCtl.ReadInProgress	TrendMsgCtl.Send	TrendMsgCtl.Resend	TrendRBCtl.ReadNextONS	TrendRBCtl.ReadNext	TrendRBCtl.ReadInProgress	TrendRBCtl.ReadComplete
/	/	/	/	ONS		L	U

Read Next
Increment Read Pointer

7

EDC Ring Buffer Control Read Next Buffer Record
TrendRBCtl.ReadNext

EDC Ring Buffer Control Next Read Record
ADD
Source A TrendRBCtl.ReadPtr 49
Source B 1
Dest TrendRBCtl.ReadPtr 49

EDC Ring Buffer Control Next Read Record
LT
Source A TrendRBCtl.ReadPtr 49
Source B 0

EDC Ring Buffer Control Next Read Record
GT
Source A TrendRBCtl.ReadPtr 49
Source B CFG.TrendRB.mi 49

EDC Ring Buffer Control Next Read Record
MOVE
Source 0
Dest TrendRBCtl.ReadPtr 49

Load Read Records from RingBuffer to Gateway Message

Read Next
Load Read Record

8

EDC Ring Buffer Control Read Next Buffer Record
TrendRBCtl.ReadNext

EDC Message Tx Data Trend Receive Message
COP
Source TrendRB[TrendRBCtl.ReadPtr]
Dest EqTxData.Trend[0]
<EQ0006_TxData.Trend[0]>
Length CFG.TrendMSG.e

EDC Gateway Trend Message Control Transaction Number
MOVE
Source TrendRB[TrendRBCtl.ReadPtr].Header.TransNum
0
Dest TrendMsgCtl.TransNum
0

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source TrendRBCtl.WritePtr
0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters
MOVE
Source TrendRBCtl.ReadPtr
49
Dest Trans.Ctl.Opt[2]
-1

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 421
Input Par 0
Input Par TrendMsgCtl.TransNum

Read Next
Set Data Ready Handshake

9

EDC Ring Buffer Control Read Next Buffer Record
TrendRBCtl.ReadNext

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send
(L)

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read
(U)

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK
(U)

EDC Gateway Trend Message Control Write Command
TrendMsgCtl.Write
(U)

EDC Gateway Trend Message Control Write Command Done
TrendMsgCtl.WriteOK
(U)

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

10

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send

EDC Gateway Trend Message Control Write Command Done
TrendMsgCtl.WriteOK

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send
(U)

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend
(U)

EDC Ring Buffer Control Read Buffer Record Complete
TrendRBCtl.ReadComplete
(L)

Load Read Records from RingBuffer to Gateway Message

Read Complete Delay
Allow Gateway processing time before attempting next record send.

11

EDC Ring Buffer Control Read Buffer Record Complete
TrendRBCtl.ReadComplete

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source 100
Dest TrendRBCtl.ReadCompleteDelayTMR.PRE 0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer TrendRBCtl.ReadCompleteDelayTMR
Preset 0
Accum 0
EN
DN

EDC Ring Buffer Control Read Complete Delay Timer
TrendRBCtl.ReadCompleteDelayTMR.DN

EDC Ring Buffer Control Read Buffer Record Complete Delay
TrendRBCtl.ReadCompleteDelay

EDC Ring Buffer Control Read Buffer Record in Progress
TrendRBCtl.ReadInProgress
U

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

12

EDC Ring Buffer Control Buffer Empty
TrendRBCtl.Empty
/

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest TrendRBCtl.NotEmptyTMR.PRE 0

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer TrendRBCtl.NotEmptyTMR
Preset 0
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A TrendRBCtl.NotEmptyTMR.ACC 0
Source B TrendRBCtl.NotEmptyMax 0

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source TrendRBCtl.NotEmptyTMR.ACC 0
Dest TrendRBCtl.NotEmptyMax 0

Load Read Records from RingBuffer to Gateway Message

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

13

EDC Ring Buffer
Control Buffer Full
TrendRBCtl.Full

EDC Ring Buffer
Control Buffer Not
Empty Alarm Timer
TrendRBCtl.NotEmptyTMR.DN

EDC Ring Buffer
Control Read Buffer
Alarm
TrendRBCtl.BufferAlarm

(End)

Trend Data Ring Buffer and Message

Equipment Trend Data
Buffered Fault Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 144
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

EDC Gateway Trend Message Control Message Transaction Timer

MOVE
Source 999999999
Dest TrendMsgCtl.Trans.TMR.PRE 0

EDC Gateway Trend Message Control Message Transaction Timer

TON
Timer TrendMsgCtl.Trans.TMR
Preset 0
Accum 0
EN
DN

Trend Data Ring Buffer and Message

Read Message Delay Timer
If driver sends data too fast, it will generate extra Read Message traffic until previous message has been processed
Optimization uses the ReadOk and ReadNotOk counts to tune the Delay Time to the Gateway response

3

EDC Gateway Trend Write Message
TrendWriteMsg.EN
<EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Message Control Read Message Ok Count

GE	
Source A	TrendMsgCtl.ReadOkCount 0
Source B	CfgRB_ReadOkLimit <CfgRB.ReadOkLimit> 5

EDC Gateway Trend Message Control Message Delay Timer

SUB	
Source A	TrendMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadOkSub <CfgRB.ReadOkSub> 10
Dest	TrendMsgCtl.DelayTMR.PRE 0

EDC Gateway Trend Message Control Message Delay Timer

LE	
Source A	TrendMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100

EDC Gateway Trend Message Control Message Delay Timer

MOVE	
Source	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100
Dest	TrendMsgCtl.DelayTMR.PRE 0

EDC Gateway Trend Message Control Read Message Not Ok Count

GE	
Source A	TrendMsgCtl.ReadNotOkCount 0
Source B	CfgRB_ReadNotOkLimit <CfgRB.ReadNotOkLimit> 2

EDC Gateway Trend Message Control Message Delay Timer

ADD	
Source A	TrendMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadNotOkAdd <CfgRB.ReadNotOkAdd> 20
Dest	TrendMsgCtl.DelayTMR.PRE 0

EDC Gateway Trend Message Control Message Delay Timer

GT	
Source A	TrendMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500

EDC Gateway Trend Message Control Message Delay Timer

MOVE	
Source	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500
Dest	TrendMsgCtl.DelayTMR.PRE 0

EDC Gateway Trend Message Control Read Message Ok Count

CLR	
Dest	TrendMsgCtl.ReadOkCount 0

EDC Gateway Trend Message Control Read Message Not Ok Count

CLR	
Dest	TrendMsgCtl.ReadNotOkCount 0

EDC Gateway Trend Message Control Message Delay Timer

TON		
Timer	TrendMsgCtl.DelayTMR	EN
Preset	0	DN
Accum	0	

Trend Data Ring Buffer and Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send

EDC Gateway Trend Write Message
TrendWriteMsg.EN
<EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Message Control Message Resend Timer

MOVE
Source CFG.BuffMsgResendTime
301000
Dest TrendMsgCtl.ResendTMR.PRE
0

EDC Gateway Trend Message Control Message Resend Timer

TON
Timer TrendMsgCtl.ResendTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Trend Message Control Message Resend Timer
TrendMsgCtl.ResendTMR.DN

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend
L

EDC Gateway Trend Message Control Message Resend Timer
TrendMsgCtl.ResendTMR
RES

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read
U

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK
U

EDC Gateway Trend Message Control Write Command
TrendMsgCtl.Write
U

EDC Gateway Trend Message Control Write Command Done
TrendMsgCtl.WriteOK
U

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 422
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Send Data Initialize

5

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send

EDC Gateway Trend Message Control Send Data 1-Shot
TrendMsgCtl.SendONS
ONS

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

EDC Gateway Trend Message Control Message Transaction Timer
TrendMsgCtl.Trans.TMR
RES

EDC Gateway Trend Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest TrendMsgCtl.Trans.Last
0

Trend Data Ring Buffer and Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

6

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK

EDC Gateway Trend Message Control Message Delay Timer
TrendMsgCtl.DelayTMR.DN

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Write Message
TrendWriteMsg.EN
<EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Read Message
TrendReadMsg.ER
<EQ0006_TrendReadMsg.ER>

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

EDC Gateway Trend Message Control Message Retry Timer
TrendMsgCtl.RetryTMR.DN

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read
(L)

EDC Gateway Trend Message Control Read Command 1-Shot
TrendMsgCtl.ReadONS
( )

EDC Gateway Trend Read Message

COP
Source UserCfgMsg.Path
<EQ0006_UserCfgMsg.Path>
Dest TrendReadMsg.Path
<EQ0006_TrendReadMsg.Path>
Length 1

EDC Gateway Trend Read Message

COP
Source CFG.MsgDest.TrendDest
Dest TrendReadMsg.RemoteElement
<EQ0006_TrendReadMsg.RemoteElement>
Length 1

EDC Gateway Trend Read Message

MOVE
Source CFG.TrendMSG.e
70
Dest TrendReadMsg.REQ_LEN
<EQ0006_TrendReadMsg.REQ_LEN>
70

EDC Gateway Trend Read Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest TrendReadMsg.UnconnectedTimeout
<EQ0006_TrendReadMsg.UnconnectedTimeout>
30000000

EDC Gateway Trend Read Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest TrendReadMsg.ConnectionRate
<EQ0006_TrendReadMsg.ConnectionRate>
7500000

EDC Gateway Trend Read Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest TrendReadMsg.TimeoutMultiplier
<EQ0006_TrendReadMsg.TimeoutMultiplier>
0

EDC Gateway Trend Read Message
TrendReadMsg.EN_CC
<EQ0006_TrendReadMsg.EN_CC>
(U)

EDC Gateway Trend Read Message

MSG
Message Control TrendReadMsg ...
<EQ0006_TrendReadMsg>
EN
DN
ER

Trend Data Ring Buffer and Message

Read Message EN Performance & Diagnostics

7

EDC Gateway Trend Message Control Read Command 1-Shot
TrendMsgCtl.ReadONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	430
Input Par	TrendMsgCtl.Trans.TMR.ACC
Input Par	TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Read Message OK - Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record)

Increment ReadOk and ReadNotOk counts
only increment ReadOk count when multiple records in Buffer (single records are always ready to send and don't test Gateway servicing speed)

8

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read

EDC Gateway Trend Read Message
TrendReadMsg.DN
<EQ0006_TrendReadMsg.DN>

EDC Message Rx Data Trend Receive Message
EQ
Source A EqRxData.Trend[0]
<EQ0006_RxData.Trend[0]>
0
Source B 0

EDC Message Rx Data Trend Receive Message
EQ
Source A EqRxData.Trend[CFG.TrendMSG.ei]
<EQ0006_RxData.Trend[CFG.TrendMSG.ei]>
0
Source B 0

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK
L

EDC Ring Buffer Control Ring Buffer Record Count
GE
Source A TrendRBCtl.RecordCount
0
Source B 2

EDC Gateway Trend Message Control Read Message Ok Count
ADD
Source A TrendMsgCtl.ReadOkCount
0
Source B 1
Dest TrendMsgCtl.ReadOkCount
0

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK
/

EDC Gateway Trend Message Control Read Message Not Ok Count
ADD
Source A TrendMsgCtl.ReadNotOkCount
0
Source B 1
Dest TrendMsgCtl.ReadNotOkCount
0

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TrendMsgCtl.Trans
PerfTrans Status.TrendTxRead.DN

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TrendMsgCtl.Trans

EDC Gateway Trend Message Control Read Command
TrendMsgCtl.Read
U

Trend Data Ring Buffer and Message

PerfTrans Status.TrendTxRead.KA

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EqRxData.Trend[0]
<EQ0006_RxData.Trend[0]> 0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source EqRxData.Trend[CFG.TrendMSG.ei]
<EQ0006_RxData.Trend[CFG.TrendMSG.ei]> 0
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 431
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest TrendMsgCtl.ReadTimeoutTMR.PRE
0

EDC Gateway Trend Message Control Message Timeout Timer

TON
Timer TrendMsgCtl.ReadTimeoutTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Trend Message Control Message Timeout Timer
TrendMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Trend Read Message
TrendReadMsg.TO
<EQ0006_TrendReadMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 434
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Read Message ER Performance & Diagnostics

10

EDC Gateway Trend Read Message
TrendReadMsg.ER
<EQ0006_TrendReadMsg.ER>

EDC Gateway Trend Message Control Read Error 1-shot
TrendMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Trend Tx ER
Status.Update.TrendTxER
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxRead.ER

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	433
Input Par	TrendMsgCtl.Trans.TMR.ACC
Input Par	TrendMsgCtl.Trans.TransNum

Trend Data Ring Buffer and Message

Write Message
Write Data is loaded by Ring Buffer routine

11

EDC Gateway Trend Message Control Send Data
TrendMsgCtl.Send

EDC Gateway Trend Message Control Read Buffer Ready
TrendMsgCtl.ReadOK

EDC Gateway Trend Message Control Write Command Done
TrendMsgCtl.WriteOK

EDC Gateway Trend Read Message
TrendReadMsg.EN
<EQ0006_TrendReadMsg.EN>

EDC Gateway Trend Write Message
TrendWriteMsg.EN
<EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend Write Message
TrendWriteMsg.ER
<EQ0006_TrendWriteMsg.ER>

EDC Gateway Trend Message Control Resend Data (Keep Alive)
TrendMsgCtl.Resend

EDC Gateway Trend Message Control Message Retry Timer
TrendMsgCtl.RetryTMR.DN

EDC Gateway Trend Message Control Write Command
TrendMsgCtl.Write
L

EDC Gateway Trend Write Message

COP	
Source	UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest	TrendWriteMsg.Path <EQ0006_TrendWriteMsg.Path>
Length	1

EDC Gateway Trend Write Message

COP	
Source	CFG.MsgDest.TrendDest
Dest	TrendWriteMsg.RemoteElement <EQ0006_TrendWriteMsg.RemoteElement>
Length	1

EDC Gateway Trend Write Message

MOVE	
Source	CFG.TrendMSG.e 70
Dest	TrendWriteMsg.REQ_LEN <EQ0006_TrendWriteMsg.REQ_LEN> 70

EDC Gateway Trend Write Message

MOVE	
Source	CFG.CLx.UC_Timeout 4000000
Dest	TrendWriteMsg.UnconnectedTimeout <EQ0006_TrendWriteMsg.UnconnectedTimeout> 30000000

EDC Gateway Trend Write Message

MOVE	
Source	CFG.CLx.ConnectionRate 1000000
Dest	TrendWriteMsg.ConnectionRate <EQ0006_TrendWriteMsg.ConnectionRate> 7500000

EDC Gateway Trend Write Message

MOVE	
Source	CFG.CLx.ConnectionMult 0
Dest	TrendWriteMsg.TimeoutMultiplier <EQ0006_TrendWriteMsg.TimeoutMultiplier> 0

EDC Gateway Trend Write Message
TrendWriteMsg.EN_CC
<EQ0006_TrendWriteMsg.EN_CC>
U

EDC Gateway Trend Write Message

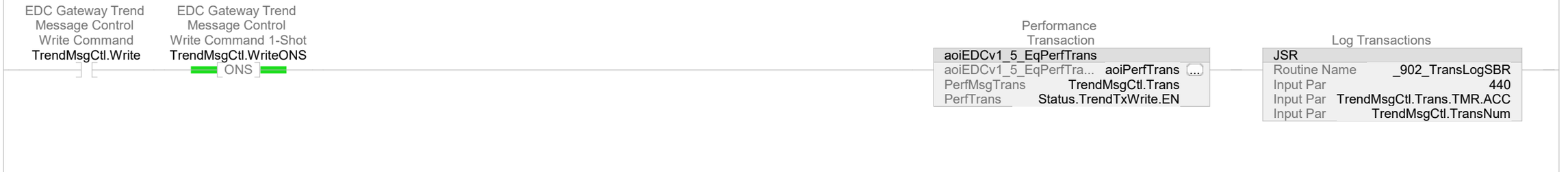
MSG	
Message Control	TrendWriteMsg <EQ0006_TrendWriteMsg> ...

EN
DN
ER

Trend Data Ring Buffer and Message

Write Message EN Performance & Diagnostics

12



Trend Data Ring Buffer and Message

Write Message Done

13

EDC Gateway Trend Message Control Write Command TrendMsgCtl.Write

EDC Gateway Trend Write Message TrendWriteMsg.DN <EQ0006_TrendWriteMsg.DN>

EDC Gateway Trend Message Control Write Command TrendMsgCtl.Write (U)

EDC Gateway Trend Message Control Write Command Done TrendMsgCtl.WriteOK (L)

EDC Gateway Trend Message Control Send Data TrendMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Trend Tx DN Status.Update.TrendTxOK (L)

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

EDC Gateway Trend Message Control Resend Data (Keep Alive) TrendMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Trend Keep Alive Tx DN Status.Update.TrendKaTxOK (L)

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxWrite.DN

EDC Gateway Trend Message Control Resend Data (Keep Alive) TrendMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TrendMsgCtl.Trans
PerfTrans	Status.TrendTxWrite.KA

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	441
Input Par	TrendMsgCtl.Trans.TMR.ACC
Input Par	TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Trend
Write Message
TrendWriteMsg.EN
<EQ0006_TrendWriteMsg.EN>

EDC Gateway Trend
Message Control
Message Timeout
Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest TrendMsgCtl.WriteTImeoutTMR.PRE
0

EDC Gateway Trend
Message Control
Message Timeout
Timer

TON
Timer TrendMsgCtl.WriteTImeoutTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Trend
Message Control
Message Timeout
Timer
TrendMsgCtl.WriteTImeoutTMR.DN

EDC Gateway Trend
Write Message
TrendWriteMsg.TO
<EQ0006_TrendWriteMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 444
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Write Message ER Performance & Diagnostics

15

EDC Gateway Trend
Write Message
TrendWriteMsg.ER
<EQ0006_TrendWriteMsg.ER>

EDC Gateway Trend
Message Control
Write Error 1-shot
TrendMsgCtl.WriteErONS
ONS

Equipment Data
Collection - Line
Equipment Status
State = Red - Trend
Tx ER
Status.Update.TrendTxER
L

Update Status State
for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance
Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TrendMsgCtl.Trans
PerfTrans Status.TrendTxWrite.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 443
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

Trend Data Ring Buffer and Message

Read/Write Message Error Retry

16

EDC Gateway Trend
Read Message
TrendReadMsg.ER
<EQ0006_TrendReadMsg.ER>

EDC Gateway Trend
Write Message
TrendWriteMsg.ER
<EQ0006_TrendWriteMsg.ER>

EDC Gateway Trend
Message Control
Message Retry Timer
TrendMsgCtl.RetryTMR.DN

EDC Gateway Trend
Message Control
Retry 1-Shot
TrendMsgCtl.RetryONS
ONS

EDC Gateway Trend
Message Control
Message Retry Timer

MOVE
Source CFG.MsgRetryTime
5000
Dest TrendMsgCtl.RetryTMR.PRE
0

EDC Gateway Trend
Message Control
Message Retry Timer

TON
Timer TrendMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Trend
Read Message
TrendReadMsg.ER
<EQ0006_TrendReadMsg.ER>

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 432
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

EDC Gateway Trend
Write Message
TrendWriteMsg.ER
<EQ0006_TrendWriteMsg.ER>

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 442
Input Par TrendMsgCtl.Trans.TMR.ACC
Input Par TrendMsgCtl.TransNum

(End)

Version Data Load for Message

Equipment Version Data
Data Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	152
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

Message Control TM Transaction Trigger

ADD	
Source A	VersionMsgCtl.TransactionNumber 677
Source B	1
Dest	VersionMsgCtl.TransactionNumber 677

Message Control TM Transaction Trigger

LT	
Source A	VersionMsgCtl.TransactionNumber 677
Source B	1

Message Control TM Transaction Trigger

GT	
Source A	VersionMsgCtl.TransactionNumber 677
Source B	CFG.MsgTransNumRollover 9999

Message Control TM Transaction Trigger

MOVE	
Source	1
Dest	VersionMsgCtl.TransactionNumber 677

Version Data Load for Message

Load Message Header, Timestamp and Footer Data

3

Equipment Data Version Message Format Header - Transaction Number (must match Footer)

MOVE
Source VersionMsgCtl.TransactionNumber 677
Dest VersionMsg.Header.TransNum 677

Equipment Data Version Message Format Header - Command Number

MOVE
Source 0
Dest VersionMsg.Header.CmdNum 0

Equipment Data Version Message Format Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE
Source UserCfg.EQID 6
Dest VersionMsg.Header.EQID 6

Equipment Data Version Message Format Header - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest VersionMsg.Header.Spare 0

Equipment Data Version Message Format Transaction Time Stamp

COP
Source CLX.TransTime
Dest VersionMsg.TransTime
Length 1

Equipment Data Version Message Format Footer - Spare (fills unused portion of DINT)

MOVE
Source 0
Dest VersionMsg.Footer.Spare 0

Equipment Data Version Message Format Footer - Transaction Number (must match Footer)

MOVE
Source VersionMsgCtl.TransactionNumber 677
Dest VersionMsg.Footer.TransNum 677

Load Logix Controller & Driver Code Versions

4

Equipment Data Version Message Format Logix Controller Version

COP
Source CLX.Version
Dest VersionMsg.LogixVersion
Length 1

Equipment Data Version Message Format Equipment Driver Program Version

COP
Source Prog.Version
Dest VersionMsg.DriverVersion
Length 1

(End)

Logix and Code Version Data Message

Equipment Version Data
Version Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 154
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Reset Delay & Transaction Timer

2

Message Control Send Data
VersionMsgCtl.Send

Message Control Send Data 1-shot
VersionMsgCtl.SendONS
ONS

Message Control Message Transaction Timer
VersionMsgCtl.Trans.TMR
RES

Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest VersionMsgCtl.Trans.Last 14

Message Transaction Timer

3

Message Control Message Transaction Timer

MOVE
Source 999999999
Dest VersionMsgCtl.Trans.TMR.PRE 999999999

Message Control Message Transaction Timer

TON
Timer VersionMsgCtl.Trans.TMR
Preset 999999999
Accum 33891337
EN
DN

Logix and Code Version Data Message

Message Delay Timer
prevent Message from sending too quickly for Gateway
Gateway processes one Line per scan -- Equipment Message could take 500-1000 msec to process
any attempt to send faster will generate extra Read Message traffic until previous message has been processed

4

EDC Gateway Version
Write Message
VersionWriteMsg.EN
<EQ0006_VersionWriteMsg.EN>

Message Control
Message Delay Timer

MOVE	
Source	CFG.MsgDelayTime
	250
Dest	VersionMsgCtl.DelayTMR.PRE
	250

Message Control
Message Delay Timer

TON		
Timer	VersionMsgCtl.DelayTMR	EN
Preset	250	DN
Accum	250	

Logix and Code Version Data Message

Write Message

5

Message Control Send Data VersionMsgCtl.Send

Message Control Message Delay Timer VersionMsgCtl.DelayTMR.DN

EDC Gateway Version Write Message VersionWriteMsg.EN <EQ0006_VersionWriteMsg.EN>

EDC Gateway Version Write Message VersionWriteMsg.ER <EQ0006_VersionWriteMsg.ER>

Message Control Message Retry Timer VersionMsgCtl.RetryTMR.DN

Message Control Write Data VersionMsgCtl.Write (L)

Message Control Message Delay Timer VersionMsgCtl.DelayTMR (RES)

EDC Gateway Version Write Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest VersionWriteMsg.Path <EQ0006_VersionWriteMsg.Path>
Length 1

EDC Gateway Version Write Message

COP
Source CFG.MsgDest.VersionDest
Dest VersionWriteMsg.RemoteElement <EQ0006_VersionWriteMsg.RemoteElement>
Length 1

EDC Gateway Version Write Message

MOVE
Source CFG.VersionMSG.e 30
Dest VersionWriteMsg.REQ_LEN <EQ0006_VersionWriteMsg.REQ_LEN> 30

EDC Gateway Version Write Message

MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest VersionWriteMsg.UnconnectedTimeout <EQ0006_VersionWriteMsg.UnconnectedTimeout> 4000000

EDC Gateway Version Write Message

MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest VersionWriteMsg.ConnectionRate <EQ0006_VersionWriteMsg.ConnectionRate> 1000000

EDC Gateway Version Write Message

MOVE
Source CFG.CLx.ConnectionMult 0
Dest VersionWriteMsg.TimeoutMultiplier <EQ0006_VersionWriteMsg.TimeoutMultiplier> 0

EDC Gateway Version Write Message VersionWriteMsg.EN_CC <EQ0006_VersionWriteMsg.EN_CC> (U)

EDC Message Tx Data Version Receive Message

COP
Source VersionMsg
Dest EqTxData.Version[0] <EQ0006_TxData.Version[0]>
Length CFG.VersionMSG.n

EDC Gateway Version Write Message

MSG
Message Control VersionWriteMsg <EQ0006_VersionWriteMsg> (...)
(EN) (DN) (ER)

Logix and Code Version Data Message

Message EN Performance & Diagnostics

6

Message Control Write Data: VersionMsgCtl.Write

Message Control Write Data 1-shot: VersionMsgCtl.WriteONS (ONS)

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	VersionMsgCtl.Trans
PerfTrans	Status.VersionTx.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	540
Input Par	VersionMsgCtl.Trans.TMR.ACC
Input Par	VersionMsgCtl.TransactionNumber

Message Done

7

Message Control Write Data: VersionMsgCtl.Write

EDC Gateway Version Write Message: VersionWriteMsg.DN <EQ0006_VersionWriteMsg.DN>

Message Control Send Data: VersionMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Version Tx DN: Status.Update.VersionTxOK (L)

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Message Control Send Data: VersionMsgCtl.Send (U)

Message Control Write Data: VersionMsgCtl.Write (U)

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	VersionMsgCtl.Trans
PerfTrans	Status.VersionTx.DN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	541
Input Par	VersionMsgCtl.Trans.TMR.ACC
Input Par	VersionMsgCtl.TransactionNumber

Logix and Code Version Data Message

Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

8

EDC Gateway Version Write Message VersionWriteMsg.EN <EQ0006_VersionWriteMsg.EN>

Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout 5000
Dest VersionMsgCtl.TimeoutTMR.PRE 5000

Message Control Message Timeout Timer

TON
Timer VersionMsgCtl.TimeoutTMR
Preset 5000
Accum 0
EN
DN

Message Control Message Timeout Timer VersionMsgCtl.TimeoutTMR.DN

EDC Gateway Version Write Message VersionWriteMsg.TO <EQ0006_VersionWriteMsg.TO>

L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 544
Input Par VersionMsgCtl.Trans.TMR.ACC
Input Par VersionMsgCtl.TransactionNumber

Message ER Performance & Diagnostics

9

EDC Gateway Version Write Message VersionWriteMsg.ER <EQ0006_VersionWriteMsg.ER>

Message Control Write Error 1-shot VersionMsgCtl.WriteErONS

ONS

Equipment Data Collection - Line Equipment Status State = Red - Version Tx ER Status.Update.VersionTxER

L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans VersionMsgCtl.Trans
PerfTrans Status.VersionTx.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 543
Input Par VersionMsgCtl.Trans.TMR.ACC
Input Par VersionMsgCtl.TransactionNumber

Logix and Code Version Data Message

Message Error Retry

10

Message Control
Write Data
VersionMsgCtl.Write

EDC Gateway Version
Write Message
VersionWriteMsg.ER
<EQ0006_VersionWriteMsg.ER>

Message Control
Message Retry Timer

MOVE
Source CFG.MsgRetryTime
5000
Dest VersionMsgCtl.RetryTMR.PRE
0

Message Control
Message Retry Timer

TON
Timer VersionMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

Message Control
Message Retry Timer
VersionMsgCtl.RetryTMR.DN

Message Control
Retry 1-shot
VersionMsgCtl.RetryONS
ONS

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 542
Input Par VersionMsgCtl.Trans.TMR.ACC
Input Par VersionMsgCtl.TransactionNumber

(End)

Time Synchronization Data Load for Message

Equipment Time Sync Data
Time Sync Data Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 162
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get

Time Sync Message Control Get Data Increment Transaction Number
TimeSyncMsgCtl.GetInc
ONS

Time Sync Message Control TM Transaction Trigger

ADD
Source A TimeSyncMsgCtl.TransactionNumber 0
Source B 1
Dest TimeSyncMsgCtl.TransactionNumber 0

Time Sync Message Control TM Transaction Trigger

LT
Source A TimeSyncMsgCtl.TransactionNumber 0
Source B 1

Time Sync Message Control TM Transaction Trigger

GT
Source A TimeSyncMsgCtl.TransactionNumber 0
Source B CFG.MsgTransNumRollover 9999

Time Sync Message Control TM Transaction Trigger

MOVE
Source 1
Dest TimeSyncMsgCtl.TransactionNumber 0

Time Synchronization Data Load for Message

Sync Time with Gateway
if Values are within limits

3

Time Sync Message Control Get Data TimeSyncMsgCtl.Get

Time Sync Message Control Read Data OK TimeSyncMsgCtl.ReadOK

Time Sync Message Control Get Data TimeSyncMsgCtl.Get

Time Sync Message Control Read Data OK TimeSyncMsgCtl.ReadOK

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	2016
Test	EqRxData.TimeSync[0] <EQ0006_RxData.TimeSync[0]> 0
High Limit	9999

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	1
Test	EqRxData.TimeSync[1] <EQ0006_RxData.TimeSync[1]> 0
High Limit	12

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	1
Test	EqRxData.TimeSync[2] <EQ0006_RxData.TimeSync[2]> 0
High Limit	31

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	0
Test	EqRxData.TimeSync[3] <EQ0006_RxData.TimeSync[3]> 0
High Limit	23

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	0
Test	EqRxData.TimeSync[4] <EQ0006_RxData.TimeSync[4]> 0
High Limit	59

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	0
Test	EqRxData.TimeSync[5] <EQ0006_RxData.TimeSync[5]> 0
High Limit	59

EDC Message Rx Data Time Sync Receive Message

LIMIT	
Low Limit	0
Test	EqRxData.TimeSync[6] <EQ0006_RxData.TimeSync[6]> 0
High Limit	999999

EDC Message Rx Data Time Sync Receive Message

SSV	
Class Name	WallClockTime
Instance Name	
Attribute Name	LocalDateTime
Source	EqRxData.TimeSync[0] <EQ0006_RxData.TimeSync[0]> 0

(End)

Time Synchronization Data Message

Equipment Time Sync Data
Time Sync Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 164
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Time Sync Message Control Message Transaction Timer

MOVE
Source 999999999
Dest TimeSyncMsgCtl.Trans.TMR.PRE 0

Time Sync Message Control Message Transaction Timer

TON
Timer TimeSyncMsgCtl.Trans.TMR
Preset 0
Accum 0
EN
DN

Get Data Initialize

3

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get

Time Sync Message Control Get Data 1-Shot
TimeSyncMsgCtl.GetONS
ONS

Time Sync Message Control Message Transaction Timer
TimeSyncMsgCtl.Trans.TMR
RES

Time Sync Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest TimeSyncMsgCtl.Trans.Last 0

Time Synchronization Data Message
Read Message

4

Time Sync Message Control Get Data TimeSyncMsgCtl.Get

EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN <EQ0006_TimeSyncReadMsg.EN>

EDC Gateway Time Sync Read Message TimeSyncReadMsg.ER <EQ0006_TimeSyncReadMsg.ER>

Time Sync Message Control Message Retry Timer TimeSyncMsgCtl.RetryTMR.DN

Time Sync Message Control Read Data TimeSyncMsgCtl.Read (L)

EDC Gateway Time Sync Read Message

COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest TimeSyncReadMsg.Path <EQ0006_TimeSyncReadMsg.Path>
Length 1

EDC Gateway Time Sync Read Message

MOVE
Source 7
Dest TimeSyncReadMsg.REQ_LEN <EQ0006_TimeSyncReadMsg.REQ_LEN>
7

EDC Gateway Time Sync Read Message

MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest TimeSyncReadMsg.UnconnectedTimeout <EQ0006_TimeSyncReadMsg.UnconnectedTimeout>
30000000

EDC Gateway Time Sync Read Message

MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest TimeSyncReadMsg.ConnectionRate <EQ0006_TimeSyncReadMsg.ConnectionRate>
7500000

EDC Gateway Time Sync Read Message

MOVE
Source CFG.CLx.ConnectionMult 0
Dest TimeSyncReadMsg.TimeoutMultiplier <EQ0006_TimeSyncReadMsg.TimeoutMultiplier>
0

EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN_CC <EQ0006_TimeSyncReadMsg.EN_CC> (U)

EDC Gateway Time Sync Read Message

MSG
Message Control TimeSyncReadMsg ... <EQ0006_TimeSyncReadMsg>
EN
DN
ER

Time Synchronization Data Message

Message EN Performance & Diagnostics

5

Time Sync Message Control Read Data
TimeSyncMsgCtl.Read

Time Sync Message Control Read Data 1-shot
TimeSyncMsgCtl.ReadONS
ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TimeSyncMsgCtl.Trans
PerfTrans	Status.TimeSyncRx.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	630
Input Par	TimeSyncMsgCtl.Trans.TMR.ACC
Input Par	TimeSyncMsgCtl.TransactionNumber

Message Done

6

Time Sync Message Control Read Data
TimeSyncMsgCtl.Read

EDC Gateway Time Sync Read Message
TimeSyncReadMsg.DN
<EQ0006_TimeSyncReadMsg.DN>

Time Sync Message Control Read Data
TimeSyncMsgCtl.Read
U

Time Sync Message Control Read Data OK
TimeSyncMsgCtl.ReadOK
L

Time Sync Message Control Get Data
TimeSyncMsgCtl.Get

Equipment Data Collection - Line Equipment Status State = Green - Time Sync Rx DN
Status.Update.TimeSyncRxOK
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate
Status	Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	TimeSyncMsgCtl.Trans
PerfTrans	Status.TimeSyncRx.DN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	631
Input Par	TimeSyncMsgCtl.Trans.TMR.ACC
Input Par	TimeSyncMsgCtl.TransactionNumber

Time Synchronization Data Message

Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

7

EDC Gateway Time Sync Read Message TimeSyncReadMsg.EN <EQ0006_TimeSyncReadMsg.EN>

Time Sync Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout 5000
Dest TimeSyncMsgCtl.TimeoutTMR.PRE 0

Time Sync Message Control Message Timeout Timer

TON
Timer TimeSyncMsgCtl.TimeoutTMR
Preset 0
Accum 0
EN
DN

Time Sync Message Control Message Timeout Timer TimeSyncMsgCtl.TimeoutTMR.DN

EDC Gateway Time Sync Read Message TimeSyncReadMsg.TO <EQ0006_TimeSyncReadMsg.TO>

L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 634
Input Par TimeSyncMsgCtl.Trans.TMR.ACC
Input Par TimeSyncMsgCtl.TransactionNumber

Message ER Performance & Diagnostics

8

EDC Gateway Time Sync Read Message TimeSyncReadMsg.ER <EQ0006_TimeSyncReadMsg.ER>

Time Sync Message Control Read Error 1-shot TimeSyncMsgCtl.ReadErONS

ONS

Equipment Data Collection - Line Equipment Status State = Red - Time Sync Rx ER Status.Update.TimeSyncRxER

L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TimeSyncMsgCtl.Trans
PerfTrans Status.TimeSyncRx.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 633
Input Par TimeSyncMsgCtl.Trans.TMR.ACC
Input Par TimeSyncMsgCtl.TransactionNumber

Time Synchronization Data Message

Message Error Retry

9

Time Sync Message Control Read Data
TimeSyncMsgCtl.Read

EDC Gateway Time Sync Read Message
TimeSyncReadMsg.ER
<EQ0006_TimeSyncReadMsg.ER>

Time Sync Message Control Message Retry Timer

MOVE
Source CFG.MsgRetryTime
5000
Dest TimeSyncMsgCtl.RetryTMR.PRE
0

Time Sync Message Control Message Retry Timer

TON
Timer TimeSyncMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

Time Sync Message Control Message Retry Timer
TimeSyncMsgCtl.RetryTMR.DN

Time Sync Message Control Retry 1-shot
TimeSyncMsgCtl.RetryONS
ONS

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 632
Input Par TimeSyncMsgCtl.Trans.TMR.ACC
Input Par TimeSyncMsgCtl.TransactionNumber

(End)

Collect Cycle Data

Equipment Cycle Data
Collect Data

0

Cycle Index
SBR
Input Par CYi

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned
MOVE
Source 171
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)
ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Exit Routine if Index is out of range

2

Cycle Index
LT
Source A CYi 4
Source B 0

Cycle Index
GT
Source A CYi 4
Source B CFG.Cycle.mi 4

lbl_Exit
JMP

Copy Cycle Arrays to local tags

3

Cycle Collect Control
COP
Source CycleCfg[CYi].CTL
Dest CycleCollectCtl
Length 1

Equipment Data Cycle Message Data
COP
Source CycleCfg[CYi].Data
Dest CycleCollect
Length 1

Collect Cycle Data

Cycle Initialize
Load Initial Step Number into StepPrev to begin Cycle Timing

4

Cycle Collect Control Cycle Start Sampling: CycleCollectCtl.Start
Cycle Collect Control Cycle Start 1-Shot: CycleCollectCtl.StartONS (ONS)
Cycle Collect Control Cycle Init Sampling: CycleCollectCtl.Init
Cycle Collect Control Cycle Done Sampling: CycleCollectCtl.Done (U)
Cycle Collect Control Cycle Save: CycleCollectCtl.Loaded (U)

Equipment Data Cycle Message Data
FLL
Source 0
Dest CycleCollect
Length 1

Cycle Collect Control Step Duration Timer: CycleCollectCtl.DurationTMR (RES)

Cycle Collect Control Duration Prev Time
CLR
Dest CycleCollectCtl.DurationPrev
36010

Cycle Collect Control Step Previous Value
MOVE
Source CycleCollectCtl.Data
31
Dest CycleCollectCtl.StepPrev
31

Cycle Collect Control Cycle Array Index
CLR
Dest CycleCollectCtl.ARRi
30

Cycle Init
JobStatus and FaultCode are updates on Start and Send

5

Cycle Collect Control Cycle Init Sampling: CycleCollectCtl.Init

Equipment Data Cycle Message Data Sequence Number
MOVE
Source CYi
4
Dest CycleCollect.SeqNum
4

Equipment Data Cycle Message Data Step Numbers Used (1-120)
MOVE
Source CFG.CycData.n
120
Dest CycleCollect.StepCount
120

Equipment Data Cycle Message Data Job Status
MOVE
Source CycleMsg.Data.JobStatus
0
Dest CycleCollect.JobStatus
1

Equipment Data Cycle Message Data Fault Code
MOVE
Source CycleMsg.Data.FaultCode
0
Dest CycleCollect.FaultCode
0

Collect Cycle Data

Cycle Vehicle Data

6

Cycle Collect
Control Cycler Load
Vehicle Data
CycleCollectCtl.LoadVeh

Equipment Data
Cycle Message Data
Program Number from
SLCP

MOVE	
Source	CycleMsg.Data.AviPrgNum 0
Dest	CycleCollect.AviPrgNum 0

Equipment Data
Cycle Message Data
Program Number
Executed

MOVE	
Source	CycleMsg.Data.ActPrgNum 0
Dest	CycleCollect.ActPrgNum 0

Equipment Data
Cycle Message Data
Assembly Body Number

MOVE	
Source	CycleMsg.Data.BodyNum 0
Dest	CycleCollect.BodyNum 0

Equipment Data
Cycle Message Data
Assembly Sequence
Number

MOVE	
Source	CycleMsg.Data.AssyNum 0
Dest	CycleCollect.AssyNum 0

Equipment Data
Cycle Message Data
VIN Number

COP	
Source	CycleMsg.Data.VinNum[0]
Dest	CycleCollect.VinNum[0]
Length	20

Equipment Data
Cycle Message Data
Spare

MOVE	
Source	CycleMsg.Data.Spare1 0
Dest	CycleCollect.Spare1 0

Equipment Data
Cycle Message Data
Spare

MOVE	
Source	CycleMsg.Data.Spare2 0
Dest	CycleCollect.Spare2 0

Collect Cycle Data

Cycle Sample TImer
Use free-running timer and samle interval to avoid accumulating Timer overruns throughout test.
Take a data sample whenever the Step Number changes

Stop suspends timiing but doesn't suspend sampling
Send stores final step and duration if array isn't full

7

Cycle Collect Control Cycle Start Sampling
CycleCollectCtl.Start

Cycle Collect Control Cycle Send
CycleCollectCtl.Send

Cycle Collect Control Cycle Done Sampling
CycleCollectCtl.Done

Cycle Collect Control Cycle Stop Sampling
CycleCollectCtl.Stop

Cycle Collect Control Step Duration Timer

MOVE
Source 99999999
Dest CycleCollectCtl.DurationTMR.PRE 99999999

Cycle Collect Control Step Duration Timer

RTO
Timer CycleCollectCtl.DurationTMR
Preset 99999999
Accum 36484
EN
DN

Cycle Collect Control Cycle Send
CycleCollectCtl.Send

Cycle Collect Control Cycle Send ONS
CycleCollectCtl.SendONS
ONS

Cycle Collect Control Step Data

NE
Source A CycleCollectCtl.Data 31
Source B CycleCollectCtl.StepPrev 31

Cycle Collect Control Duration Accumulation

SUB
Source A CycleCollectCtl.DurationTMR.ACC 36484
Source B CycleCollectCtl.DurationPrev 36010
Dest CycleCollectCtl.DurationAcc 1200

Cycle Collect Control Cycle Sample
CycleCollectCtl.Sample

Cycle Collect Control Cycle Send
CycleCollectCtl.Send

Collect Cycle Data

Cycle Sample
store previous Step Number and Accum after the Step Number changes

8

Cycle Collect
Control Cycle Sample
CycleCollectCtl.Sample

Equipment Data
Cycle Message Data
Step Number
(0-65,536)

MOVE
Source CycleCollectCtl.StepPrev 31
Dest CycleCollect.Step[CycleCollectCtl.ARRi] 0

Equipment Data
Cycle Message Data
Duration (0-655.36
seconds)

DIV
Source A CycleCollectCtl.DurationAcc 1200
Source B 10
Dest CycleCollect.Duration[CycleCollectCtl.ARRi] 0

Cycle Collect
Control Cycle Array
Index

ADD
Source A CycleCollectCtl.ARRi 30
Source B 1
Dest CycleCollectCtl.ARRi 30

Cycle Collect
Control Cycle Array
Index

LT
Source A CycleCollectCtl.ARRi 30
Source B 0

Cycle Collect
Control Cycle Array
Index

GT
Source A CycleCollectCtl.ARRi 30
Source B CFG.CycData.mi 119

Cycle Collect
Control Cycle Done
Sampling
CycleCollectCtl.Done
L

Cycle Collect
Control Step
Previous Value

MOVE
Source CycleCollectCtl.Data 31
Dest CycleCollectCtl.StepPrev 31

Cycle Collect
Control Duration
Prev Time

ADD
Source A CycleCollectCtl.DurationPrev 36010
Source B CycleCollectCtl.DurationAcc 1200
Dest CycleCollectCtl.DurationPrev 36010

Collect Cycle Data

Cycle Samples Done
Update Status/Fault and Job Data
Load RIng Buffer

9

Cycle Collect Control Cycle Send CycleCollectCtl.Send

Cycle Collect Control Cycle Save CycleCollectCtl.Loaded /

Cycle Collect Control Cycle Send CycleCollectCtl.Send

Equipment Data Cycle Message Data Job Status

MOVE
Source CycleMsg.Data.JobStatus
0
Dest CycleCollect.JobStatus
1

Equipment Data Cycle Message Data Fault Code

MOVE
Source CycleMsg.Data.FaultCode
0
Dest CycleCollect.FaultCode
0

Equipment Data Cycle Message Data Step Numbers Used (1-120)

MOVE
Source CycleCollectCtl.ARRi
30
Dest CycleCollect.StepCount
120

Load Cycle Data record to Ring Buffer

JSR
Routine Name _172_CycleDataLoadRB

Cycle Collect Control Cycle Save CycleCollectCtl.Loaded (L)

Cycle Collect Control Cycle Start Sampling CycleCollectCtl.Start (U)

Cycle Collect Control Cycle Send CycleCollectCtl.Send (U)

Copy local tags to Cycle Array

10

Cycle Data Collection Cycle Control

COP
Source CycleCollectCtl
Dest CycleCfg[CYi].CTL
Length 1

Cycle Data Collection Cycle Data

COP
Source CycleCollect
Dest CycleCfg[CYi].Data
Length 1

Exit Routine

11

lbl_Exit
LBL

NOP

(End)

Load Cycle Data record to Ring Buffer

Equipment Cycle Data
Ring Buffer Load

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 172
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Increment Transaction Number

2

EDC Ring Buffer Control Transaction Number

ADD
Source A CycleRBCtl.TransactionNumber 0
Source B 1
Dest CycleRBCtl.TransactionNumber 0

EDC Ring Buffer Control Transaction Number

LT
Source A CycleRBCtl.TransactionNumber 0
Source B 1

EDC Ring Buffer Control Transaction Number

GT
Source A CycleRBCtl.TransactionNumber 0
Source B CFG.MsgTransNumRollover 9999

EDC Ring Buffer Control Transaction Number

MOVE
Source 1
Dest CycleRBCtl.TransactionNumber 0

Load Cycle Data record to Ring Buffer

Load Message Header, Timestamp, Cycle Data and Footer Data

3

Equipment Data Cycle Message Header - Transaction Number (must match Footer)

MOVE	
Source	CycleRBCtl.TransactionNumber 0
Dest	CycleMsg.Header.TransNum 0

Equipment Data Cycle Message Header - Command Number

MOVE	
Source	0
Dest	CycleMsg.Header.CmdNum 0

Equipment Data Cycle Message Header - Equipment ID Number (0102 = Line 1, Equip 2)

MOVE	
Source	UserCfg.EQID 6
Dest	CycleMsg.Header.EQID 0

Equipment Data Cycle Message Header - Spare (fills unused portion of DINT)

MOVE	
Source	0
Dest	CycleMsg.Header.Spare 0

Equipment Data Cycle Message Transaction Time Stamp

COP	
Source	CLX.TransTime
Dest	CycleMsg.TransTime
Length	1

Equipment Data Cycle Message Data - Cycle

COP	
Source	CycleCollect
Dest	CycleMsg.Data
Length	1

Equipment Data Cycle Message Footer - Spare (fills unused portion of DINT)

MOVE	
Source	0
Dest	CycleMsg.Footer.Spare 0

Equipment Data Cycle Message Footer - Transaction Number (must match Footer)

MOVE	
Source	CycleRBCtl.TransactionNumber 0
Dest	CycleMsg.Footer.TransNum 0

Load Cycle Data record to Ring Buffer

RB Write -- Reset Write Pointer if outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

4

EDC Ring Buffer Control Next Write Record

LT
Source A CycleRBCtl.WritePtr 0
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A CycleRBCtl.WritePtr 0
Source B CFG.CycleRB.mi 19

EDC Ring Buffer Control Diagnostic - Manual Reset

CycleRBCtl.WritePtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source CycleRBCtl.WritePtr 0
Dest CycleRBCtl.WritePtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.WritePtr 0
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.ReadPtr 19
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 713
Input Par 0
Input Par CycleMsg.Header.TransNum

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest CycleRBCtl.WritePtr 0

Load Cycle Data record to Ring Buffer

RB Write - Ring Buffer Full
Exit without loading Message to Ring Buffer
Manually Reset Error Indicator & Tags

5

EDC Ring Buffer Control Next Write Record

EQ
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadPtr 19

EDC Ring Buffer Control Buffer Full - Block Writes
CycleRBCtl.FullBlockWrite

EDC Ring Buffer Control Diagnostic - Manual Reset
CycleRBCtl.WriteBlockErr
L

EDC Ring Buffer Control Diagnostic - Manual Reset Value

ADD
Source A CycleRBCtl.WriteBlockErrCnt 0
Source B 1
Dest CycleRBCtl.WriteBlockErrCnt 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.WritePtr 0
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.ReadPtr 19
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 714
Input Par 0
Input Par CycleMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans TempMsgTrans
PerfTrans Status.CycleRx.ER

lbl_Exit
JMP

Load Cycle Data record to Ring Buffer

RB Write -- Load Message into Ring Buffer

6

Equipment Data Cycle Message

COP
Source CycleMsg
Dest CycleRB[CycleRBCtl.WritePtr]
Length 1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.WritePtr
0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.ReadPtr
19
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 711
Input Par 0
Input Par CycleMsg.Header.TransNum

Message Control Transaction Timing

FLL
Source 0
Dest TempMsgTrans
Length 1

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans TempMsgTrans
PerfTrans Status.CycleRx.DN

RB Write -- Increment Write Pointer

7

EDC Ring Buffer Control Next Write Record

ADD
Source A CycleRBCtl.WritePtr
0
Source B 1
Dest CycleRBCtl.WritePtr
0

EDC Ring Buffer Control Next Write Record

LT
Source A CycleRBCtl.WritePtr
0
Source B 0

EDC Ring Buffer Control Next Write Record

GT
Source A CycleRBCtl.WritePtr
0
Source B CFG.CycleRB.mi
19

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest CycleRBCtl.WritePtr
0

Exit Routine

8

lbl_Exit
LBL

NOP

Load Cycle Data record to Ring Buffer

(End)

Load Read Records from RingBuffer to Gateway Message

Equipment Cycle Data
Ring Buffer Unload to Gateway Message

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 173
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Load Read Records from RingBuffer to Gateway Message

Reset Pointers if Read Pointer outside limits
(this should never happen - but protects the Array Indexes)

Manually Reset Error Indicator & Tags

2

EDC Ring Buffer Control Next Read Record

LT
Source A CycleRBCtl.ReadPtr 19
Source B 0

EDC Ring Buffer Control Next Read Record

GT
Source A CycleRBCtl.ReadPtr 19
Source B CFG.CycleRB.mi 19

EDC Ring Buffer Control Diagnostic - Manual Reset

CycleRBCtl.ReadPtrErr (L)

EDC Ring Buffer Control Diagnostic - Manual Reset

MOVE
Source CycleRBCtl.ReadPtr 19
Dest CycleRBCtl.ReadPtrErrPtr 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.WritePtr 0
Dest Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source CycleRBCtl.ReadPtr 19
Dest Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 723
Input Par 0
Input Par CycleRBCtl.TransactionNumber

EDC Ring Buffer Control Next Write Record

MOVE
Source 0
Dest CycleRBCtl.WritePtr 0

EDC Ring Buffer Control Next Read Record

MOVE
Source CFG.CycleRB.mi 19
Dest CycleRBCtl.ReadPtr 19

Load Read Records from RingBuffer to Gateway Message

Next Read Location
ReadPointer = Current/Last Read Record (Incremented before Read)
ReadNextPointer = Next Read Record (if/when Written)

3

EDC Ring Buffer Control Next Read Record

ADD
Source A CycleRBCtl.ReadPtr 19
Source B 1
Dest CycleRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Next Read Record

GT
Source A CycleRBCtl.ReadNextPtr 0
Source B CFG.CycleRB.mi 19

EDC Ring Buffer Control Next Read Record

MOVE
Source 0
Dest CycleRBCtl.ReadNextPtr 0

Buffer Full - Read Pointer = Write Pointer
WritePointer = Next Write Record (Incremented after Write)

Buffer Empty -- Read Pointer next to Write Pointer
no more messages in Ring Buffer
(RB Empty when Read complete - don't increment Read Pointer)

4

EDC Ring Buffer Control Next Read Record

EQ
Source A CycleRBCtl.ReadPtr 19
Source B CycleRBCtl.WritePtr 0

EDC Ring Buffer Control Buffer Full
CycleRBCtl.Full

EDC Ring Buffer Control Read Buffer Record in Progress
CycleRBCtl.ReadInProgress

EDC Ring Buffer Control Next Read Record

EQ
Source A CycleRBCtl.ReadNextPtr 0
Source B CycleRBCtl.WritePtr 0

EDC Ring Buffer Control Buffer Empty
CycleRBCtl.Empty

Load Read Records from RingBuffer to Gateway Message Buffer Count

5

EDC Ring Buffer Control Read Buffer Record in Progress
CycleRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
CycleRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record
GT
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadPtr 19

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest CycleRBCtl.RecordCount 0
Expression CycleRBCtl.WritePtr-CycleRBCtl.ReadPtr

EDC Ring Buffer Control Next Write Record
LT
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadPtr 19

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest CycleRBCtl.RecordCount 0
Expression CFG.CycleRB.n-(CycleRBCtl.ReadPtr-CycleRBCtl.WritePtr)

EDC Ring Buffer Control Read Buffer Record in Progress
CycleRBCtl.ReadInProgress

EDC Ring Buffer Control Read Buffer Record Complete
CycleRBCtl.ReadComplete

EDC Ring Buffer Control Next Write Record
GE
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest CycleRBCtl.RecordCount 0
Expression CycleRBCtl.WritePtr-CycleRBCtl.ReadNextPtr

EDC Ring Buffer Control Next Write Record
LT
Source A CycleRBCtl.WritePtr 0
Source B CycleRBCtl.ReadNextPtr 0

EDC Ring Buffer Control Ring Buffer Record Count
CPT
Dest CycleRBCtl.RecordCount 0
Expression CFG.CycleRB.n-(CycleRBCtl.ReadNextPtr-CycleRBCtl.WritePtr)

EDC Ring Buffer Control Ring Buffer Record Count
GT
Source A CycleRBCtl.RecordCount 0
Source B CycleRBCtl.RecordCountMax 0

EDC Ring Buffer Control Ring Buffer Record Count Max
MOVE
Source CycleRBCtl.RecordCount 0
Dest CycleRBCtl.RecordCountMax 0

Load Read Records from RingBuffer to Gateway Message

Read Next
Ready for next data to Gateway Message

6

EDC Ring Buffer Control Buffer Empty: CycleRBCtl.Empty
EDC Ring Buffer Control Read Buffer Record in Progress: CycleRBCtl.ReadInProgress
Ring Buffer Message Control Send Data: CycleMsgCtl.Send
Ring Buffer Message Control Resend Data (Keep Alive): CycleMsgCtl.Resend
EDC Ring Buffer Control Read Next Buffer Record 1-shot: CycleRBCtl.ReadNextONS
EDC Ring Buffer Control Read Next Buffer Record: CycleRBCtl.ReadNext
EDC Ring Buffer Control Read Buffer Record in Progress: CycleRBCtl.ReadInProgress
EDC Ring Buffer Control Read Buffer Record Complete: CycleRBCtl.ReadComplete

Read Next
Increment Read Pointer

7

EDC Ring Buffer Control Read Next Buffer Record: CycleRBCtl.ReadNext

ADD
Source A: CycleRBCtl.ReadPtr 19
Source B: 1
Dest: CycleRBCtl.ReadPtr 19

EDC Ring Buffer Control Next Read Record

LT
Source A: CycleRBCtl.ReadPtr 19
Source B: 0

EDC Ring Buffer Control Next Read Record

GT
Source A: CycleRBCtl.ReadPtr 19
Source B: CFG.CycleRB.mi 19

EDC Ring Buffer Control Next Read Record

MOVE
Source: 0
Dest: CycleRBCtl.ReadPtr 19

Read Next
Load Read Record

8

EDC Ring Buffer Control Read Next Buffer Record: CycleRBCtl.ReadNext

EDC Message Tx Data Cycle Receive Message

COP
Source: CycleRB[CycleRBCtl.ReadPtr]
Dest: EqTxData.Cycle[0] <EQ0006_TxData.Cycle[0]>
Length: CFG.CycleMSG.e

Ring Buffer Message Control Transaction Number

MOVE
Source: CycleRB[CycleRBCtl.ReadPtr].Header.TransNum 0
Dest: CycleMsgCtl.TransNum 0

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: CycleRBCtl.WritePtr 0
Dest: Trans.Ctl.Opt[1] -1

Transaction Log Control SBR External Load -- Optional Parameters

MOVE
Source: CycleRBCtl.ReadPtr 19
Dest: Trans.Ctl.Opt[2] -1

Log Transactions

JSR
Routine Name: _902_TransLogSBR
Input Par: 721
Input Par: 0
Input Par: CycleMsgCtl.TransNum

Load Read Records from RingBuffer to Gateway Message

Read Next
Set Data Ready Handshake

9

EDC Ring Buffer Control Read Next Buffer Record
CycleRBCtl.ReadNext

Ring Buffer Message Control Send Data
CycleMsgCtl.Send (L)

Ring Buffer Message Control Read Command
CycleMsgCtl.Read (U)

Ring Buffer Message Control Read Buffer Ready
CycleMsgCtl.ReadOK (U)

Ring Buffer Message Control Write Command
CycleMsgCtl.Write (U)

Ring Buffer Message Control Write Command Done
CycleMsgCtl.WriteOK (U)

Read Complete
Wait for TM Handshake to Acknowledge Read Complete

10

Ring Buffer Message Control Send Data
CycleMsgCtl.Send

Ring Buffer Message Control Write Command Done
CycleMsgCtl.WriteOK

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Ring Buffer Message Control Send Data
CycleMsgCtl.Send (U)

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend (U)

EDC Ring Buffer Control Read Buffer Record Complete
CycleRBCtl.ReadComplete (L)

Read Complete Delay
Allow Gateway processing time before attempting next record send.

11

EDC Ring Buffer Control Read Buffer Record Complete
CycleRBCtl.ReadComplete

EDC Ring Buffer Control Read Complete Delay Timer

MOVE
Source 100
Dest CycleRBCtl.ReadCompleteDelayTMR.PRE
0

EDC Ring Buffer Control Read Complete Delay Timer

TON
Timer CycleRBCtl.ReadCompleteDelayTMR
Preset 0
Accum 0
EN
DN

EDC Ring Buffer Control Read Complete Delay Timer
CycleRBCtl.ReadCompleteDelayTMR.DN

EDC Ring Buffer Control Read Buffer Record Complete Delay
CycleRBCtl.ReadCompleteDelay ( )

EDC Ring Buffer Control Read Buffer Record in Progress
CycleRBCtl.ReadInProgress (U)

Load Read Records from RingBuffer to Gateway Message

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

12

EDC Ring Buffer Control Buffer Empty
CycleRBCtl.Empty

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

MOVE
Source 300000
Dest CycleRBCtl.NotEmptyTMR.PRE 0

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

TON
Timer CycleRBCtl.NotEmptyTMR
Preset 0
Accum 0
EN
DN

EDC Ring Buffer Control Buffer Not Empty Alarm Timer

GT
Source A CycleRBCtl.NotEmptyTMR.ACC 0
Source B CycleRBCtl.NotEmptyMax 0

EDC Ring Buffer Control Buffer Not Empty Max Accum

MOVE
Source CycleRBCtl.NotEmptyTMR.ACC 0
Dest CycleRBCtl.NotEmptyMax 0

RingBuffer Alarm

Alarm if the RingBuffer is not Empty for 5 minutes.
(This will provide time to resolve the problem BEFORE the RingBuffer is full and data is lost.)

13

EDC Ring Buffer Control Buffer Full
CycleRBCtl.Full

EDC Ring Buffer Control Buffer Not Empty Alarm Timer
CycleRBCtl.NotEmptyTMR.DN

EDC Ring Buffer Control Read Buffer Alarm
CycleRBCtl.BufferAlarm

(End)

Trend Data Ring Buffer and Message

Equipment Cycle Data
Buffered Fault Message to Gateway

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	174
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Message Transaction Timer

2

Ring Buffer Message Control Message Transaction Timer

MOVE	
Source	999999999
Dest	CycleMsgCtl.Trans.TMR.PRE 0

Ring Buffer Message Control Message Transaction Timer

TON		
Timer	CycleMsgCtl.Trans.TMR	EN
Preset	0	DN
Accum	0	

Trend Data Ring Buffer and Message

Read Message Delay Timer
If driver sends data too fast, it will generate extra Read Message traffic until previous message has been processed
Optimization uses the ReadOk and ReadNotOk counts to tune the Delay Time to the Gateway response

3

EDC Gateway Cycle Write Message
CycleWriteMsg.EN
<EQ0006_CycleWriteMsg.EN>

EDC Gateway Cycle Read Message
CycleReadMsg.EN
<EQ0006_CycleReadMsg.EN>

Ring Buffer Message Control Read Message Ok Count

GE	
Source A	CycleMsgCtl.ReadOkCount 0
Source B	CfgRB_ReadOkLimit <CfgRB.ReadOkLimit> 5

Ring Buffer Message Control Message Delay Timer

SUB	
Source A	CycleMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadOkSub <CfgRB.ReadOkSub> 10
Dest	CycleMsgCtl.DelayTMR.PRE 0

Ring Buffer Message Control Message Delay Timer

LE	
Source A	CycleMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100

Ring Buffer Message Control Message Delay Timer

MOVE	
Source	CfgRB_ReadDelayMin <CfgRB.ReadDelayMin> 100
Dest	CycleMsgCtl.DelayTMR.PRE 0

Ring Buffer Message Control Read Message Not Ok Count

GE	
Source A	CycleMsgCtl.ReadNotOkCount 0
Source B	CfgRB_ReadNotOkLimit <CfgRB.ReadNotOkLimit> 2

Ring Buffer Message Control Message Delay Timer

ADD	
Source A	CycleMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadNotOkAdd <CfgRB.ReadNotOkAdd> 20
Dest	CycleMsgCtl.DelayTMR.PRE 0

Ring Buffer Message Control Message Delay Timer

GT	
Source A	CycleMsgCtl.DelayTMR.PRE 0
Source B	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500

Ring Buffer Message Control Message Delay Timer

MOVE	
Source	CfgRB_ReadDelayMax <CfgRB.ReadDelayMax> 500
Dest	CycleMsgCtl.DelayTMR.PRE 0

Ring Buffer Message Control Read Message Ok Count

CLR	
Dest	CycleMsgCtl.ReadOkCount 0

Ring Buffer Message Control Read Message Not Ok Count

CLR	
Dest	CycleMsgCtl.ReadNotOkCount 0

Ring Buffer Message Control Message Delay Timer

TON		
Timer	CycleMsgCtl.DelayTMR	EN
Preset	0	DN
Accum	0	

Trend Data Ring Buffer and Message

Message Resend Timer
Keep Alive - resends last data without increment of Transaction Number

4

Ring Buffer Message Control Send Data
CycleMsgCtl.Send
/

EDC Gateway Cycle Write Message
CycleWriteMsg.EN
<EQ0006_CycleWriteMsg.EN>
/

EDC Gateway Cycle Read Message
CycleReadMsg.EN
<EQ0006_CycleReadMsg.EN>
/

Ring Buffer Message Control Message Resend Timer

MOVE
Source CFG.BuffMsgResendTime
301000
Dest CycleMsgCtl.ResendTMR.PRE
0

Ring Buffer Message Control Message Resend Timer

TON
Timer CycleMsgCtl.ResendTMR
Preset 0
Accum 0
EN
DN

Ring Buffer Message Control Message Resend Timer
CycleMsgCtl.ResendTMR.DN

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend
L

Ring Buffer Message Control Message Resend Timer
CycleMsgCtl.ResendTMR
RES

Ring Buffer Message Control Read Command
CycleMsgCtl.Read
U

Ring Buffer Message Control Read Buffer Ready
CycleMsgCtl.ReadOK
U

Ring Buffer Message Control Write Command
CycleMsgCtl.Write
U

Ring Buffer Message Control Write Command Done
CycleMsgCtl.WriteOK
U

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 722
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Send Data Initialize

5

Ring Buffer Message Control Send Data
CycleMsgCtl.Send

Ring Buffer Message Control Send Data 1-Shot
CycleMsgCtl.SendONS
ONS

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Ring Buffer Message Control Message Transaction Timer
CycleMsgCtl.Trans.TMR
RES

Ring Buffer Message Control Transaction Elapsed Time for Last Event (used to calculate differential)

CLR
Dest CycleMsgCtl.Trans.Last
0

Trend Data Ring Buffer and Message

Read Message
Check Gateway Buffer for Empty Record
(Gateway clears Buffer after processing Record)

6

Ring Buffer Message Control Send Data
CycleMsgCtl.Send

Ring Buffer Message Control Read Buffer Ready
CycleMsgCtl.ReadOK

Ring Buffer Message Control Message Delay Timer
CycleMsgCtl.DelayTMR.DN

EDC Gateway Cycle Read Message
CycleReadMsg.EN
<EQ0006_CycleReadMsg.EN>

EDC Gateway Cycle Write Message
CycleWriteMsg.EN
<EQ0006_CycleWriteMsg.EN>

EDC Gateway Cycle Read Message
CycleReadMsg.ER
<EQ0006_CycleReadMsg.ER>

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Ring Buffer Message Control Message Retry Timer
CycleMsgCtl.RetryTMR.DN

Ring Buffer Message Control Read Command
CycleMsgCtl.Read
L

Ring Buffer Message Control Read Command 1-Shot
CycleMsgCtl.ReadONS

EDC Gateway Cycle Read Message

COP
Source UserCfgMsg.Path
<EQ0006_UserCfgMsg.Path>
Dest CycleReadMsg.Path
<EQ0006_CycleReadMsg.Path>
Length 1

EDC Gateway Cycle Read Message

COP
Source CFG.MsgDest.CycleDest
Dest CycleReadMsg.RemoteElement
<EQ0006_CycleReadMsg.RemoteElement>
Length 1

EDC Gateway Cycle Read Message

MOVE
Source CFG.CycleMSG.e
134
Dest CycleReadMsg.REQ_LEN
<EQ0006_CycleReadMsg.REQ_LEN>
134

EDC Gateway Cycle Read Message

MOVE
Source CFG.CLx.UC_Timeout
4000000
Dest CycleReadMsg.UnconnectedTimeout
<EQ0006_CycleReadMsg.UnconnectedTimeout>
30000000

EDC Gateway Cycle Read Message

MOVE
Source CFG.CLx.ConnectionRate
1000000
Dest CycleReadMsg.ConnectionRate
<EQ0006_CycleReadMsg.ConnectionRate>
7500000

EDC Gateway Cycle Read Message

MOVE
Source CFG.CLx.ConnectionMult
0
Dest CycleReadMsg.TimeoutMultiplier
<EQ0006_CycleReadMsg.TimeoutMultiplier>
0

EDC Gateway Cycle Read Message
CycleReadMsg.EN_CC
<EQ0006_CycleReadMsg.EN_CC>
U

EDC Gateway Cycle Read Message

MSG
Message Control CycleReadMsg ...
<EQ0006_CycleReadMsg>
EN
DN
ER

Trend Data Ring Buffer and Message

Read Message EN Performance & Diagnostics

7

Ring Buffer Message Control Read Command 1-Shot

CycleMsgCtl.ReadONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CycleMsgCtl.Trans
PerfTrans	Status.CycleTxRead.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	730
Input Par	CycleMsgCtl.Trans.TMR.ACC
Input Par	CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message

Read Message OK
Gateway Buffer has Empty Record
(Gateway clears Buffer after processing Record)

8

Ring Buffer Message Control Read Command CycleMsgCtl.Read

EDC Gateway Cycle Read Message CycleReadMsg.DN <EQ0006_CycleReadMsg.DN>

Ring Buffer Message Control Read Command CycleMsgCtl.Read (U)

EDC Message Rx Data Cycle Receive Message

EQ
Source A EqRxData.Cycle[0]
<EQ0006_RxData.Cycle[0]>
0
Source B 0

EDC Message Rx Data Cycle Receive Message

EQ
Source A EqRxData.Cycle[CFG.CycleMSG.ei]
<EQ0006_RxData.Cycle[CFG.CycleMSG.ei]>
0
Source B 0

Ring Buffer Message Control Read Buffer Ready CycleMsgCtl.ReadOK (L)

EDC Ring Buffer Control Ring Buffer Record Count

GE
Source A CycleRBCtl.RecordCount
0
Source B 2

Ring Buffer Message Control Read Message Ok Count

ADD
Source A CycleMsgCtl.ReadOkCount
0
Source B 1
Dest CycleMsgCtl.ReadOkCount
0

Ring Buffer Message Control Read Buffer Ready CycleMsgCtl.ReadOK (/)

Ring Buffer Message Control Read Message Not Ok Count

ADD
Source A CycleMsgCtl.ReadNotOkCount
0
Source B 1
Dest CycleMsgCtl.ReadNotOkCount
0

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans (...)
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxRead.DN

Ring Buffer Message Control Resend Data (Keep Alive) CycleMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans (...)
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxRead.KA

Trend Data Ring Buffer and Message

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source EqRxData.Cycle[0]
<EQ0006_RxData.Cycle[0]>
0
Dest Trans.Ctl.Opt[1]
-1

Transaction Log
Control SBR External
Load -- Optional
Parameters

MOVE
Source EqRxData.Cycle[CFG.CycleMSG.ei]
<EQ0006_RxData.Cycle[CFG.CycleMSG.ei]>
0
Dest Trans.Ctl.Opt[2]
-1

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 731
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Read Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

9

EDC Gateway Cycle
Read Message
CycleReadMsg.EN
<EQ0006_CycleReadMsg.EN>

Ring Buffer Message
Control Message
Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest CycleMsgCtl.ReadTimeoutTMR.PRE
0

Ring Buffer Message
Control Message
Timeout Timer

TON
Timer CycleMsgCtl.ReadTimeoutTMR
Preset 0
Accum 0
EN
DN

Ring Buffer Message
Control Message
Timeout Timer
CycleMsgCtl.ReadTimeoutTMR.DN

EDC Gateway Cycle
Read Message
CycleReadMsg.TO
<EQ0006_CycleReadMsg.TO>
L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 734
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message
Read Message ER Performance & Diagnostics

10

EDC Gateway Cycle Read Message
CycleReadMsg.ER
<EQ0006_CycleReadMsg.ER>

Ring Buffer Message Control Read Error 1-shot
CycleMsgCtl.ReadErONS
ONS

Equipment Data Collection - Line Equipment Status State = Red - Cycle Tx ER
Status.Update.CycleTxER
L

Update Status State for Transactions
aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate
Status Status

Performance Transaction
aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxRead.ER

Log Transactions
JSR
Routine Name _902_TransLogSBR
Input Par 733
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message
Write Data is loaded by Ring Buffer routine

11

Ring Buffer Message Control Send Data CycleMsgCtl.Send
Ring Buffer Message Control Read Buffer Ready CycleMsgCtl.ReadOK
Ring Buffer Message Control Write Command Done CycleMsgCtl.WriteOK
EDC Gateway Cycle Read Message CycleReadMsg.EN <EQ0006_CycleReadMsg.EN>
EDC Gateway Cycle Write Message CycleWriteMsg.EN <EQ0006_CycleWriteMsg.EN>
EDC Gateway Cycle Write Message CycleWriteMsg.ER <EQ0006_CycleWriteMsg.ER>
Ring Buffer Message Control Resend Data (Keep Alive) CycleMsgCtl.Resend
Ring Buffer Message Control Message Retry Timer CycleMsgCtl.RetryTMR.DN

Ring Buffer Message Control Write Command CycleMsgCtl.Write (L)

EDC Gateway Cycle Write Message
COP
Source UserCfgMsg.Path <EQ0006_UserCfgMsg.Path>
Dest CycleWriteMsg.Path <EQ0006_CycleWriteMsg.Path>
Length 1

EDC Gateway Cycle Write Message
COP
Source CFG.MsgDest.CycleDest
Dest CycleWriteMsg.RemoteElement <EQ0006_CycleWriteMsg.RemoteElement>
Length 1

EDC Gateway Cycle Write Message
MOVE
Source CFG.CycleMSG.e 134
Dest CycleWriteMsg.REQ_LEN <EQ0006_CycleWriteMsg.REQ_LEN> 134

EDC Gateway Cycle Write Message
MOVE
Source CFG.CLx.UC_Timeout 4000000
Dest CycleWriteMsg.UnconnectedTimeout <EQ0006_CycleWriteMsg.UnconnectedTimeout> 30000000

EDC Gateway Cycle Write Message
MOVE
Source CFG.CLx.ConnectionRate 1000000
Dest CycleWriteMsg.ConnectionRate <EQ0006_CycleWriteMsg.ConnectionRate> 7500000

EDC Gateway Cycle Write Message
MOVE
Source CFG.CLx.ConnectionMult 0
Dest CycleWriteMsg.TimeoutMultiplier <EQ0006_CycleWriteMsg.TimeoutMultiplier> 0

EDC Gateway Cycle Write Message CycleWriteMsg.EN_CC <EQ0006_CycleWriteMsg.EN_CC> (U)

EDC Gateway Cycle Write Message
MSG
Message Control CycleWriteMsg <EQ0006_CycleWriteMsg> (...)
EN
DN
ER

Trend Data Ring Buffer and Message

Write Message EN Performance & Diagnostics

12

Ring Buffer Message Control Write Command
CycleMsgCtl.Write

Ring Buffer Message Control Write Command 1-Shot
CycleMsgCtl.WriteONS
ONS

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans
PerfMsgTrans	CycleMsgCtl.Trans
PerfTrans	Status.CycleTxWrite.EN

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	740
Input Par	CycleMsgCtl.Trans.TMR.ACC
Input Par	CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message Done

13

Ring Buffer Message Control Write Command
CycleMsgCtl.Write

EDC Gateway Cycle Write Message
CycleWriteMsg.DN
<EQ0006_CycleWriteMsg.DN>

Ring Buffer Message Control Write Command
CycleMsgCtl.Write
U

Ring Buffer Message Control Write Command Done
CycleMsgCtl.WriteOK
L

Ring Buffer Message Control Send Data
CycleMsgCtl.Send

Equipment Data Collection - Line Equipment Status State = Green - Cycle Tx DN
Status.Update.CycleTxOK
L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate	
aoiEDCv2_00_EqStatus...	aoiEqStatusUpdate ...
Status	Status

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Equipment Data Collection - Line Equipment Status State = Green - Cycle Keep Alive Tx DN
Status.Update.CycleKaTxOK
L

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	CycleMsgCtl.Trans
PerfTrans	Status.CycleTxWrite.DN

Ring Buffer Message Control Resend Data (Keep Alive)
CycleMsgCtl.Resend

Performance Transaction

aoiEDCv1_5_EqPerfTrans	
aoiEDCv1_5_EqPerfTra...	aoiPerfTrans ...
PerfMsgTrans	CycleMsgCtl.Trans
PerfTrans	Status.CycleTxWrite.KA

Log Transactions

JSR	
Routine Name	_902_TransLogSBR
Input Par	741
Input Par	CycleMsgCtl.Trans.TMR.ACC
Input Par	CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message

Write Message Timeout
(this ensures Message doesn't lock up connection if timeout settings too high)

14

EDC Gateway Cycle Write Message CycleWriteMsg.EN <EQ0006_CycleWriteMsg.EN>

Ring Buffer Message Control Message Timeout Timer

MOVE
Source CFG.CLx.MsgTimeout
5000
Dest CycleMsgCtl.WriteTImeoutTMR.PRE
0

Ring Buffer Message Control Message Timeout Timer

TON
Timer CycleMsgCtl.WriteTImeoutTMR
Preset 0
Accum 0
EN
DN

Ring Buffer Message Control Message Timeout Timer CycleMsgCtl.WriteTImeoutTMR.DN

EDC Gateway Cycle Write Message CycleWriteMsg.TO <EQ0006_CycleWriteMsg.TO>

L

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 744
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Write Message ER Performance & Diagnostics

15

EDC Gateway Cycle Write Message CycleWriteMsg.ER <EQ0006_CycleWriteMsg.ER>

Ring Buffer Message Control Write Error 1-shot CycleMsgCtl.WriteErONS

ONS

Equipment Data Collection - Line Equipment Status State = Red - Cycle Tx ER Status.Update.CycleTxER

L

Update Status State for Transactions

aoiEDCv2_00_EqStatusUpdate
aoiEDCv2_00_EqStatus... aoiEqStatusUpdate ...
Status Status

Performance Transaction

aoiEDCv1_5_EqPerfTrans
aoiEDCv1_5_EqPerfTra... aoiPerfTrans ...
PerfMsgTrans CycleMsgCtl.Trans
PerfTrans Status.CycleTxWrite.ER

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 743
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

Trend Data Ring Buffer and Message
Read/Write Message Error Retry

16

EDC Gateway Cycle Read Message
CycleReadMsg.ER
<EQ0006_CycleReadMsg.ER>

EDC Gateway Cycle Write Message
CycleWriteMsg.ER
<EQ0006_CycleWriteMsg.ER>

Ring Buffer Message Control Message Retry Timer
CycleMsgCtl.RetryTMR.DN

Ring Buffer Message Control Retry 1-Shot
CycleMsgCtl.RetryONS
ONS

Ring Buffer Message Control Message Retry Timer

MOVE
Source CFG.MsgRetryTime
5000
Dest CycleMsgCtl.RetryTMR.PRE
0

Ring Buffer Message Control Message Retry Timer

TON
Timer CycleMsgCtl.RetryTMR
Preset 0
Accum 0
EN
DN

EDC Gateway Cycle Read Message
CycleReadMsg.ER
<EQ0006_CycleReadMsg.ER>

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 732
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

EDC Gateway Cycle Write Message
CycleWriteMsg.ER
<EQ0006_CycleWriteMsg.ER>

Log Transactions

JSR
Routine Name _902_TransLogSBR
Input Par 742
Input Par CycleMsgCtl.Trans.TMR.ACC
Input Par CycleMsgCtl.TransNum

(End)

Simulation of Fault Data

Simulation
Fault Data

Modified to use Scan Interval for Timing
v1.3 Dec2016 - Modified to send 2 Packets

0

NOP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

1

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE
Source 800
Dest Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD
Source A Prog.RoutineScan[Prog.Routine] 1106519820
Source B 1
Dest Prog.RoutineScan[Prog.Routine] 1106519820

Configuration
Toggle BIts to Enable Simulation and Tests

EN - enables Simulation and blocks User Mapping of Data
Test1EN - enables Test 1 - each bit turned on/off in order
Test2EN - enables Test 2 - each bit turned on in order then off in order

2

Simultation for Fault Data Enable Simulation
SimFault.EN

Simultation for Fault Data Test 1 Enable
SimFault.Test1EN

Simultation for Fault Data Test 2 Enable
SimFault.Test2EN

NOP

Simulation of Fault Data

Test 1 Delay Timer

3

Simultation for Fault Data Enable Simulation
SimFault.EN

Simultation for Fault Data Test 1 Enable
SimFault.Test1EN

Simultation for Fault Data Test 1 Delay Timer

MOVE
Source 60000
Dest SimFault.Test1DelayTMR.PRE
60000

Simultation for Fault Data Test 1 Delay Timer

TON
Timer SimFault.Test1DelayTMR
Preset 60000
Accum 0
EN
DN

Simultation for Fault Data Test 1 Enable 1-shot
SimFault.Test1EN_ONS
ONS

Simultation for Fault Data Test 1 Done
SimFault.Test1DN
U

Simultation for Fault Data Test 2 Bit Set Done
SimFault.Test2SetDN
U

Simultation for Fault Data Test 2 Done
SimFault.Test2DN
U

Simultation for Fault Data Fault Packet Index

CLR
Dest SimFault.PKTi
0

Simultation for Fault Data Fault Word Index

CLR
Dest SimFault.WDi
0

Simultation for Fault Data Fault Bit Index

CLR
Dest SimFault.BITi
0

Equipment Data Fault/Status Message Data - Predefined Status

FLL
Source 0
Dest FaultMsg.PreDef
Length 1

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined

FLL
Source 0
Dest FaultMsgCtl.EqDef[0]
Length CFG.FaultEq.n

Simultation for Fault Data Test 1 Interval Preset Time

MOVE
Source 1000
Dest SimFault.Test1IntervalPre
0

Simultation for Fault Data Test 1 Interval Last Time

CLR
Dest SimFault.Test1IntervalLast
0

Simulation of Fault Data

Test 1 Signal Test Timer
Use free-running timer and 1-second interval to avoid accumulating Timer overruns throughout test.

4

Simultation for Fault Data Test 1 Delay Timer
SimFault.Test1DelayTMR.DN

Simultation for Fault Data Test 1 Interval Timer

MOVE	
Source	999999999
Dest	SimFault.Test1IntervalTMR.PRE 999999999

Simultation for Fault Data Test 1 Interval Timer

TON	
Timer	SimFault.Test1IntervalTMR
Preset	999999999
Accum	0

EN
DN

Simultation for Fault Data Test 1 Interval Accum Time

SUB	
Source A	SimFault.Test1IntervalTMR.ACC 0
Source B	SimFault.Test1IntervalLast 0
Dest	SimFault.Test1IntervalAcc 0

Simultation for Fault Data Test 1 Interval Accum Time

GE	
Source A	SimFault.Test1IntervalAcc 0
Source B	SimFault.Test1IntervalPre 0

Simultation for Fault Data Test 1 Interval Last Time

ADD	
Source A	SimFault.Test1IntervalLast 0
Source B	SimFault.Test1IntervalPre 0
Dest	SimFault.Test1IntervalLast 0

Simultation for Fault Data Test 1 Interval
SimFault.Test1Interval

Simulation of Fault Data

Test 1 - Set Next Bit

5

Simultation for Fault Data Test 1 Interval: SimFault.Test1Interval

Equipment Data Fault/Status Message Data - Predefined Status

FLL
- Source: 0
- Dest: FaultMsg.PreDef
- Length: 1

Fault Ring Buffer Message Control for multiplexed packets Equipment Data Fault/Status Equipment Defined

FLL
- Source: 0
- Dest: FaultMsgCtl.EqDef[0]
- Length: CFG.FaultEq.n

Simultation for Fault Data Test 1 Done: SimFault.Test1DN (/)

Simultation for Fault Data Temp DINT

CLR
- Dest: SimFault.TempDINT
- 16#0000_0000

Simultation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (L)

Simultation for Fault Data Fault Word Index

EQ
- Source A: SimFault.WDi
- 0
- Source B: 0

Equipment Data Fault/Status Message Data - Predefined Status

COP
- Source: SimFault.TempDINT
- Dest: FaultMsg.PreDef
- Length: 1

Simultation for Fault Data Fault Word Index

NE
- Source A: SimFault.WDi
- 0
- Source B: 0

Simultation for Fault Data Equipment Defined Fault Word Index

SUB
- Source A: SimFault.WDi
- 0
- Source B: 1
- Dest: SimFault.EqWDi
- 7

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

COP
- Source: SimFault.TempDINT
- Dest: FaultMsgCtl.EqDef[SimFault.PKTi].Word[SimFault.EqWDi]
- Length: 1

Test 1 - Increment Indexes
All Bits Set & Last Bit Reset - Test Complete

6

Simultation for Fault Data Test 1 Interval: SimFault.Test1Interval

Simultation for Fault Data Test 1 Done: SimFault.Test1DN

Simultation for Fault Data Test 1 Enable: SimFault.Test1EN (U)

Simulation of Fault Data

Test 1 - Increment Indexes using Packet/DataCount configuration for limits
Check if all Test Bits have been Set
(skip Word 0 when PKTi>0)

7

Simulation for Fault Data Test 1 Interval
SimFault.Test1Interval

Simulation for Fault Data Test 1 Done
SimFault.Test1DN

Simulation for Fault Data Fault Bit Index
ADD
Source A SimFault.BITi 0
Source B 1
Dest SimFault.BITi 0

Simulation for Fault Data Fault Bit Index
GT
Source A SimFault.BITi 0
Source B 31

Simulation for Fault Data Fault Bit Index
MOVE
Source 0
Dest SimFault.BITi 0

Simulation for Fault Data Fault Word Index
ADD
Source A SimFault.WDi 0
Source B 1
Dest SimFault.WDi 0

Simulation for Fault Data Fault Word Index
GT
Source A SimFault.WDi 0
Source B UserCfg.EqDefDataCount[SimFault.PKTi] 8

Simulation for Fault Data Fault Word Index
MOVE
Source 1
Dest SimFault.WDi 0

Simulation for Fault Data Fault Packet Index
ADD
Source A SimFault.PKTi 0
Source B 1
Dest SimFault.PKTi 0

Simulation for Fault Data Fault Packet Index
GT
Source A SimFault.PKTi 0
Source B CFG.FaultEq.ei 2

Simulation for Fault Data Fault Packet Index
CLR
Dest SimFault.PKTi 0

Simulation for Fault Data Fault Word Index
CLR
Dest SimFault.WDi 0

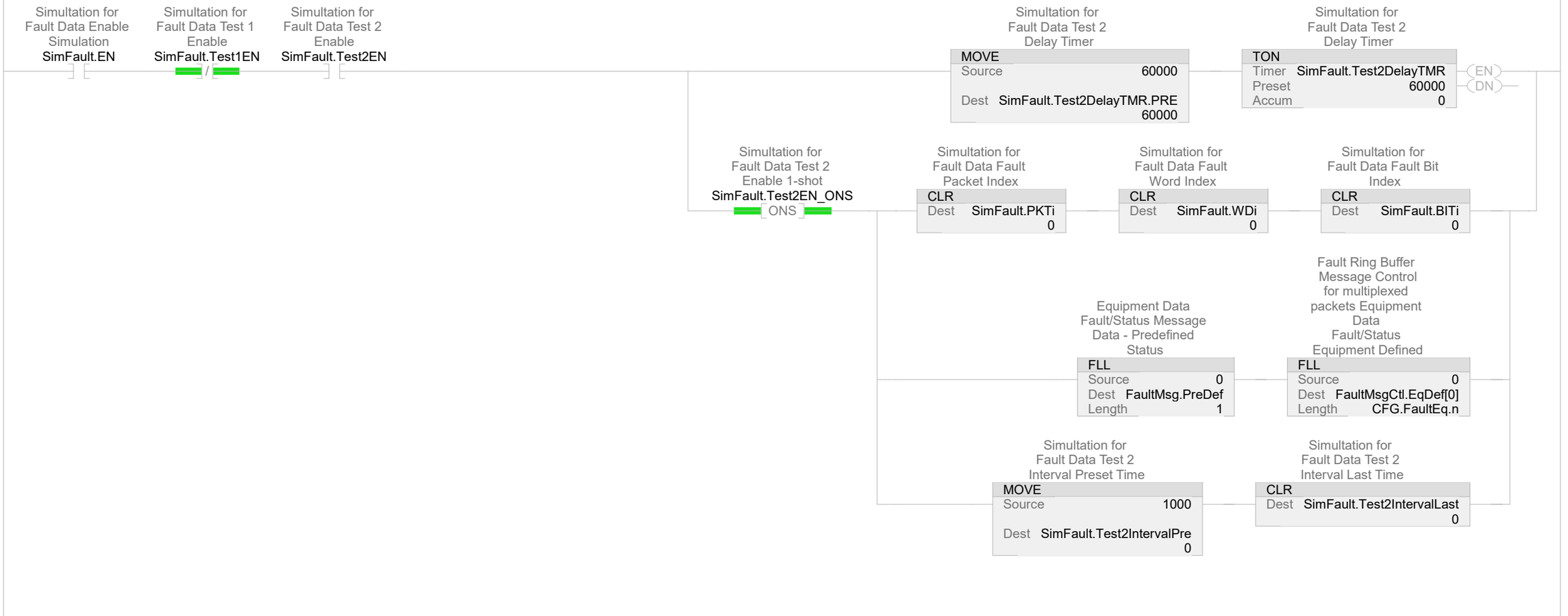
Simulation for Fault Data Fault Bit Index
CLR
Dest SimFault.BITi 0

Simulation for Fault Data Test 1 Done
SimFault.Test1DN
(L)

Simulation of Fault Data

Test 2 Delay Timer

8



Simulation of Fault Data

Test 2 Signal Interval Timer
Use free-running timer and 1-second interval to avoid accumulating Timer overruns throughout test.

9

Simultation for Fault Data Test 2 Delay Timer
SimFault.Test2DelayTMR.DN

Simultation for Fault Data Test 2 Interval Timer

MOVE	
Source	999999999
Dest	SimFault.Test2IntervalTMR.PRE 999999999

Simultation for Fault Data Test 2 Interval Timer

TON		
Timer	SimFault.Test2IntervalTMR	EN
Preset	999999999	DN
Accum	0	

Simultation for Fault Data Test 2 Interval Accum Time

SUB	
Source A	SimFault.Test2IntervalTMR.ACC 0
Source B	SimFault.Test2IntervalLast 0
Dest	SimFault.Test2IntervalAcc 0

Simultation for Fault Data Test 2 Interval Accum Time

GE	
Source A	SimFault.Test2IntervalAcc 0
Source B	SimFault.Test2IntervalPre 0

Simultation for Fault Data Test 2 Interval Last Time

ADD	
Source A	SimFault.Test2IntervalLast 0
Source B	SimFault.Test2IntervalPre 0
Dest	SimFault.Test2IntervalLast 0

Simultation for Fault Data Test 2 Interval
SimFault.Test2Interval

Simulation of Fault Data

Test 2 - Set Next Bit without Reset of Previous Bits

10

Simulation for Fault Data Test 2 Interval: SimFault.Test2Interval

Simulation for Fault Data Test 2 Done: SimFault.Test2DN

Simulation for Fault Data Fault Word Index

EQ
Source A: SimFault.WDi 0
Source B: 0

Simulation for Fault Data Temp DINT

COP
Source: FaultMsg.PreDef
Dest: SimFault.TempDINT
Length: 1

Simulation for Fault Data Test 2 Bit Set Done: SimFault.Test2SetDN

Simulation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (L)

Simulation for Fault Data Test 2 Bit Set Done: SimFault.Test2SetDN

Simulation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (U)

Equipment Data Fault/Status Message Data - Predefined Status

COP
Source: SimFault.TempDINT
Dest: FaultMsg.PreDef
Length: 1

Simulation for Fault Data Fault Word Index

NE
Source A: SimFault.WDi 0
Source B: 0

Simulation for Fault Data Equipment Defined Fault Word Index

SUB
Source A: SimFault.WDi 0
Source B: 1
Dest: SimFault.EqWDi 7

Simulation for Fault Data Temp DINT

COP
Source: FaultMsgCtl.EqDef[SimFault.PKTi].Word[SimFault.EqWDi]
Dest: SimFault.TempDINT
Length: 1

Simulation for Fault Data Test 2 Bit Set Done: SimFault.Test2SetDN

Simulation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (L)

Simulation for Fault Data Test 2 Bit Set Done: SimFault.Test2SetDN

Simulation for Fault Data Temp DINT: SimFault.TempDINT.[SimFault.BITi] (U)

Fault Ring Buffer Message Control for multiplexed packets Equipment Defined Status Words

COP
Source: SimFault.TempDINT
Dest: FaultMsgCtl.EqDef[SimFault.PKTi].Word[SimFault.EqWDi]
Length: 1

Test 2 - Increment Indexes
All Bits Set & Reset - Test Complete

11

Simulation for Fault Data Test 2 Interval: SimFault.Test2Interval

Simulation for Fault Data Test 2 Done: SimFault.Test2DN

Simulation for Fault Data Test 2 Enable: SimFault.Test2EN (U)

Simulation of Fault Data

Test 2 - Increment Indexes using Packet/DataCount configuration for limits
Check if all Test Bits have been Set
(skip Word 0 when PKTi>0)

12

Simultation for Fault Data Test 2 Interval
SimFault.Test2Interval

Simultation for Fault Data Test 2 Done
SimFault.Test2DN

Simultation for Fault Data Fault Bit Index
ADD
Source A SimFault.BITi 0
Source B 1
Dest SimFault.BITi 0

Simultation for Fault Data Fault Bit Index
GT
Source A SimFault.BITi 0
Source B 31

Simultation for Fault Data Fault Bit Index
MOVE
Source 0
Dest SimFault.BITi 0

Simultation for Fault Data Fault Word Index
ADD
Source A SimFault.WDi 0
Source B 1
Dest SimFault.WDi 0

Simultation for Fault Data Fault Word Index
GT
Source A SimFault.WDi 0
Source B UserCfg.EqDefDataCount[SimFault.PKTi] 8

Simultation for Fault Data Fault Word Index
MOVE
Source 1
Dest SimFault.WDi 0

Simultation for Fault Data Fault Packet Index
ADD
Source A SimFault.PKTi 0
Source B 1
Dest SimFault.PKTi 0

Simultation for Fault Data Fault Packet Index
GT
Source A SimFault.PKTi 0
Source B CFG.FaultEq.ei 2

Simultation for Fault Data Test 2 Bit Set Done
SimFault.Test2SetDN

Simultation for Fault Data Test 2 Done
SimFault.Test2DN
L

Simultation for Fault Data Fault Packet Index
CLR
Dest SimFault.PKTi 0

Simultation for Fault Data Fault Word Index
CLR
Dest SimFault.WDi 0

Simultation for Fault Data Fault Bit Index
CLR
Dest SimFault.BITi 0

Simultation for Fault Data Test 2 Bit Set Done
SimFault.Test2SetDN
L

Simulation of Fault Data

Disable Simulation when Tests Complete

13

Simultation for Fault Data Test 1 Enable
SimFault.Test1EN

Simultation for Fault Data Test 2 Enable
SimFault.Test2EN

Simultation for Fault Data All Tests Done Timer

MOVE
Source 60000
Dest SimFault.TestsDoneTMR.PRE
60000

Simultation for Fault Data All Tests Done Timer

TON
Timer SimFault.TestsDoneTMR
Preset 60000
Accum 0
EN
DN

Simultation for Fault Data All Tests Done Timer
SimFault.TestsDoneTMR.DN

Simultation for Fault Data Enable Simulation
SimFault.EN
U

Simultation for Fault Data All Tests Done Timer
SimFault.TestsDoneTMR
RES

(End)

Log Transactions

Transaction Log

Capture Event for each Transaction
Code = Transaction Code (see Excel worksheet for definitions)
Time = Transaction Time (msec)
Num = Transaction Number

0

Transaction Log Control SBR Input Parameter -- Transaction Code

SBR	
Input Par	Trans.Ctl.Code
Input Par	Trans.Ctl.Time
Input Par	Trans.Ctl.Num

NOP

Exit if Transaction Log is not Enabled

1

EDC Config Enable Transaction Logging
CFG.TransLogEN

lblExit
JMP

Load Current Routine Number for Fault Routine diagnostics
Increment Routine Scan Counter (activity indicator for debug)

2

Local Program Scan Time & Fault Routine Routine Number currently Scanned

MOVE	
Source	902
Dest	Prog.Routine 154

Local Program Scan Time & Fault Routine Routine Scan Counters (Indexed by Routine Number)

ADD	
Source A	Prog.RoutineScan[Prog.Routine] 1106519820
Source B	1
Dest	Prog.RoutineScan[Prog.Routine] 1106519820

Log Transactions

Exit if Transaction Code, LNi and EQi are not within filtered logging values.

3

Transaction Log Control Enable Code Filters
Trans.Ctl.Filter.CodeEN
/

Transaction Log Control Log Record
Trans.Ctl.Filter.Log

Transaction Log Control SBR Input Parameter -- Transaction Code

LIMIT
Low Limit: Trans.Ctl.Filter.CodeLow 0
Test: Trans.Ctl.Code 141
High Limit: Trans.Ctl.Filter.CodeHigh 0

Transaction Log Control Log Record
Trans.Ctl.Filter.Log
/

lblExit
JMP

Increment Log Index and check for legal value

4

Transaction Log Control -- Transaction Log Index

ADD
Source A: Trans.Ctl.LOGi 0
Source B: 1
Dest: Trans.Ctl.LOGi 0

Transaction Log Control -- Transaction Log Index

LT
Source A: Trans.Ctl.LOGi 0
Source B: 0

Transaction Log Control -- Transaction Log Index

GE
Source A: Trans.Ctl.LOGi 0
Source B: CFG.TransLOG.n 150

Transaction Log Control -- Transaction Log Index

CLR
Dest: Trans.Ctl.LOGi 0

Log Transactions

Generate Log Index for clearing Next Transaction

5

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

ADD
Source A: Trans.Ctl.LOGi 0
Source B: 1
Dest: Trans.Ctl.LOGx 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

LT
Source A: Trans.Ctl.LOGx 0
Source B: 0

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

GE
Source A: Trans.Ctl.LOGx 0
Source B: CFG.TransLOG.n 150

Transaction Log Control -- Transaction Log Index +1 (for clearing next record)

CLR
Dest: Trans.Ctl.LOGx 0

Initialize Transaction and Next Transaction

6

Transaction Log Control Transaction Log

FLL
Source: 0
Dest: Trans.Log[Trans.Ctl.LOGi]
Length: 1

Transaction Log Control Transaction Log

FLL
Source: 0
Dest: Trans.Log[Trans.Ctl.LOGx]
Length: 1

Log Transactions

Store Transaction Header
-- Transaction Code, Time, Point Index
-- DateStamp (YYMMDD), TimeStamp (HHMMSS), TimeMicroSec (microseconds), Scan Counter

7

Transaction Log Control Transaction Code

MOVE
Source Trans.Ctl.Code
141
Dest Trans.Log[Trans.Ctl.LOGi].TransCode
0

Transaction Log Control Transaction Time

MOVE
Source Trans.Ctl.Time
15
Dest Trans.Log[Trans.Ctl.LOGi].TransTime
0

Transaction Log Control Transaction Number

MOVE
Source Trans.Ctl.Num
9993
Dest Trans.Log[Trans.Ctl.LOGi].Num
0

Transaction Log Control Transaction Date (YYMMDD)

MOVE
Source CLX.SysTime[8]
20261030
Dest Trans.Log[Trans.Ctl.LOGi].DateStamp
0

Transaction Log Control Transaction Time (HHMMSS)

MOVE
Source CLX.SysTime[9]
152452
Dest Trans.Log[Trans.Ctl.LOGi].TimeStamp
0

Transaction Log Control Transaction Time (microseconds)

MOVE
Source CLX.SysTime[6]
273675
Dest Trans.Log[Trans.Ctl.LOGi].TimeMicroSec
0

Transaction Log Control Transaction Scan Couner

MOVE
Source CLX.ScanCount
1106690484
Dest Trans.Log[Trans.Ctl.LOGi].ScanCount
0

Store Optional Transaction Details
(loaded before calling Log Routine)

8

Transaction Log Control Optional Parameter 1 (defined by Transaction)

MOVE
Source Trans.Ctl.Opt[1]
-1
Dest Trans.Log[Trans.Ctl.LOGi].Opt1
0

Transaction Log Control Optional Parameter 2 (defined by Transaction)

MOVE
Source Trans.Ctl.Opt[2]
-1
Dest Trans.Log[Trans.Ctl.LOGi].Opt2
0

Transaction Log Control Optional Parameter 3 (defined by Transaction)

MOVE
Source Trans.Ctl.Opt[3]
-1
Dest Trans.Log[Trans.Ctl.LOGi].Opt3
0

Transaction Log Control Optional Parameter 4 (defined by Transaction)

MOVE
Source Trans.Ctl.Opt[4]
-1
Dest Trans.Log[Trans.Ctl.LOGi].Opt4
0

Transaction Log Control Optional Parameter 5 (defined by Transaction)

MOVE
Source Trans.Ctl.Opt[5]
-1
Dest Trans.Log[Trans.Ctl.LOGi].Opt5
0

Log Transactions

Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

Events codes should be assigned as follows:
xx0 = Enable -- Yellow Status for Normal Operation in Process
xx1 = Done -- Green Status for Normal Operation Completed
xx2 = Retry -- Orange Status for Operation in Retry
xx3 = Error -- Red Status for Operation Failed
xx4-xx9 = Other Errors -- see TransLog Definitions

9xx = Initialization Events
v1.3 Block Retry events from Error Log

9

Transaction Log Control -- Transaction Code Last Digit (used to determine Fault Severity)

MOD
Source A Trans.Ctl.Code
141
Source B 10
Dest Trans.Ctl.Digit
0

Transaction Log Control -- Transaction Code Last Digit (used to determine Fault Severity)

GE
Source A Trans.Ctl.Digit
0
Source B 3

LogErr

Transaction Log Control SBR Input Parameter -- Transaction Code

GE
Source A Trans.Ctl.Code
141
Source B 900

LogErr
/

lblExit
JMP

Error Transaction Log

Capture Event for each Error Transaction in separate Error Transaction Log
(preserves errors from being overwritten by normal transactions)

10

NOP

Log Transactions

Increment Error Log Index and check for legal value

11

Transaction Log Control -- Transaction Error Log Index

ADD
Source A: Trans.Ctl.ERRi 0
Source B: 1
Dest: Trans.Ctl.ERRi 0

Transaction Log Control -- Transaction Error Log Index

LT
Source A: Trans.Ctl.ERRi 0
Source B: 0

Transaction Log Control -- Transaction Error Log Index

GE
Source A: Trans.Ctl.ERRi 0
Source B: CFG.TransERR.n 50

Transaction Log Control -- Transaction Error Log Index

CLR
Dest: Trans.Ctl.ERRi 0

Generate Error Log Index for clearing Next Transaction

12

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

ADD
Source A: Trans.Ctl.ERRi 0
Source B: 1
Dest: Trans.Ctl.ERRx 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

LT
Source A: Trans.Ctl.ERRx 0
Source B: 0

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

GE
Source A: Trans.Ctl.ERRx 0
Source B: CFG.TransERR.n 50

Transaction Log Control -- Transaction Error Log Index +1 (for clearing next record)

CLR
Dest: Trans.Ctl.ERRx 0

Log Transactions

Copy Transaction to Transaction Error Log and Initialize Next Transaction Error Log

13

Transaction Log Control Transaction Log

COP
Source Trans.Log[Trans.Ctl.LOGi]
Dest Trans.Err[Trans.Ctl.ERRi]
Length 1

Transaction Log Control Transaction Log

FLL
Source 0
Dest Trans.Err[Trans.Ctl.ERRx]
Length 1

Exit Routine

14

lblExit
LBL

NOP

Clear Optional Transaction Details
Clear i to avoid bad indexes on next Transaction

15

Transaction Log Control SBR External Load -- Optional Parameters

FLL
Source -1
Dest Trans.Ctl.Opt[0]
Length Trans.Ctl.OptN

(End)

eAssy_EDC_Gway
MainTask
EDC_EqDriver
_001_Main
Ladder Diagram 1
_002_Fault
Ladder Diagram 15
_009_ConfigInit
Ladder Diagram 19
_010_Reset
Ladder Diagram 21
_011_Config
Ladder Diagram 28
_012_Initialize
Ladder Diagram 57
_030_CLxStatus
Ladder Diagram 59
_031_AsciiTable
Ladder Diagram 70
_050_DuplicateMsgDetect
Ladder Diagram 72
_090_USER_Config
Ladder Diagram 80
_092_USER_LogixRemote
Ladder Diagram 83
_093_USER_LogixFaults
Ladder Diagram 85
_094_USER_LogixTools
Ladder Diagram 123
_095_USER_LogixTrends
Ladder Diagram 139
_096_USER_LogixCycles
Ladder Diagram 140
_100_PreDefFaultData
Ladder Diagram 141
_112_CalloutDataLoad
Ladder Diagram 150
_114_CalloutDataMsg
Ladder Diagram 155
_115_CalloutDataMask
Ladder Diagram 166
_121_FaultDataCollect
Ladder Diagram 179
_122_FaultDataLoadRB
Ladder Diagram 184
_123_FaultRingBuffer
Ladder Diagram 190
_124_FaultDataMsg

Ladder Diagram 201
_132_QualityDataLoadRB
Ladder Diagram 216
_133_QualityRingBuffer
Ladder Diagram 222
_134_QualityDataMsg
Ladder Diagram 229
_141_TrendDataCollect
Ladder Diagram 242
_142_TrendDataLoadRB
Ladder Diagram 249
_143_TrendRingBuffer
Ladder Diagram 255
_144_TrendDataMsg
Ladder Diagram 263
_152_VersionDataLoad
Ladder Diagram 276
_154_VersionDataMsg
Ladder Diagram 278
_162_TimeSyncDataLoad
Ladder Diagram 284
_164_TimeSyncMsg
Ladder Diagram 286
_171_CycleDataCollect
Ladder Diagram 291
_172_CycleDataLoadRB
Ladder Diagram 297
_173_CycleRingBuffer
Ladder Diagram 303
_174_CycleDataMsg
Ladder Diagram 310
_800_SimFault
Ladder Diagram 323
_902_TransLogSBR
Ladder Diagram 333